

AMD730

General-Purpose Vector AC Drive

User Guide

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Legal Information Statement:

- The product described in this documentation may be operated only by personnel qualified for the specific task in accordance with the relevant documentation.
- All operations on the product must follow respective descriptions provided in the documentation, in particular, its warning notices and safety instructions.
- Damage caused by improper use is not covered by warranty.
- The company will disclaim any legal liability for any personal injury or property damage caused by improper usage.

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About the Guide

Scope of Application

This guide describes the specifications, installation, commissioning, operation and maintenance of the AMD730 General-Purpose Vector AC Drive, helping you understand the safety information and get familiar with the functions and usage of the drive.

Intended Users

This guide is applicable to industrial personal computer (IPC) operators and qualified electrical technicians.

- They should have certain professional knowledge of electronics, electrical wiring and machinery, and be familiar with electrical and mechanical schematic diagram.
- They should have received professional training related to the installation and commissioning of electrical equipment.
- They should be able to respond to dangers or unexpected situations in the process of installation or trial operation.
- They should be familiar with local standards and related safety specifications of electrical systems.
- Read this guide completely and master the safety matters related to operation.

Usage of this Guide

Read this user guide before use.

- This guide may be updated from time to time due to product upgrade or other reasons. The user guide delivered with your product shall prevail.
- Unless otherwise agreed, this guide is only used as an instruction, and all statements, information and suggestions in this guide do not constitute any express or implied warranty.

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Safety Information and Precautions

To avoid personal injury or damage to the equipment, matters to be followed are stated as follows:

- Read and follow the Safety Information and Precautions before use.
- Use this product according to the designated environment requirements.
- Follow all safety information and precautions described in the product identification and manual.

The degree of injury and damage caused by improper use of this product is distinguished and described as follows:

 DANGER	This mark indicates that failure to comply with the notice will result in severe personal injury or even death.
 WARNING	This mark indicates that failure to comply with the notice may result in severe personal injury or even death.
 CAUTION	This mark indicates that failure to comply with the notice may result in minor or moderate personal injury or damage to the equipment.
NOTICE	If rated precautions are not taken, it may cause undesirable result or state.

Matters to be followed are described using the following graphic marks:

	This graphic mark indicates contents that must be performed.		This graphic mark indicates contents that must not be performed.
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DANGER

- Install this product on non-combustible materials such as metal.
- Set up the product in a clean place where it does not contact water or oil.
- Installation and wiring must be performed by qualified electricians.
- Installation personnel must be familiar with product installation requirements and relevant technical materials.
- The moving, installation, wiring, and inspection of this product can be performed only after you cut off the power supply, wait at least 10 minutes, and determine that there is no risk of electric shock.
- Follow the proper electrostatic discharge (ESD) procedures and wear an anti-static wrist strap to perform wiring.
- The cables should be properly connected. The energized part must be properly insulated using an insulator.



- Do not place any combustible material around this product.
- Do not place this product around heating elements such as heaters and large wire-wound resistors.
- Do not use this product in a corrosive and inflammable gas environment or in a place close to combustible materials.
- Do not use this product in a place with strong vibration or impact.
- Do not use this product after the cables are immersed in oil or water.
- Do not perform wiring at power-on.
- Do not damage the cables or apply any excessive external force, weight, or pinch to them.
- Do not connect this product directly to the commercial power supply.
- Do not perform installation and wiring in a place with strong electric or magnetic field.
- Do not perform wiring and equipment operations with wet hands.
- Do not reach your hands into this product.



WARNING



- Specialized loading and unloading equipment must be used to handle the product.
- When handling the equipment with bare hands, hold the equipment casing firmly with care to prevent parts from falling.
- Handle the equipment with care during transportation and mind your steps.
- When this product is installed in a terminal device, the terminal device must be equipped with protection. The protection class must comply with relevant IEC standards and local regulations.
- Cables used for wiring must meet cross sectional area and shielding requirements. The shield of the cable must be reliably grounded at one end.



- Do not install the equipment if you find damage, rust, or signs of use on the equipment or accessories upon unpacking.
- Do not install the equipment if you find water seepage or missing or damaged components upon unpacking.
- Do not install the equipment if you find the packing list does not conform to the equipment you received.
- When the product is lifted by a crane, personnel cannot stand or stay under the product.
- Do not modify this product.
- Do not fiddle with the bolts used to fix equipment components or the bolts marked in red.
- Do not connect the input power supply to the output end of the equipment.

CAUTION



- Check whether the equipment or accessories show the evidence of damage, rust, impact, or dampness.
- Check whether the package contents are consistent with the packing list.
- After wiring is completed, ensure that there are no screws fallen or cables exposed in the equipment.
- Make sure that the temperature around the equipment is within the range of temperature and humidity.
- Dispose of the equipment as industrial waste during discarding.



- Do not stand on the equipment or place a weight on it.
- Do not let the equipment fall or invert it during the handling or setup.
- Do not place any barriers around the product and peripheral equipment to hinder ventilation.
- Do not let the equipment suffer from any strong impact.

Safety Label:



- Conduct protective grounding to prevent electric shock.



- Do not touch the device during operation and within a short time after shutdown. Failure to comply may cause burns.



- Do not touch terminals with power-on or within 15 minutes after disconnecting the power supply to prevent the risk of electric shock.



- Before using or maintaining the device, read this guide carefully and follow the device labels and all safety precautions in the guide.



Environmental Protection



Reuse

- Some components of the product can be reused due to high metal content. Dismantle the product into individual components to improve the metal recycling efficiency. Electrical and electronic components contain metal materials that can also be recycled through a specific separation process.



Disposal

- Discard components that cannot be degraded and recycled as industrial wastes according to local regulations.

Chapter 1

Product Information

1.1 Description of the Product Model

AMD730 - 4T 3R7 G - B - **

AMD730 - 4T 3R7 G - B - **	Model ID		N/A: Standard **: Non-standard
	Product configuration		N/A: No built-in braking B: Standard braking
	Product type		G: Heavy-duty P: Light load
	Rated power	SIZE B	3R7: 3.7kW 5R5: 5.5kW 7R5: 7.5kW
		SIZE C	011: 11 kW 015: 15 kW
		SIZE D	018: 18.5 kW 022: 22 kW 030: 30 kW
		SIZE E	037: 37 kW 045: 45 kW
		SIZE G	055: 55 kW 075: 75 kW 090: 90 kW
		SIZE H	110: 110 kW 132: 132 kW
Voltage rating		2T (Three-phase)/2S (Single-phase): 220V 4T: 380V	
Product series		AMD730 General-Purpose Vector AC Drive	

Figure 1-1 Product model description

Table 1-1 Product model list

	Model	Power
SIZE B	AMD730-4T3R7G/5R5P-B	3.7 kW
	AMD730-4T5R5G/7R5P-B	5.5 kW
	AMD730-4T7R5G-B	7.5 kW
SIZE C	AMD730-4T011G/015P-B	11 kW
	AMD730-4T015G/018P-B	15 kW
SIZE D	AMD730-4T018G/022P-B	18.5 kW
	AMD730-4T022G/030P-B	22 kW
	AMD730-4T030G/037P-B	30 kW
SIZE E	AMD730-4T037G/045P-B	37 kW
	AMD730-4T045G-B	45 kW
SIZE G	AMD730-4T055G/075P- (B)	55 kW
	AMD730-4T075G/090P- (B)	75 kW
	AMD730-4T090G/110P- (B)	90 kW
SIZE H	AMD730-4T110G/132P	110 kW
	AMD730-4T132G/160P	132 kW

NOTICE

- In the product model list, models with a pair of parentheses have optional specifications.

1.2 Nameplate

The product model and related information can be seen on the model label and nameplate attached to the outer packaging of the product.

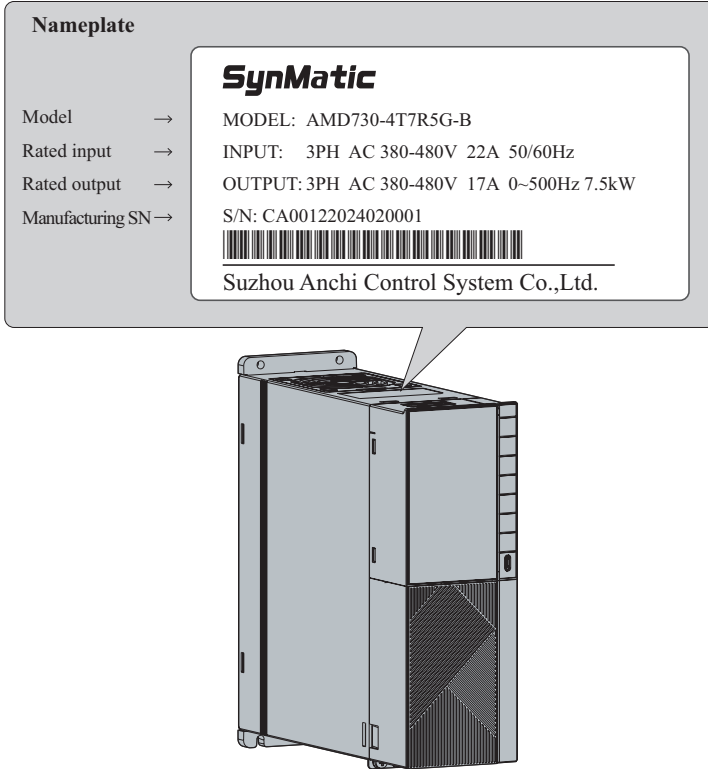


Figure 1-2 Product nameplate

1.3 Components

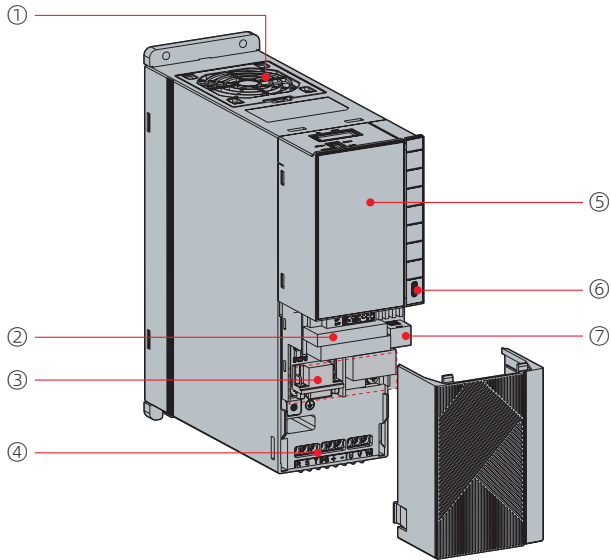


Figure 1-3 Main components of the product

No.	Name	No.	Name
①	Fan	⑤	Display and operation panel
②	Controller wiring terminal	⑥	Commissioning interface
③	Expansion card (Optional)	⑦	Interface for the external operating panel
④	Main circuit wiring terminals	-	-

NOTICE

- The above figure describes the component layout of the SIZE B drive. Component layout of other models may be different.

1.4 Ratings

Power (kW)		Rated Input Current (A)	Rated Output Current (A)
SIZE B	3.7	11.4	9
	5.5	16.7	13
	7.5	22	17
SIZE C	11	32.2	25.0
	15	41.3	32.0
SIZE D	18.5	49.5	37
	22	59	45
	30	70	60
SIZE E	37	80	75
	45	95	91
SIZE G	55	125	112
	75	160	150
	90	190	176
SIZE H	110	220	210
	132	260	253

1.5 Technical Specifications

Item		Specifications
Power input and output	Input voltage	AC: Three-phase 380V (-15%) to 480V (+10%)
	Input current	See the electrical specification list.
	Input frequency	50/60 Hz, actual allowed range: 47.5 Hz to 63 Hz
	Output voltage	0 to input voltage
	Output power	See the electrical specification list.
	Output frequency	0 Hz to 600 Hz
	Output frequency resolution	Digital setting: 0.01 Hz
		Analog setting: Maximum frequency x 0.025%

Item		Specifications
Basic control function	Control mode	V/f control, SVC
	Motor type	Asynchronous motor, permanent magnet synchronous motor, synchronous reluctance motor
	Speed regulation range	1:50 (Asynchronous motor, V/f control)
		1:500 (Asynchronous motor, SVC)
	Speed control accuracy	±1% (V/f control)
		±0.05% (SVC)
	Speed fluctuation	±0.5% (SVC)
	Torque response	< 2 ms (SVC)
	Torque control accuracy	±3% (SVC, Above 5 Hz)
	Starting torque	Speed control: SVC 3Hz, 150% Torque control: SVC 0.3Hz, 200%
	Torque boost	Auto torque boost
		Manual torque boost: 0.1% to 30%
	Overload capacity	SIZE G: 150% 60 s, SIZE P: 110% 60 s
	RUN command source	Three sources: Local/External operating panel, control terminal, and serial port. Multiple ways of switching.
	Frequency source	Eight frequency sources: Digital setting, voltage analog input, current analog input, pulse, communication, PID, multi-speed setting, and built-in PLC. Multiple ways of switching and combination.
	Acceleration/Deceleration curve	Straight line, S-curve mode
	Speed tracking	Inrush-free smooth start of running motor.
	Output voltage self-regulation	Automatic guarantee of constant output voltage upon fluctuations of the mains voltage.
	Auto current limiting	Limiting on output current to prevent frequent trips triggered by overcurrent.
	Fault protection	Protection against short to ground, phase-to-phase short circuit, drive overcurrent, drive overload, drive overvoltage, drive undervoltage, drive over-temperature, input/output phase loss, and so on

Item		Specifications
Input and output functions	Analogy reference power supply	10 V - GND/20 mA Generally used to power an external potentiometer with resistance in the range: 1 k Ω to 5 k Ω
	Digital control power supply	24 V - COM / 200 mA Generally used to power the DI/DO terminals and external sensors.
	Analog input	Two analog inputs: ● AI1-GND: -10 V to +10 V, 12-bit resolution, correction accuracy of 0.3%, input impedance of 22 kΩ for voltage input ● AI2-GND: 0-10 V/0-20 mA/0-40 mA, 12-bit resolution, correction accuracy of 0.3%, input impedance of 22 kΩ for voltage input and 500 Ω for current input, You can use the P4-45 parameter to switch between AI signal types.
	Analog output	Two analog outputs AO-GND Output voltage: 0-10 V; Output current: 0-20 mA You can switch the AO through gears 1 and 2 of the DIP switch S1.
	Digital input	Five ordinary digital inputs (DIO1 can be set as an ordinary digital input using function code P4-04.) ● DIO1-COM and DI1-COM to DI6-COM: response time of 10 ms, high-speed pulse input not supported, input frequency below 100 Hz, optocoupler isolation, compatible with dual-polarity inputs, input impedance of 3.3 kΩ, input voltage range at effective level: 15-30 V ● HDI-COM: configurable as a high-speed pulse input of 100 kHz NPN or PNP input selected using the J9 jumper.
	Digital output	Two digital outputs (DIO1 can be set as an ordinary DO using function code P4-24.) ● DIO1-COM: Ordinary collector open-drain outputs, which must be connected to a pull-up resistor with resistance matching the load before connected to the power supply; maximum output capacity: 24 V DC/50 mA ● DO1-COM: configurable as a high-speed pulse input of 100 kHz, or an ordinary DO with the same specifications as DIO1
	Relay output	One relay output TA-TB: normally closed; TA-TC: normally open Contact capacity: 30 V DC / 3A; 250 V AC / 3A ($\cos\phi = 0.4$)
	Communication interfaces	One RS-485 interface, complying with the standard Modbus protocol. Baud rates supported: 2400/4800/9600/19200/38400/57600/119200 bps
	Expansion interfaces	Two-channel non-discriminatory expansion interfaces, which can be used to support various types of I/O expansion cards and CAN communication cards.

Item		Specifications
Others	Mounting mode	Wall mounting and backplane mounting
	Operating temperature	Operating temperature range:-10 °C to +50 °C Current derated by 1.5% for every temperature rise of 1 °C when the temperature is above 40 °C; maximum temperature: 50 °C
	IP rating	IP20
	Pollution degree	Specifications defined in IEC 60721-3-3:
		Chemical gas: Class 3C2
		Solid particle: Class 3S2
	Cooling mode	Forced air cooling
	Braking unit	Built-in standard for 45 kW and below models; built-in or external optional for 55 kW to 90 kW models; external optional for 110 kW and above models.
	Human-machine interaction	Standard configuration: operating panel with LED segment displays and tri-color light.
		Type-C interface: connected to a backend computer for software debugging and program updating.
		RJ45 interface: connected to the external operating panel for device commissioning.

1.6 Options

Name		Model	Ordering Code	Description
Functional expansion card	I/O expansion card (standard)	AMD760-IO-A1	CA002037	5 DIs, 2 DOs, 1 AI, and 2 AOs
	Relay I/O expansion card (push-in terminal wiring)	AMD760-IO-A3	CA001891	5 DIs, 1 DO, 1 AI, 1 AO, and 1 relay output
	Relay I/O expansion card (screw terminal wiring)	AMD760-IO-A4	CA002110	4 DIs, 1 AO, and 1 relay output
Communication expansion card	CAN communication card	AMD760-CAN-A1	CA001895	1 CAN, 1 PTC temperature, 1 KTY84 temperature, and 2 AIs

Chapter 2

Mechanical Installation



CAUTION



Before
installation

- Check that the packing box is intact and not damped.
- Unpack the box and check for any abnormalities of the equipment, such as deformation, paint loss, cracks, and water stains.

During
installation

- Ensure that the installation site meets the environmental requirements, and that there are no flammable or combustible materials near the equipment.
- Ensure that the ground has sufficient mechanical strength to bear the weight of the equipment.
- Leave sufficient space for heat dissipation of the equipment.
- Use an appropriate covering to protect the equipment from ingress of metal shavings, oil, water, and other substances.
- Tighten all screws with the specified torque. Failure to comply may result in a fire or electric shock.
- Use a flame-retardant mounting bracket if necessary.

After
installation

- Make sure that the equipment has been installed properly and securely, and eliminate the possibility that the fixed parts will fall off due to uneven distribution of force.
- Remove the covering to prevent the equipment from overheating due to poor ventilation.

2.1 Installation Environment

Item	Requirements
Site	Indoors
Relative humidity	Operating: 5% to 95%, non-condensing At most 60% RH allowed in corrosive gases
Altitude	≤ 1000 m, derated by 1% for every rise of 100 m when the altitude is above 1000 m; up to 4000 m altitude allowed for a star power grid and 2000 m for a triangle power grid
Vibration	Operating: 0.5 g
Overvoltage category	OVC III

NOTICE

- When installing the drive in a control cabinet, take into consideration the temperature change of the cooling air. Rapid temperature drop of the cooling air is not allowed, and the air temperature change rate should be below 0.5°C/min.

2.2 Installation Orientation

The AMD730 series drive can only be installed vertically, with the top facing up. A wrong orientation may cause over-temperature.

**CAUTION**

- The AMD730 multidrive product must be installed vertically to facilitate heat dissipation.
- Do not tilt or invert it or install it horizontally. Improper installation orientation may cause over-temperature and then damage to the device.

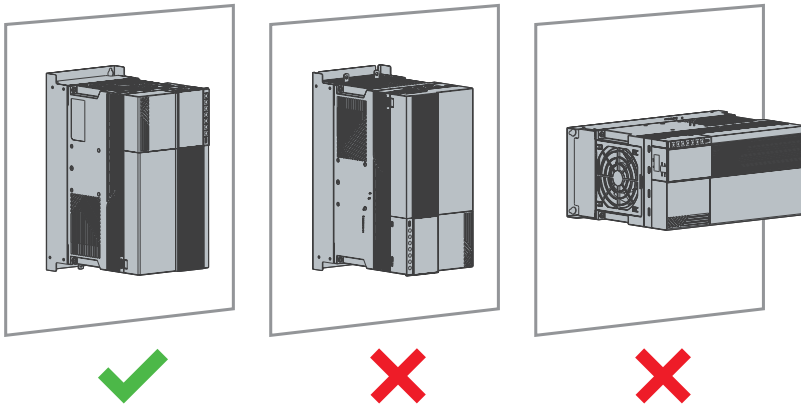


Figure 2-1 Correct and incorrect mounting orientations

2.3 Mounting Clearance

The AMD730 series product can only be installed in an enclosed shell or control cabinet with sufficient space reserved for maintenance and heat dissipation.



CAUTION

- To ensure the cooling effect and uniform temperature distribution in the control cabinet, reserve sufficient space and install a fan.
- When installing a fan in the control cabinet, ensure that the fan can exhaust hot air in the cabinet to the outside. Failure to comply may cause accumulation of hot air, leading to over-temperature or damage to the drive system.
- Clean the air inlet and outlet regularly to avoid the air inlet being blocked by impurities.
- The mounting clearances required vary depending on the product model. Note the following requirements before the installation.

■ Clearances required for single-device installation

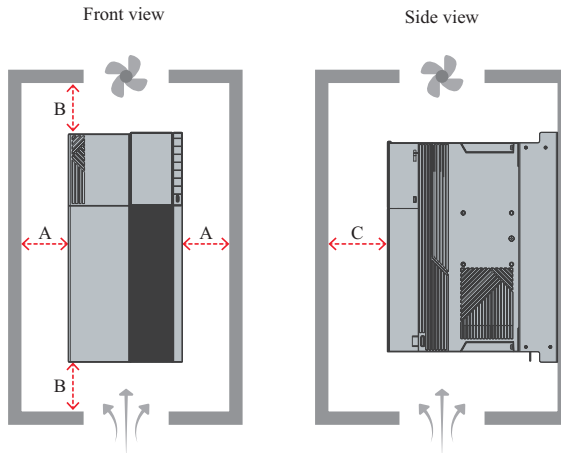


Figure 2-2 Illustration of clearances for single-device installation

Model		Clearance Required (mm)		
		A	B	C
SIZE B	3.7 kW to 7.5 kW	≥10	≥100	≥40
SIZE C	11 kW to 15 kW	≥10	≥200	≥40
SIZE D	18.5 kW to 30 kW	≥10	≥200	≥40

Model		Clearance Required (mm)		
		A	B	C
SIZE E	37 kW to 45 kW	≥50	≥300	≥40
SIZE G	55 kW to 90 kW	≥50	≥300	≥40
SIZE H	110 kW to 132 kW	≥50	≥300	≥40

■ Clearances required for multi-device installation

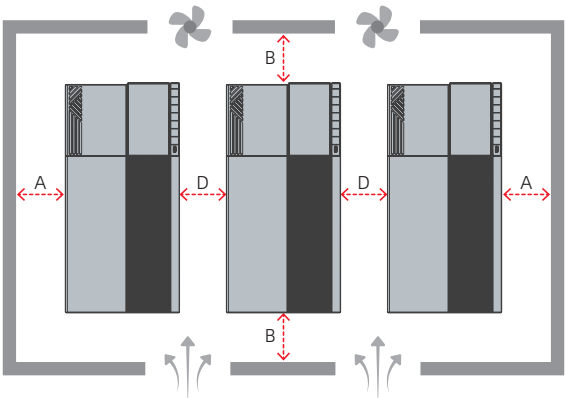


Figure 2-3 Illustration of clearances for multi-device installation

Model		Clearance Required (mm)		
		A	B	C
SIZE B	3.7 kW to 7.5 kW	≥10	≥100	≥30
SIZE C	11 kW to 15 kW	≥10	≥200	≥30
SIZE D	18.5 kW to 30 kW	≥10	≥200	≥30
SIZE E	37 kW to 45 kW	≥50	≥300	≥50
SIZE G	55 kW to 90 kW	≥50	≥300	≥50
SIZE H	110 kW to 132 kW	≥50	≥300	≥50

■ Clearances required for vertically serial installation

When a drive is installed above another drive, heat dissipated from the lower drive will cause the temperature of the upper one to rise. To prevent over-temperature conditions on the upper drive, install a heat insulation baffle between the two drives.

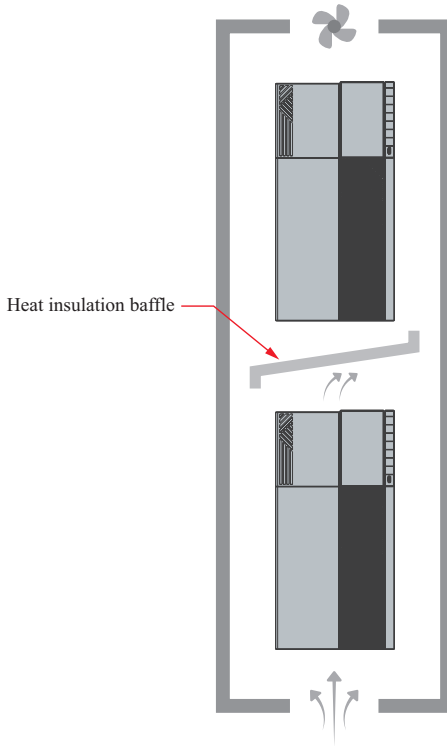


Figure 2-4 Illustration of clearances for vertically serial installation

2.4 Mounting Dimensions

■ SIZE B (3.7 kW to 7.5 kW)

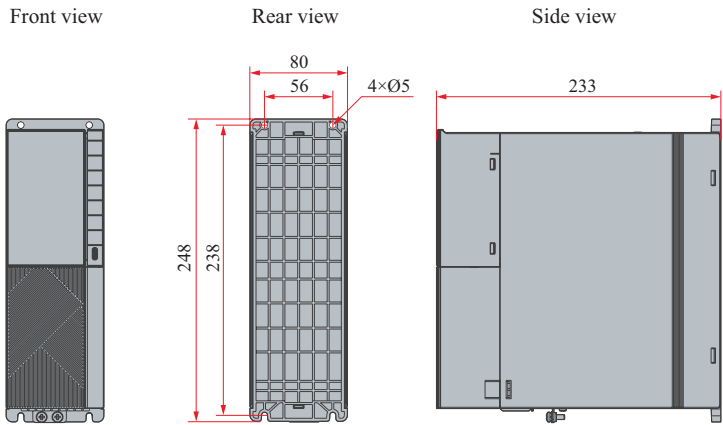


Figure 2-5 Diagram of SIZE B (unit: mm)

■ SIZE C (11 kW to 15 kW)

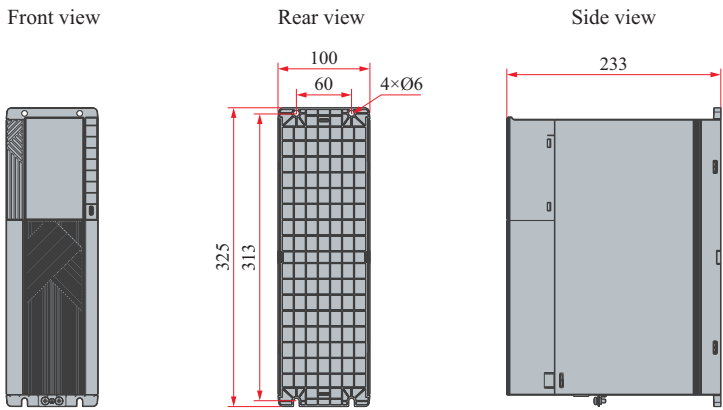


Figure 2-6 Diagram of SIZE C (unit: mm)

■ **SIZE D (18.5 kW to 30 kW)**

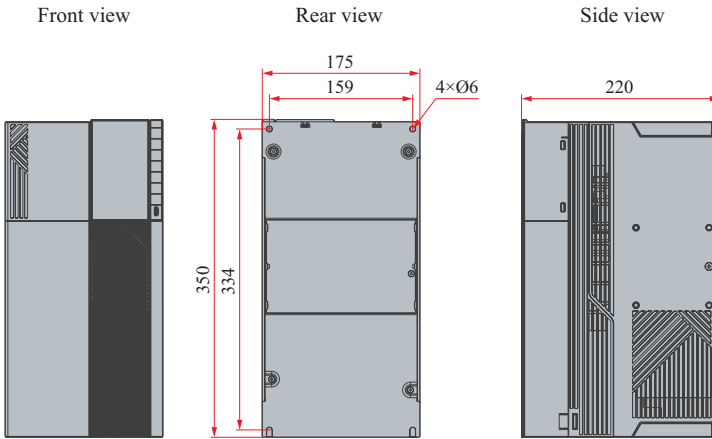


Figure 2-7 Diagram of SIZE D (unit: mm)

■ **SIZE E (37 kW to 45 kW)**

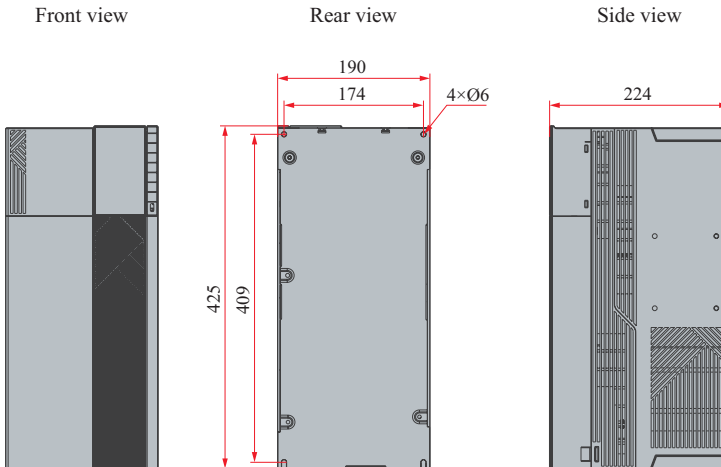


Figure 2-8 Diagram of SIZE E (unit: mm)

■ SIZE G (55 kW to 90 kW)

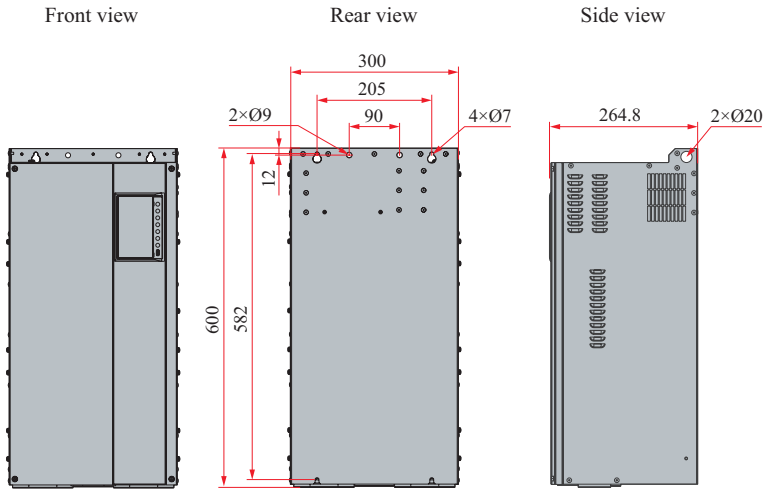


Figure 2-9 Diagram of SIZE G (unit: mm)

■ SIZE H (110 kW to 132 kW)

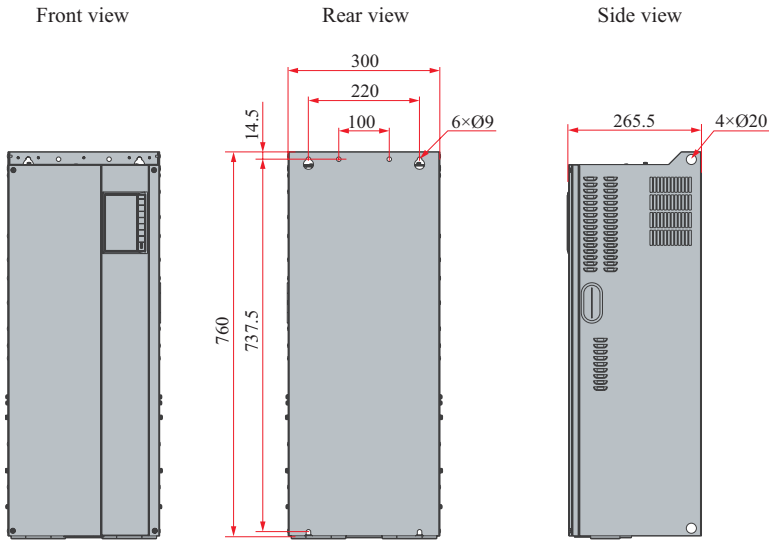


Figure 2-10 Diagram of SIZE H (unit: mm)

2.5 Installation Tools

The following installation tools are recommended. Other auxiliary tools can be used if necessary.

Electric hand drill 	Drilling bit 	Slotted insulation torque screwdriver 	Phillips insulation torque screwdriver 
Marker 	Measuring tape 	Protective gloves 	SEMS screw 

2.6 Mounting Modes

Table 2-1 Mounting modes recommended for various models

Model	Wall Mounting	Backplane Mounting
SIZE B	√	-
SIZE C	√	-
SIZE D	√	√
SIZE E	√	√
SIZE G	√	√
SIZE H	√	√

Refer to corresponding installation instructions in this section based on your product model and its application environment.

CAUTION

- Determine the tightening torque of the mounting screws based on the strength of the screws used and the material of the mounting position. Ensure that no screw is loose or damaged.
- Both the top and bottom of the drive must be secured with screws. If only the top is secured, the drive may fall down due to uneven stress after running for a long time.
- Use a flame-retardant mounting frame if necessary.
- If the drive needs to be used in an environment with metal dust, you are advised to install it in a closed cabinet to protect it from metal dust. In this case, install the radiator outside the cabinet.

2.6.1 Wall Mounting

Before the installation, mark the hole positions on the mounting plate based on the dimensions for mounting holes, and then drill holes.

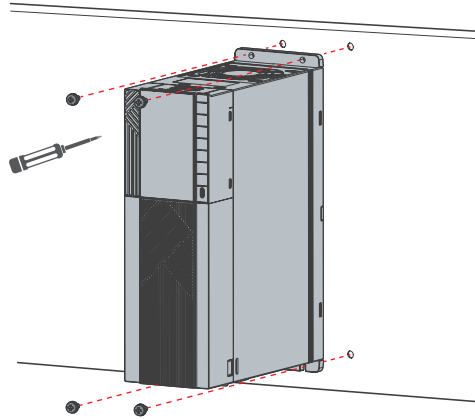


Figure 2-11 Mounting a model of SIZE B/C/D/E on a wall

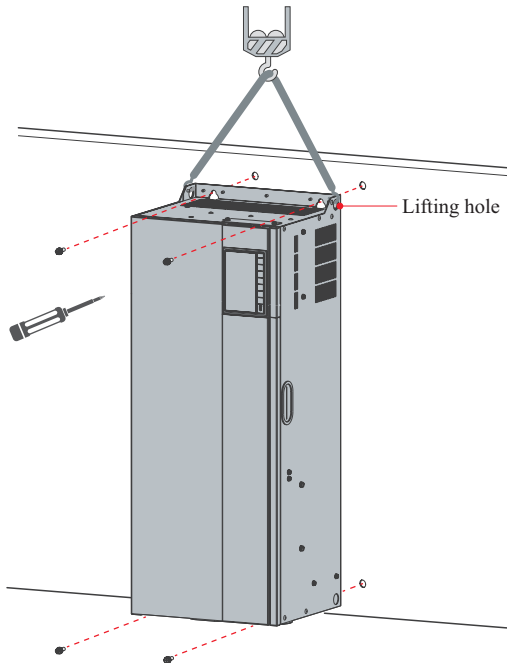


Figure 2-12 Mounting a model of SIZE G/H on a wall

Table 2-2 Screw types

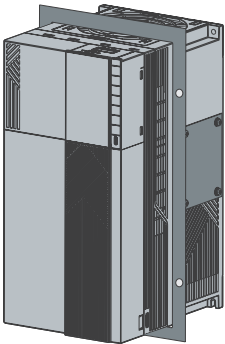
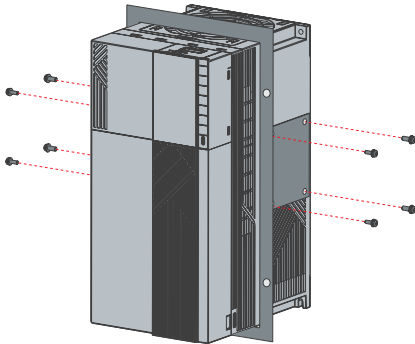
Model	Screw Type	Quantity	Torque
SIZE B	M4	4	1.3 N·m to 1.6 N·m
SIZE C	M5	4	2.5 N·m to 3.0 N·m
SIZE D	M5	4	2.5 N·m to 3.0 N·m
SIZE E	M6	4	3.5 N·m to 4.5 N·m
SIZE G	M6	4	3.5 N·m to 4.5 N·m
SIZE H	M8	4	11 N·m to 13 N·m

2.6.2 Backplane Mounting

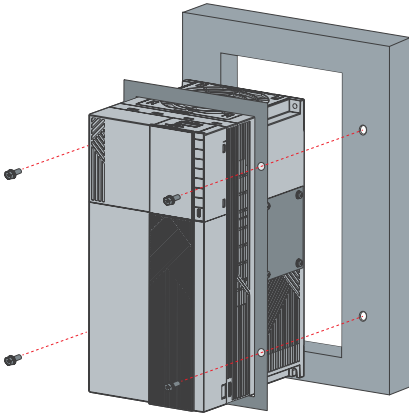
■ SIZE D/E

① Secure the mounting frame on the drive.

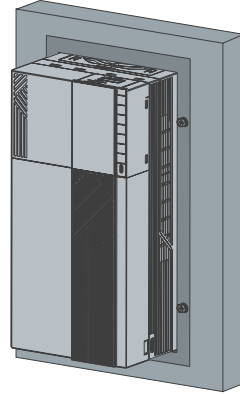
② Complete installation of the mounting frame.



- ③ Embed the drive into the backplane and tighten the screws on the mounting frame.

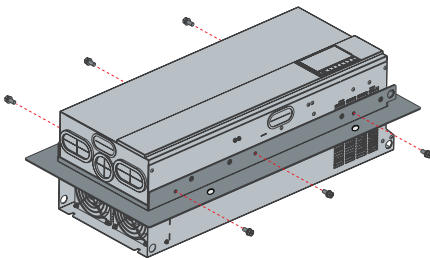


- ④ Complete backplane mounting.

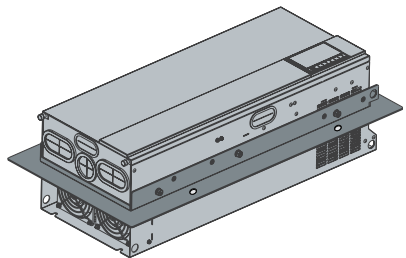


■ SIZE G/H

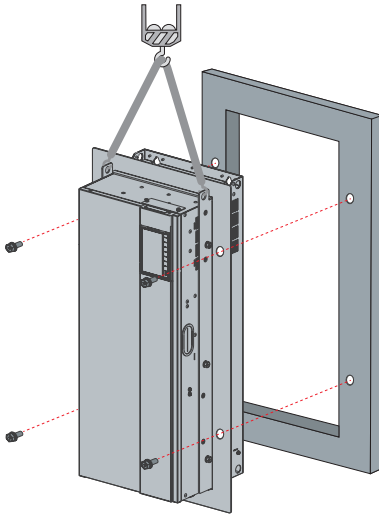
- ① Secure the mounting frame on the drive.



- ② Complete installation of the mounting frame.



- ③ Lift the drive through the lifting holes (2xØ20 mm) on the mounting frame, embed the drive into the backplane, and tighten the screws on the mounting frame.



- ④ Complete backplane mounting.

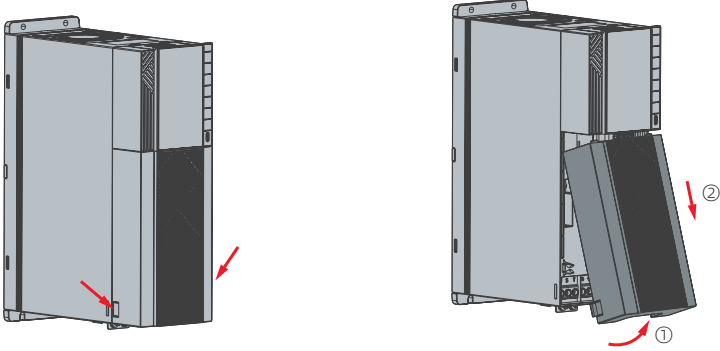


NOTICE

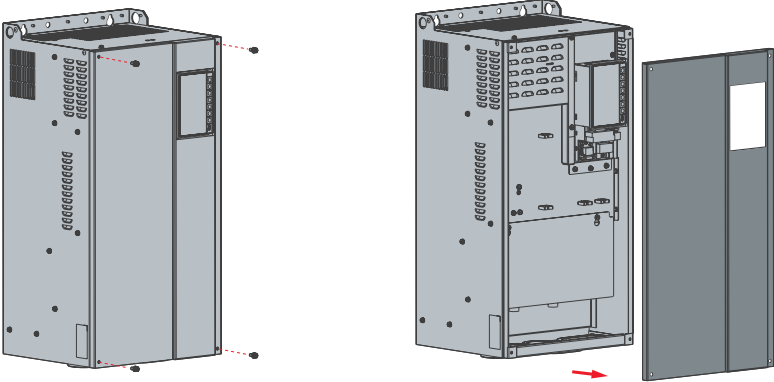
- The mounting frame needs to be self-provided.

2.7 Cover Removal and Installation

■ SIZE B/C/D/E

Removal	① Hold down the clips at both sides of the cover. ② When the clips are released, lift the bottom of the cover, and remove the cover downwards.
	
Installation	Embed top of the cover into the drive and insert the clips at both sides into the slots to lock the cover.

■ SIZE G/H

Removal	① Remove the four M5 combination bolts from the cover. ② Lift the bottom of the cover and remove the cover downwards.
	
Installation	Attach the cover to the device and tighten the combination bolts on the cover.

Chapter 3

Electrical Connection



WARNING



- Do not power on the device before wiring is completed. Failure to comply may result in an electric shock.
- Do not connect the output terminals U, V, and W of the controller to a three-phase power supply. Failure to comply may result in physical injury or a fire.
- Do not connect the motor terminals U, V, and W to a mains power supply. Failure to comply may result in physical injury or a fire.



- Only electrical engineering specialists can perform wiring.
- Ensure that the input voltage of the controller is within the allowable range. Failure to comply may result in product faults.
- Connect an electromagnetic contactor between the input power supply and the main circuit of the controller, to form a structure that can cut off the power supply on the power side of the controller. If no electromagnetic contactor is connected, continuous large current upon controller faults may cause a fire.
- Insulate the connection part of power supply terminals during wiring of the power supply and main circuit. Failure to comply may result in an electric shock.
- Ground the entire system. Failure to comply may result in damage or accidents.
- After power-off, wait at least 15 minutes before further wiring operations because residual voltage exists after power-off. Failure to comply may result in an electric shock.



CAUTION



- Never place cables under heavy objects or drag cables vigorously. Failure to comply may result in an electric shock due to cable damage.
- Protect the controller from ingress of metal shavings, screws, or liquid when connecting the cables. Failure to comply will result in an insulation failure or short circuit.



- Protect external wiring, branches, and short circuits according to local regulations.
- Ensure that the cross-sectional area and voltage withstanding capacity of power cables and control power input cables meet the requirements.
- Strictly follow the specified torque when tightening the screws.
- When using peripheral devices, read the user guide for each component and use it properly after fully confirming the precautions.
- Route the device properly. Improper wiring may cause damage to the controller and motor.

3.1 System Wiring

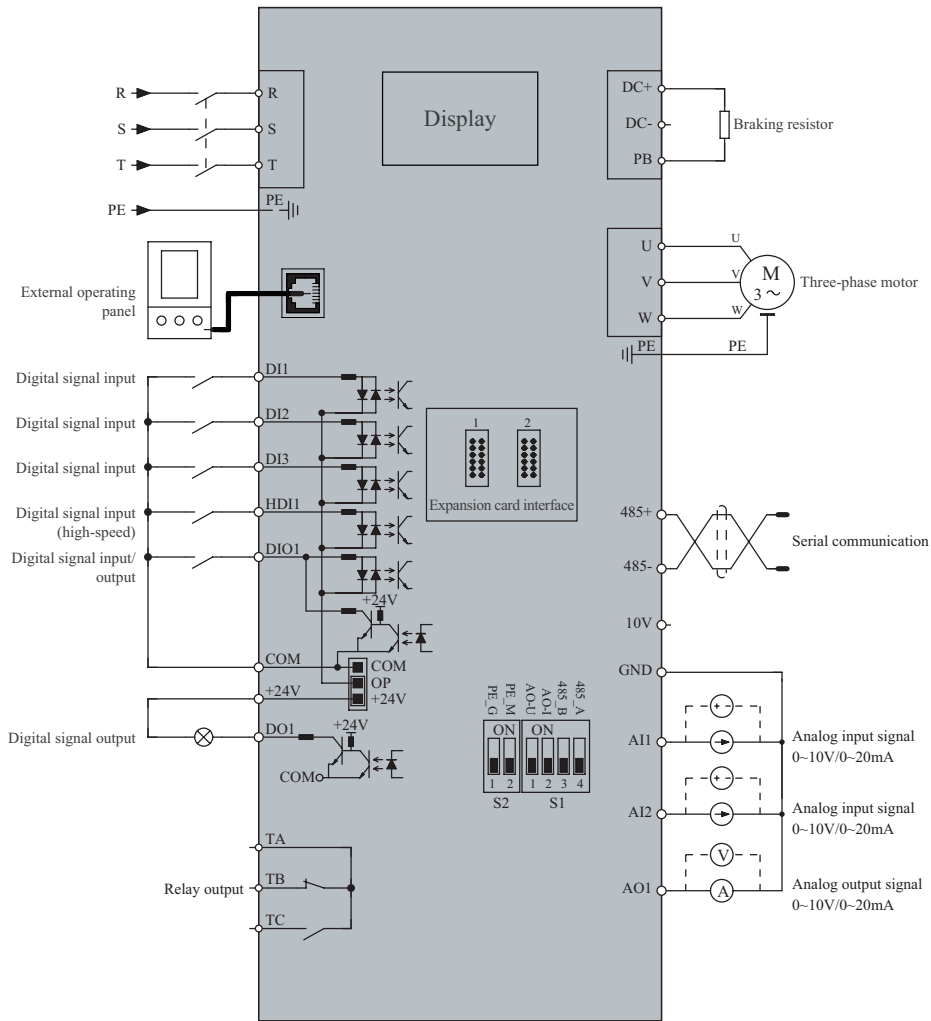


Figure 3-1 System wiring diagram

3.2 Main Circuit Wiring

3.2.1 Main Circuit Terminals

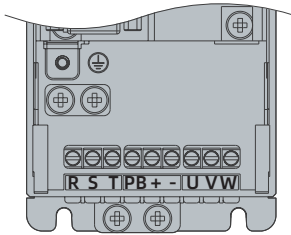


Figure 3-2 Layout of main circuit terminals on
SIZE B

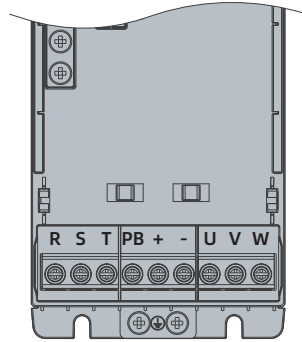


Figure 3-3 Layout of main circuit terminals on
SIZE C

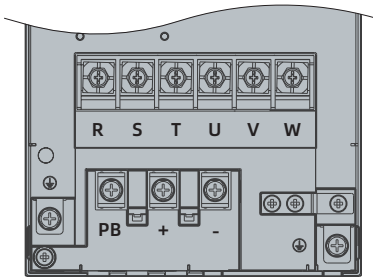


Figure 3-4 Layout of main circuit terminals on
SIZE D/E

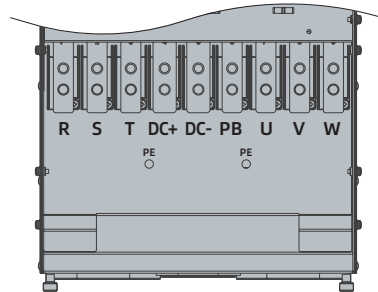


Figure 3-5 Layout of main circuit terminals on
SIZE G

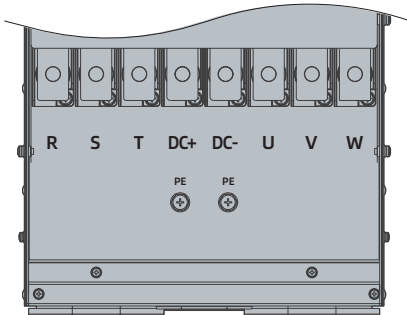


Figure 3-6 Layout of main circuit terminals on
SIZE H

Table 3-1 Description of main circuit terminals

Terminal Symbol	Function Description
R, S, T	Main power input terminal
U, V, W	Controller output terminal
PB, +	Braking resistor wiring terminal
DC+, DC-	DC bus terminal
 , PE	Ground terminal

3.2.2 Main Circuit Cables

■ Cable specifications

Table 3-2 Recommended specifications for main circuit cables

Model	Power (kW)	Input Cable (mm ²)	Output Cable (mm ²)	PE Cable (mm ²)	Motor Ground Cable (mm ²)	Braking Cable (mm ²)
SIZE B	5.5	2.5	1.5	2.5	1.5	1
	7.5	2.5	2.5	2.5	2.5	1.5
SIZE C	11	4	4	4	4	2.5
	15	6	6	6	6	2.5
SIZE D	18.5	10	6	10	6	4
	22	16	10	16	10	4
	30	16	16	16	16	6
SIZE E	37	25	16	25	16	10
	45	25	25	25	25	10
SIZE G	55	50	35	50	35	16
	75	70	50	70	50	25
	90	70	70	70	70	25
SIZE H	110	95	95	95	95	35
	132	120	120	120	120	50



CAUTION

- Observe national or regional regulations when selecting power cables.
- It is recommended that you use an insulated, voltage withstanding 1000 V cable as the braking cable.

■ Lug models

Table 3-3 Recommended lug models for main circuit cables

Model	Power (kW)	Power Input	Power Output	PE	Motor Ground	Braking Terminal	Screw Type	Tightening Torque (N·m)
SIZE B	3.7	E1508	E1008	RNYDS1-4	RNYDS1-4	E1008	M3	0.5 - 0.6
	5.5	E2508	E1508	RNYDS2-4	RNYDS1-4	E1008		
	7.5	E2508	E2508	RNYDS2-4	RNYDS1-4	E1508		
SIZE C	11	E4010	E4010	RNYDS5-4	RNYDS5-4	E2510	M4	1.2
	15	E6010	E6010	RNYDS5-4	RNYDS5-4	E2510		
SIZE D	18.5	SC10-6	SC6-6	SC10-6	SC6-6	SC4-6	M6	4.8
	22	SC16-6	SC10-6	SC16-6	SC10-6	SC4-6		
	30	SC16-6	SC16-6	SC16-6	SC16-6	SC6-6		
SIZE E	37	SC25-6	SC16-6	SC25-5	SC16-5	SC10-5	M6	4.8
	45	SC25-6	SC25-6	SC25-5	SC25-5	SC10-5		
SIZE G	55	SC50-8	SC35-8	SC50-8	SC35-8	SC16-8	M8	13
	75	SC70-8	SC50-8	SC70-8	SC50-8	SC25-8		
	90	SC70-8	SC70-8	SC70-8	SC70-8	SC25-8		
SIZE H	110	SC95-12	SC95-12	SC95-12	SC95-12	SC35-12	M12	35
	132	SC120-12	SC120-12	SC120-12	SC120-12	SC50-12		

NOTICE

- TLC or KST lugs are recommended.

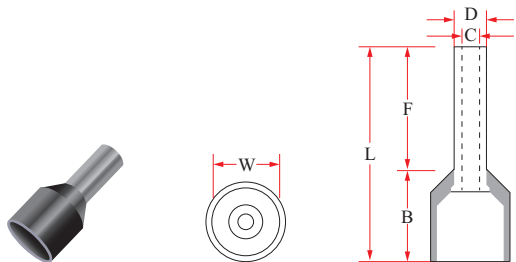


Table 3-4 Specifications and dimensions of cord end terminals

Conductor Size (mm ²)	Terminal Model	Dimensions (mm)					
		F	L	W	B	D	C
1.0	E 1008	8.0	14.3	3.0	6.3	1.7	1.4
1.5	E 1508	8.0	14.5	3.5	6.3	2.0	1.7
2.5	E 2508	8.0	15.4	4.0	7.4	2.6	2.3
2.5	E 2510	10.0	17.4	4.0	7.4	2.6	2.3
4.0	E 4010	10.0	17.4	4.5	7.4	3.2	2.8
6.0	E 6010	10.0	19.5	6.0	8.5	3.9	3.5

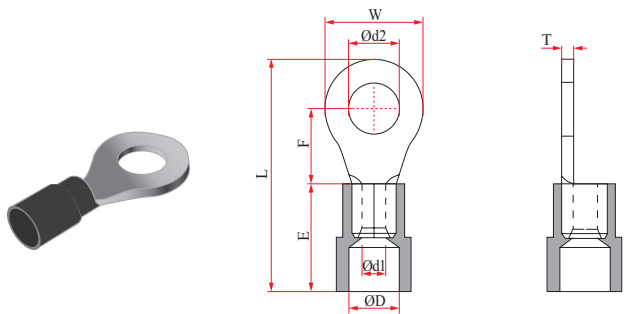


Table 3-5 Specifications and dimensions of ring terminals

Conductor Size (mm ²)	Terminal Model	Dimensions (mm)							
		d2	W	F	L	E	D	d1	T
0.5 to 1.5	RNYDS 1-4	4.3	6.6	6.3	20.4	11.0	4.5	1.7	0.75
1.5 to 2.5	RNYDS 2-4	4.3	6.6	6.3	20.4	11.0	5.2	2.3	0.8
4.0 to 6.0	RNYDS 5-4	4.3	7.2	6.1	22.7	13.0	7.0	3.4	1.0

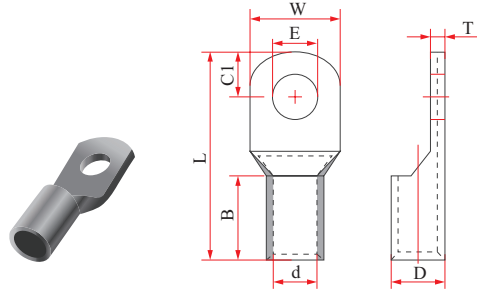


Table 3-6 Specifications and dimensions of SC terminals

Conductor Size (mm ²)	Terminal Model	Dimensions (mm)							
		E	D	d	W	B	C1	L	T
0.5 to 1.5	SC 1.5-4	4.3	4.0	2.0	8.0	5.0	4.5	16.5	1.00
1.5 to 2.5	SC 2.5-4	4.3	4.5	2.9	8.0	7.0	4.5	19.0	1.10
2.5 to 4	SC 4-4	4.3	5.2	3.6	10.0	7.0	5.0	21.0	1.00
2.5 to 4	SC 4-6	6.5	5.2	3.6	10.0	7.0	5.0	21.0	1.00
4 to 6	SC 6-4	4.3	5.5	3.8	10.5	10.0	5.5	24.0	1.15
4 to 6	SC 6-6	6.4	5.5	3.8	11.0	10.0	5.5	24.5	1.10
10	SC 10-5	5.3	6.5	4.9	12.0	10.0	5.5	25.0	1.20
10	SC 10-6	6.4	6.5	4.9	12.0	10.0	5.5	25.0	1.20
16	SC 16-6	6.4	7.5	5.8	12.2	11.4	6.5	30.0	1.60
16	SC 16-8	8.4	7.5	5.8	12.2	11.4	6.5	30.0	1.60
25	SC 25-5	5.3	9.5	7.5	13.8	12.0	7.8	35.0	1.90
25	SC 25-6	6.4	9.5	7.5	13.8	12.0	7.8	35.0	1.90
25	SC 25-8	8.4	9.5	7.5	15.0	12.0	8.0	35.0	1.8
35	SC 35-8	8.4	10.8	8.5	15.7	14.0	8.0	37.0	2.2
35	SC 35-12	10.5	10.5	8.3	18.7	15.5	9.0	38.0	1.7
50	SC 50-8	8.4	12.6	9.9	18.0	16.0	10.0	45.0	2.5
70	SC 70-8	8.4	15.0	12.0	21.8	19.0	12.0	50.5	3.0
95	SC 95-12	13.0	17.4	13.5	25.5	23.0	12.5	58.0	3.8
120	SC 120-12	13.0	19.5	15.0	28.0	22.0	15.0	63.0	4.4

3.2.3 Main Circuit Wiring Requirements

■ Power input terminals (R, S, T)

- There is no specific requirement for the phase sequence of the wires connected to the input terminals.
- Specifications and installation of all external power cables must comply with local safety regulations and relevant IEC standards.
- Select copper wires of appropriate sizes according to recommendations provided in Table 3-2.
- Install the filter close to power input terminals of the drive, and make sure the cable between the filter and drive is shorter than 30 cm. Connect the ground terminals of the filter and drive together, and keep the filter and drive on the same conducting plane, which is connected to the main ground terminal of the cabinet.

■ DC bus terminals (+, -)

- DC bus terminals, labeled + and -, carry residual voltage for a period after the drive is powered off. To prevent electric shock, wait for at least 15 minutes before connecting cables to the terminals after the drive is powered off.
- When you install an external braking unit on the drive, do not reverse the positive and negative poles, as doing so will damage the drive and braking unit, and even cause fire.
- Use a cable no more than 10 m long to connect the braking unit. Use a twisted pair or tight pair for parallel connection.
- To prevent equipment damage or fire, do not connect the braking resistor directly to the DC bus.

■ Braking resistor connection terminals (PB, +)

- The braking resistor connection terminals are effective only for product models with a built-in braking unit (standard component for models of 45 kW or below, optional for models of 55 kW to 90 kW, and replaced by an external braking unit for models of 110 kW or above).
- Select a braking resistor with recommended specifications and use a cable no more than 5 m long to connect it. Otherwise, the drive may be damaged.
- Do not place any combustible objects around the braking resistor. Otherwise, adjacent components may be burned once the braking resistor overheats.
- If your drive is equipped with a built-in braking unit, set the braking unit utilization rate and braking action starting voltage after connecting the braking resistor.

■ Power output terminals (U, V, W)

- Specifications and installation of all external power cables must comply with local safety regulations and relevant IEC standards.
- Select copper wires of appropriate sizes.
- Do not connect any capacitor or surge protector to the output terminals, as such device can trigger protection actions frequently or even damage the drive.
- Long motor cables can cause electrical resonance due to distributed capacitance. Electrical

resonance may damage the motor insulator or produce high leakage current to trigger overcurrent protection. Therefore, if the cable length exceeds 100 m, install an AC output reactor close to the drive.

■ Ground terminal (PE)

- Use a proper yellow-green copper cable as the protective earthing conductor. Do not connect the cable to a circuit breaker or any other switching device.
- Do not use the PE terminal as a neutral wire terminal (N terminal) of the power supply, or vice versa.
- Ground the PE terminal properly. Otherwise, the drive may be unable to work normally or even be damaged.
- It is recommended that the drive be installed on a conductive metal plane, and ensure that the bottom of the drive is closely attached to the plane.
- Install the filter and drive on the same plane to ensure the filtering effect.

■ Main circuit cable protection

- For main circuit cables, cover the copper tube of the lug and core wires with a heat shrink sleeve, and ensure that the sleeve completely covers the cable conductor.

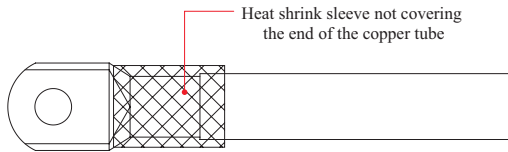


Figure 3-7 Heat shrink sleeve for a cable

■ Power input protection

- Install appropriate protection devices on the power input lines of the drive. The protection devices must provide protection capabilities, such as overcurrent protection, short-circuit protection, and electric isolation.
- Cables and protection devices on power input must be suitably rated for the power and voltage class of the drive under normal conditions, and under possible fault conditions such as system overload and short-circuit on the power input.

■ Requirements for the mains power supply system

- This product is applicable to a mains power system with an earthed neutral point. If it is used in an IT power system with a non-earthed neutral point, remove its grounding varistor (VDR) and grounding safety capacitor (EMC), and do not install a filter. Failure to comply with this requirement may result in personal injury or damage to the drive.
- If a leakage circuit breaker is used and it trips during startup, remove the safety capacitor (EMC).

3.3 Control Circuit Wiring

3.3.1 Control Circuit Terminals

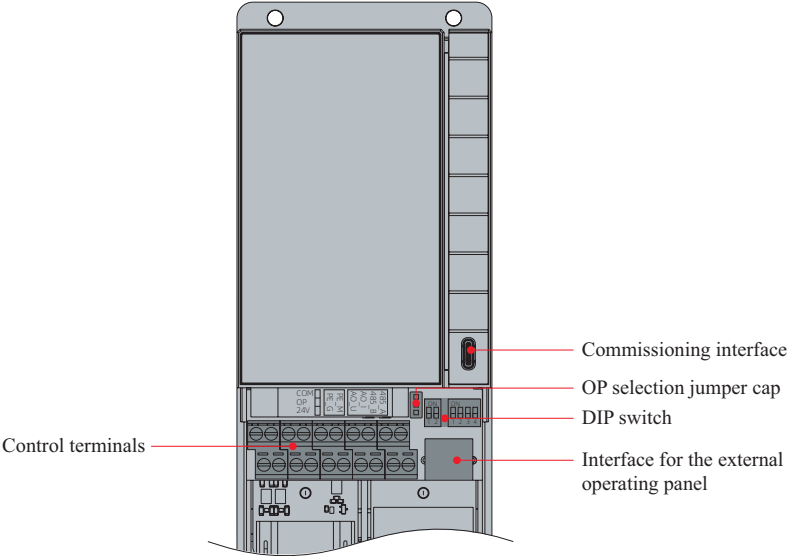
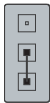
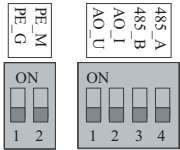


Figure 3-8 Layout of control circuit terminals

NOTICE

- Control circuit terminals are distributed basically the same way on different product models. This figure uses a size B model as an example, which can be used as a reference for cable connections on other models.

Terminal		Description	
		+10V	10 V power output
		AI1	Analog input 1
		AI2	Analog input 2
		AO	Analog output
		GND	Analog ground
		485+	RS-485 positive
		485-	RS-485 negative

Terminal	Description	
	COM	Digital ground
	HDI1	High-speed DI
	TC	Relay output TC
	TB	Relay output TB
	TA	Relay output TA
	DI1	Digital input 1
	DI2	Digital input 2
	DI3	Digital input 3
	DIO1	Digital input/output 1
	DO1	Digital output 1
	COM	Digital ground
	+24V	Digital 24 V power output
 Commissioning interface	Type-C USB, used to connect to a computer for software upgrading and debugging	
 OP selection jumper cap		PNP input on DI
		NPN input on DI
 DIP switch	PE_G	It is in the ON position by default. If a fault is reported mistakenly due to strong interference, turn it to the OFF position.
	PE_M	It is in the ON position by default. If a fault is reported mistakenly due to strong interference, turn it to the OFF position.
	AO_U	It is in the ON position when the analog output is voltage, in which case AO_I needs to be turned OFF.

Terminal	Description	
	AO_I	It is in the ON position when the analog output is current, in which case AO_U needs to be turned OFF.
	485_B/ 485_A	They are used to select terminal resistors and in the ON position by default. When multiple drives are connected in a row, keep these terminals in the ON position on the drives at two ends of the row and turn the terminals OFF on the other drives.
 Interface for the external operating panel	RJ45 interface, reserved for the external operating panel	

3.3.2 Control Circuit Cables

■ Cable specifications

Table 3-7 Recommended specifications for control circuit cables

Function	Cable
Analog input (AI1, AI2)	Shielded twisted-pair cable
Analog output (AO)	Shielded twisted-pair cable
Communication (485+, 485-)	Shielded twisted-pair cable
Control signal	0.5/0.75 mm ² annealed copper cable

■ Terminal models

Cord end terminals with an insulating sleeve are recommended for signal cables. The recommended terminal models are E0508 and E7508.

- Crimp each signal cable into a UL-certified cord end terminal.
- Slip a UL- and CSA-certified insulating heat shrink sleeve (withstanding at least 50 V DC) onto the seam between the cable and the terminal.
- The length of exposed core wires must be within 6 mm.

NOTICE

- For the appearance and dimensions of the terminals needed, see Table 3-4 "Specifications and dimensions of cord end terminals."

3.3.3 Description of Control Circuit Wiring

■ Wiring of analog terminals

Analog voltage signals are susceptible to external interference. Therefore, you are advised to select shielded twisted-pair cables and keep the cables as short as possible (within 20 m). The shield of a cable must be grounded on one side.

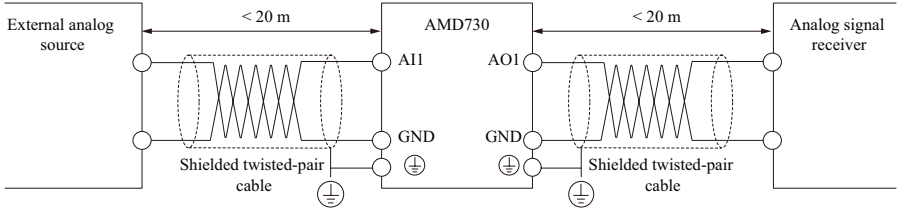


Figure 3-9 Analog input/output terminal wiring diagram

In scenarios where analog signals experience severe external interference, install a filter capacitor or a ferrite core on the twisted-pair cables.

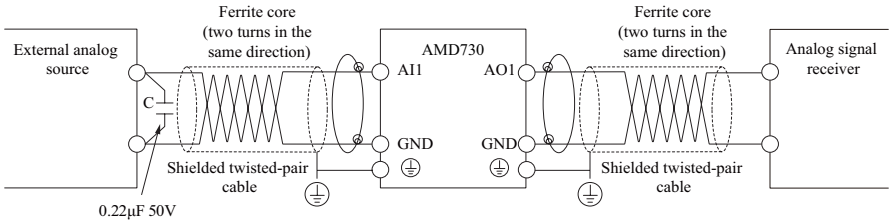


Figure 3-10 Analog input/output terminal wiring diagram

The analog input/output signal type can be set to voltage or current on the back end.

Table 3-8 Analog input parameters

Parameter	Communication Address	Name	Setting Range	Default	Unit	Modification
P4-44	0xA42C	Type of AI1 (AI1)	0: Voltage signal (0-10 V)	0	-	●
P4-45	0xA42D	Type of AI2 (AI2)	0: Voltage signal (0-10 V) 1: Current signal 1 (0-20 mA) 2: Current signal 2 (0-40 mA)	0	-	★
P4-46	0xA42E	Type of AI3 (expansion AI1-I/O card)	0: Voltage signal (0-10 V) 1: Current signal (0-20 mA) 2: PT100 3: PT1000	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Modification
P4-47	0xA42F	Type of AI4 (expansion AI2-I/O card)	0: Voltage signal (0-10 V) 1: Current signal (0-20 mA) 2: PT100 3: PT1000	0	-	★
P4-48	0xA430	Type of AI5 (expansion AI3-resolver PG card)	0: PT100 1: PT1000 2: KTY84	0	-	★
P4-49	0xA431	Type of AO1	0: Voltage signal (0-10 V) 1: Current signal (0-20 mA)	0	-	★
P4-50	0xA432	Type of AO2	0: Voltage signal (0-10 V) 1: Current signal (0-20 mA)	0	-	★

■ Wiring of digital terminals

DI terminals fall into two types: three ordinary DI terminals, with the common terminal named OP, and one HDI1 terminal (which can be used as an ordinary DI terminal), with the common terminal named OP. Select cables for the terminals based on the types of sensors used.

DI signals can be provided by a dry contact or an active switching device (NPN or PNP mode). The controller supports dry contact, NPN, and PNP signals, but NPN and PNP signals cannot coexist in a system.

Single-core or multi-core cables can be used as DI signal cables. In an environment with strong interference, use shielded cables and minimize the cable length (within 50 m). When active devices are used, take necessary filtering measures against crosstalk between power supply units.

Connection of DI terminals:

- Common cathode connection for the dry contact or active device

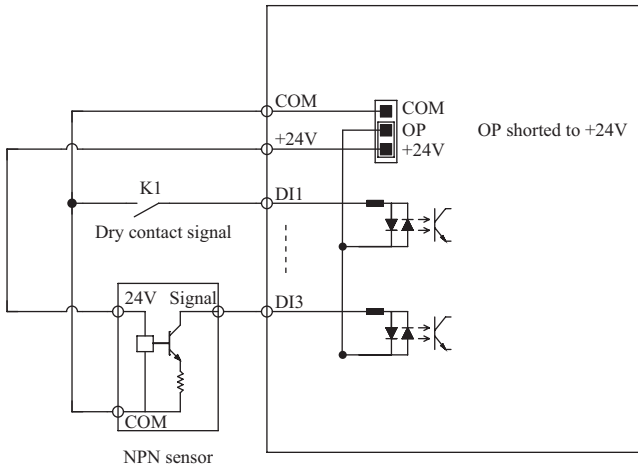


Figure 3-11 Common cathode connection for the dry contact or active device (internal power supply)

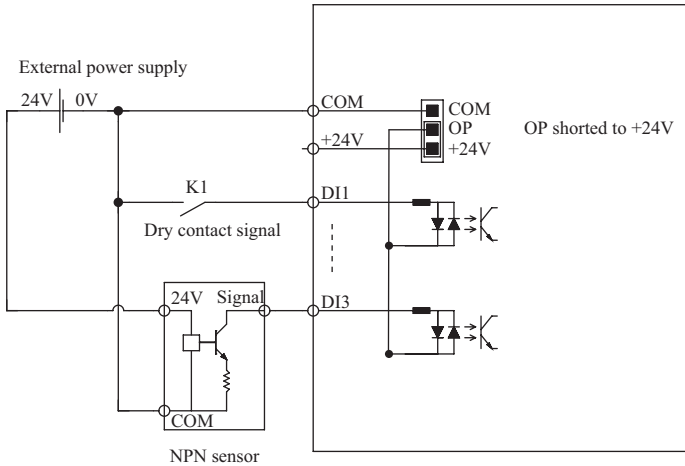


Figure 3-12 Common cathode connection for the dry contact or active device (external power supply)

to control devices such as LED alarm indicators, LED lights, relays, and solenoid valves (snubber diodes required for inductive loads).

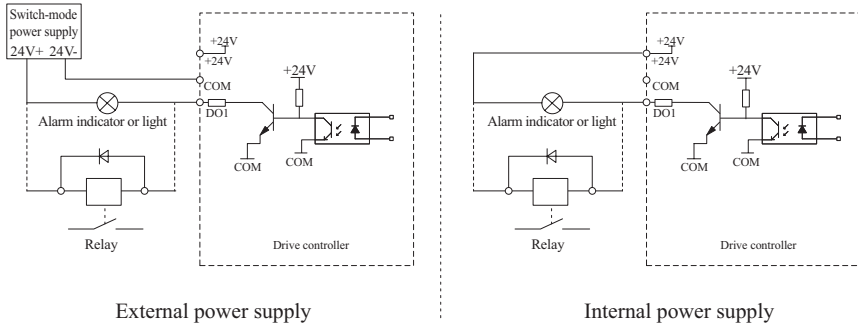


Figure 3-15 DO terminal wiring diagrams

3.3.4 Control Circuit Wiring Requirements

■ Separation of power and signal cables

To reduce interference to control signals, ensure that:

- Signal cables are separated from power cables (power input, drive output, and braking resistor connection cables).
- Signal cables are more than 50 cm away from power cables.
- Signal cables are kept vertical to power cables when their routes overlap.

■ Requirements for analog control signal cables

To prevent interference caused by contact between shielded cables and other devices, ensure that:

- Shielded twisted-pair cables are used as analog control signal cables.
- The stripped part of each cable is 5-7 mm long, and the exposed shield is wrapped with insulating tape.

■ Requirements for motor cables

To reduce interference caused by rapid changes of output voltage from the drive, ensure that:

- The distance between the drive speed regulation cabinet and motor is as short as possible.
- Motor cables are routed separately from other cables and not parallel to other cables over a long distance.

3.4 Braking Resistor Selection

■ Selecting braking resistor resistance

During a braking process, almost all regenerative energy of the motor is consumed by the braking resistor. The resistance of a braking resistor is calculated using the following formula:

$$U \times U / R = P_b$$

U: Braking voltage that ensures stable braking of the system (The value varies with systems, within the range of 380 V to 440 V for three-phase systems. The default braking voltage of a drive is 720 V, which can be adjusted using the parameter P5-65.)

P_b: Braking power

■ Selecting braking resistor power

Theoretically, the power of a braking resistor is the same as the braking power, but the derating coefficient K needs to be considered.

So, the following equation is formed: **K x Pr = P_b x D**

K: Around 50%

Pr: Power of the braking resistor

D: Braking frequency, that is, the percentage of the regenerative process to the whole working process

The following equations can be derived from the equations above:

$$K \times Pr = P_b \times D = U \times U / R \times D$$

$$Pr = (U \times U \times D) / (R \times K)$$

The braking resistor power Pr can be calculated accordingly.

K is the derating coefficient of the braking resistor. A small value of K prevents the braking resistor from overheating. You can increase the value of K properly if heat can be well dissipated, but in no case should K exceed 50%. Failure to comply may result in a fire due to braking resistor overheating.

The braking frequency D is determined by applications. Typical braking frequencies in different applications are listed in the following table.

Application	Occasional braking load	General application (e.g., translational conveying)	Vertical lifting	Winding and unwinding	Machine tool spindle	Centrifuge
Braking frequency	5%	10%	20% to 30%	20% to 30%	30% to 50%	50% to 60%

■ Recommended specifications of braking components

Drive Model	Braking Unit	Total Power	125% Braking Torque (10% ED; max. 10s)	Min. Braking Resistance
AMD730-4T3R7G/5R5P-B	Built-in (standard)	3.7 kW	0.8 kW 140 Ω	53 Ω
AMD730-4T5R5G/7R5P-B		5.5 kW	1.2 kW 95 Ω	
AMD730-4T7R5G-B		7.5 kW	1.6 kW 70 Ω	32 Ω
AMD730-4T011G/015P-B		11 kW	2.2 kW 50 Ω	
AMD730-4T015G/018P-B		15 kW	3.0 kW 38 Ω	20 Ω
AMD730-4T018G/022P-B		18.5 kW	3.8 kW 30 Ω	16 Ω
AMD730-4T022G/030P-B		22 kW	4.6 kW 25 Ω	
AMD730-4T030G/037P-B		30 kW	5.7 kW 20 Ω	12 Ω
AMD730-4T037G/045P-B		37 kW	7.7 kW 15 Ω	
AMD730-4T045G-B		45 kW	9.6 kW 12 Ω	10 Ω
AMD730-4T055G/075P-B	Built-in (optional)	55 kW	11 kW 10 Ω	8 Ω
AMD730-4T075G/090P-B		75 kW	16.5 kW 7 Ω	6 Ω
AMD730-4T090G/110P-B		90 kW	19 kW 6 Ω	4 Ω

NOTICE

- You can adjust the initial braking voltage based on the mains voltage. Higher initial braking voltage requires higher braking resistance. For three-phase power systems of 380 V to 440 V, the default initial braking voltage is 720 V.
- Data in the preceding table is for reference only. You can select the resistance and power of the braking resistor as required. (Note that the resistance cannot be lower than the recommended minimum value, whereas the power can exceed the recommended value.) You need to select the braking resistor model based on the generation power of the motor system inertia, deceleration time, and potential energy load. A larger system inertia requires shorter deceleration time and more frequent braking, which means higher power and lower resistance of the braking resistor.

Chapter 4

System Commissioning

4.1 Operating Panel

The operating panel consists of a display and multiple buttons. You can view information about the drive and enter simple commands on the operating panel.

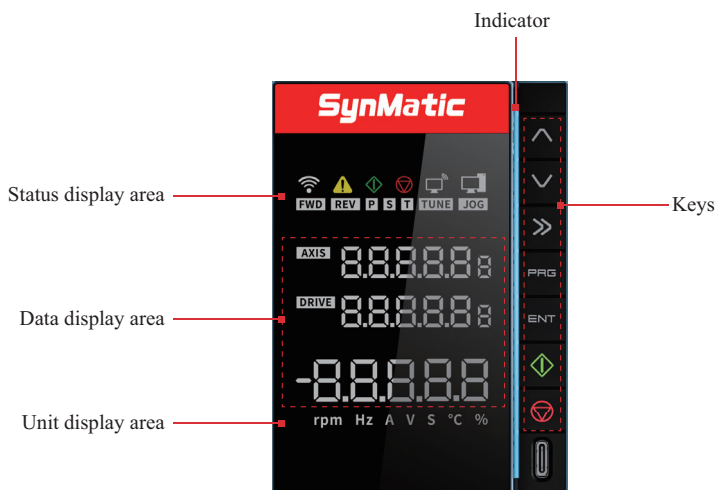
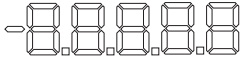
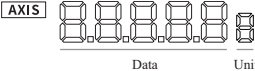
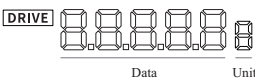


Figure 4-1 LED operating panel

■ Display

Area	Illustration	Name	Description
Status display	FWD	Forward (FWD)	Forward run
	REV	Reverse (REV)	Reverse run
	P	Position mode	Currently run in the position mode.
	S	Speed mode	Currently run in the speed mode.
	T	Torque mode	Currently run in the torque mode.
	TUNE	Auto-tuning	Operating in auto-tuning state

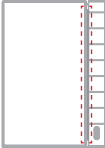
Area	Illustration	Name	Description
		Jog	Operating in jog mode
		USB connection	On: Normal USB connection on the back end Off: No USB connection on the back end
		Alarm	Steady ON: The device fails. Flashing: The device has an alarm.
		Running	Running status
		Stop	Stop status
		Control mode	On: Local control Off: Terminal control Blinking: Communication control
		Master station indicator	On: Drive as the master station
Data display		Primary display area	Negative sign on: A negative value displayed in this area 5-digit LED display: Frequency reference, output frequency, monitoring data, error code, and so on
		Secondary display area 1	5-digit value set for parameter P5-63, which defaults to 5 (bus voltage) Other options described in section 5.3.6 "Group P5–Auxiliary Function Group" Unit display modes:  Current unit  Speed unit  Frequency unit  Voltage unit
		Secondary display area 2	5-digit value set for parameter P5-64, which defaults to 3 (output current) Other options described in section 5.3.6 "Group P5–Auxiliary Function Group" Unit display modes:  Current unit  Speed unit  Frequency unit  Voltage unit

Area	Illustration	Name	Description
Unit display		Speed unit	-
		Frequency unit	-
		Current unit	-
		Voltage unit	-
		Second	-
		Temperature unit	-
		Percentage	-

■ Button description

Button	Name	Function
	Increment key	Increase parameter number or value.
	Decrement key	Decrease parameter number or value.
	Shift key	Select the displayed parameters in turn, and select the digit to be modified when modifying parameters.
	Programming	Back to the previous page. Access a Level-1 menu.
	Confirm key	Enter the next page. Confirm the parameter setting.
	RUN	Start the device in the operating panel control mode.
	Stop/Reset	Stop the device when the device is running. Reset the alarm upon fault/alarm.

■ Indicator description

Symbol	Description
	Off: The drive has stopped. Steady green: The drive is running. Steady red: The drive is faulty. Steady yellow: An alarm is reported.

■ Parameter settings

The operating panel provides three levels of menus for parameter settings.

- Level-1 menu: parameter group
- Level-2 menu: parameter
- Level-3 menu: parameter setpoint

In a menu, when a digit is flashing, you can press ,  or  to set this digit.

In a level-3 menu, you can press **PRG** or **ENT** to return to the upper level-2 menu.

NOTICE

- **PRG**: Saves the current setpoint and returns to the level-2 menu, where the next parameter is automatically displayed.
- **ENT**: Cancels the change to the parameter and returns to the level-2 menu.

NOTICE

If no digit of a parameter flashes, the parameter cannot be modified because:

- The parameter is a non-modifiable parameter, such as a product type, actual measurement, or operation log.
- The parameter is non-modifiable in running state. In this case, you can modify the parameter after stopping the drive.

■ Status parameter display

You can check the status parameters by pressing  when the system is in running or stop state.

The following status parameters are displayed by default: running frequency, frequency reference, bus voltage, output voltage, and output current.

■ Error code display

When the product is faulty, the red indicator turns on, and the operating panel displays an error code, as shown in the following figure.



Figure 4-2 Error code display

Find the cause of the fault based on the fault information displayed on the operating panel, and rectify the fault according to Chapter 6 "Troubleshooting." Then reset the drive.

If the fault persists after the reset, seek technical support.

■ Using the operating panel to drive the motor

Set relevant parameters and press  and  to make the motor jog forwards, jog backwards, start, and stop.

Step 1: Verify that the system is installed and connected properly.

Step 2: Press the power button to power on the drive.

Step 3: Check the bottom LED display on the operating panel. If 0 flashes on the LED display, the drive has been powered on.

Step 4: Set PE-00 to 10 to restore all parameters to factory defaults.

Step 5: Set motor parameters in group P1 based on the nameplate of the motor.

Step 6: Set P1-00 and press  to confirm the setting. When TUNE is displayed, press  on the operating panel to start auto-tuning. When TUNE disappears from the operating panel about 1 minute later, auto-tuning is completed.

Step 7: Set PE-03 to 1, and then press  to make the motor jog forwards.

Step 8: Set PE-03 to 0, and then press  to start the motor. The motor shaft starts accelerative rotation, with the real-time rotation speed displayed on the operating panel. After the acceleration stage ends, the rated speed is displayed as the frequency value. Press  to switch to other status parameters.

Step 9: Press  to stop the drive. The motor then decelerates to stop.

4.2 Commissioning Process

4.2.1 Basic Commissioning Process

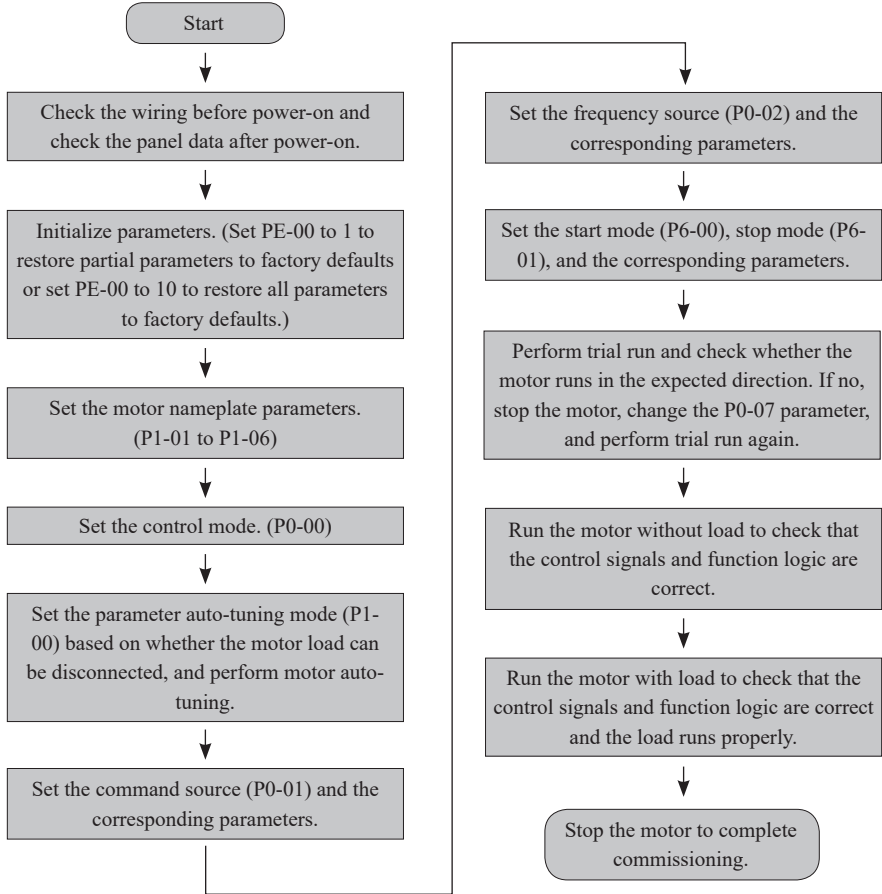


Figure 4-3 Basic commissioning flowchart

4.2.2 Parameter Auto-tuning Process

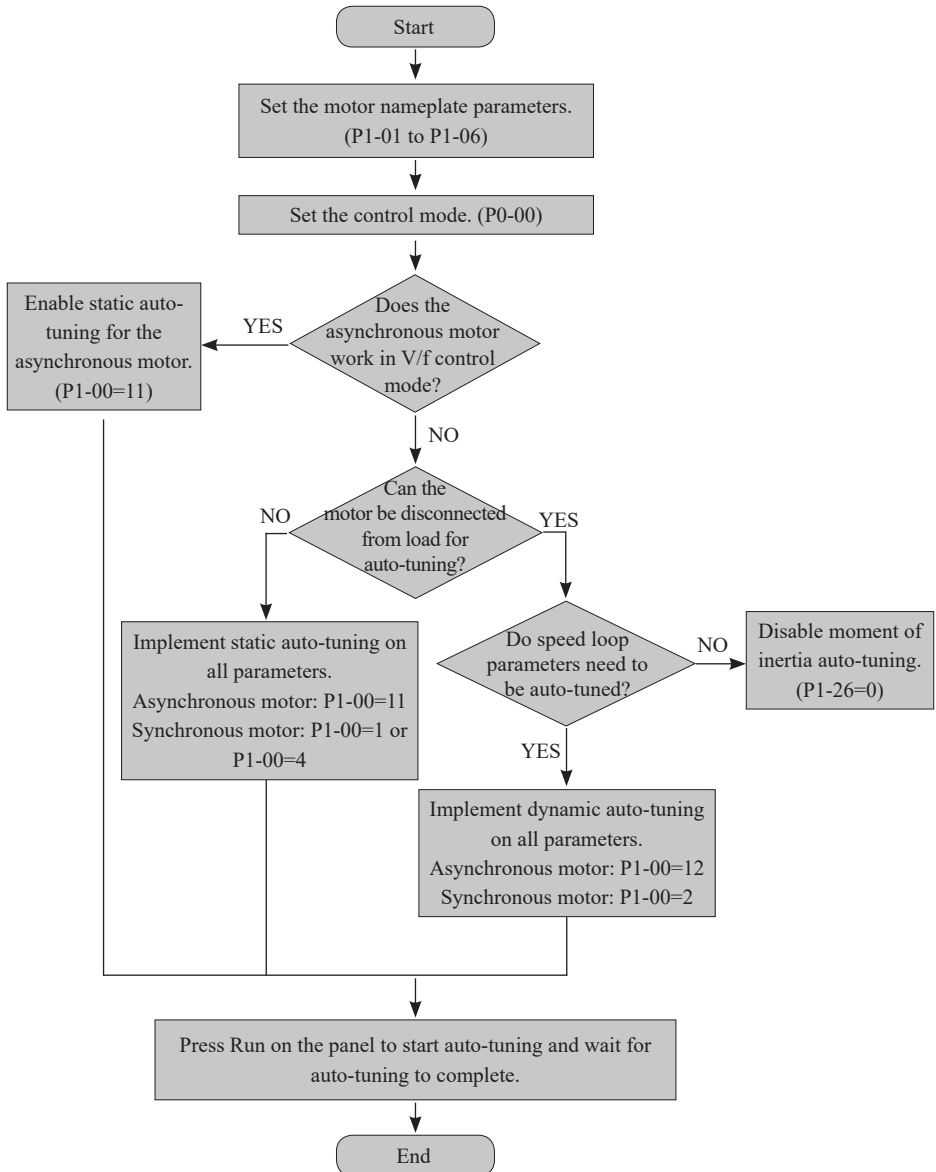


Figure 4-4 Parameter auto-tuning flowchart

Chapter 5

Parameter List

5.1 Parameter Settings

Meaning of each column in a parameter table.

Parameter: Function parameter group and parameter No.

Communication Address: Address used to control the product running status, and obtain the status information and function parameter settings.

Name: Full name of a function parameter.

Setting Range: Effective value range of a function parameter, displayed on the LED display of the operating panel.

Default: Factory default of a function parameter.

Unit: Unit of a function parameter.

Operation: Whether a function parameter can be modified (including the conditions for modification).

☆	Modifiable in the running or stop state: The parameter value can be changed when the drive has stopped or is running.
★	Modifiable in the stop state: The parameter value cannot be changed when the drive is running.
●	Non-modifiable: The parameter value is measured and recorded by the system, and cannot be changed.

(The system automatically restricts modifiability of all parameters to prevent misoperation.)

5.2 Parameter Group Description

Parameter Group	Name	Parameter Group	Name
P0	Basic function group	PD	Multi-segment references of simple PLC group
P1	Motor parameter group	PE	Parameter management group
P2	Vector control group	B0	Process data address mapping group
P3	V/f control group	B1	Tension control group
P4	DIO parameter group	B2	Tension setting group
P5	Auxiliary function group	B3	Tension taper group
P6	Start and stop parameter group	B5	Load synchronization group

Parameter Group	Name	Parameter Group	Name
P7	AI parameter group	B6	Droop control group
P8	Torque control group	B9	Virtual DIO parameter group
P9	Fault and protection group	BA	User-defined parameter group
PA	Auxiliary control group	C1	Drive parameter group
PB	PID parameter group	U0	Observation group
PC	Communication parameter group	U2	Observation group

5.3 Parameter Description

5.3.1 Group P0–Basic Function Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P0-00	0xA000	Motor control mode	0: Sensorless vector control (SVC) 1: Reserved 2: V/f control 3: Reserved 4: Sensorless vector control (SVC) 2	2	-	★
P0-01	0xA001	Command source	0: Operating panel or backend computer 1: DI 2: Communication	0	-	★
P0-02	0xA002	Main frequency reference source	0: Digital setting (P0-06: non-retentive upon power failure) 1: Digital setting (retentive upon power failure) 2: AI1 3: AI2 4: AI3 5: AI4 6: AI5 7: Pulse reference 8: Multi-segment reference 9: Simple PLC 10: PID 11: Communication 12: Expansion communication setting	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P0-03	0xA003	Auxiliary frequency setting source	0: Digital setting (P0-06: non-retentive upon power failure) 1: Digital setting (retentive upon power failure) 2: AI1 3: AI2 4: AI3 5: AI4 6: AI5 7: Pulse reference 8: Multi-segment reference 9: Simple PLC 10: PID 11: Communication 12: Expansion communication setting	0	-	★
P0-04	0xA004	Select main and auxiliary frequency sources	0: Main frequency source 1: Auxiliary frequency source 2: Coupling operation of main and auxiliary frequency sources 3: Switching between main and auxiliary frequency sources 4: Switching between main and operational frequency sources 5: Switching between auxiliary and operational frequency sources	0	-	★
P0-05	0xA005	Main and auxiliary frequency coupling type	0: Main frequency source + auxiliary frequency source 1: Main frequency source - auxiliary frequency source 2: Max. (main frequency source, auxiliary frequency source) 3: Min. (main frequency source, auxiliary frequency source) 4: Main frequency source x auxiliary frequency source	0	-	★
P0-06	0xA006	Local frequency reference	P0-09 - P0-08	50.00	Hz	☆
P0-07	0xA007	Running direction	0: Forward 1: Reverse	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P0-08	0xA008	Maximum frequency	5.00 - 300.00	50.00	Hz	★

NOTICE

- When the Frequency Resolution (P0-23) is set to 0.01 Hz, the Maximum Frequency (P0-08) can be set to 300.00 Hz.
- When the Frequency Resolution (P0-23) is set to 0.1 Hz, the Maximum Frequency (P0-08) can be set to 600.0 Hz.

P0-09	0xA009	Maximum frequency of reverse rotation	-P0-09 - 0.00	-50.00	Hz	☆
P0-10	0xA00A	Upper-limit frequency source	0: Digital setting (P0-11) 1: AI1 2: AI2 3: AI3 4: AI4 5: AI5 6: Communication (0x7709) 7: Multi-segment reference	0	-	☆
P0-11	0xA00B	Upper-limit frequency	P0-13 - P0-09	50.00	Hz	☆
P0-12	0xA00C	Upper-limit frequency offset	0 - P0-09	0.00	Hz	☆
P0-13	0xA00D	Lower-limit frequency	0.00 - P0-11	0.00	Hz	☆
P0-14	0xA00E	Carrier frequency	1.2 - 12.0	4.0	kHz	☆
P0-15	0xA00F	Acceleration time 1	0.00 - 655.35	20.00	s	☆
P0-16	0xA010	Deceleration time 1	0.00 - 655.35	20.00	s	☆
P0-17	0xA011	Acceleration/Deceleration time resolution	0: 1s 1: 0.1s 2: 0.01s	2	-	★
P0-18	0xA012	Acceleration/Deceleration time base frequency	0: Maximum frequency (P0-08) 1: Frequency reference 2: 100Hz	0	-	☆
P0-19	0xA013	Select auxiliary frequency source range during superposition	0: Relative to maximum frequency 1: Relative to main frequency	0	-	☆
P0-20	0xA014	Auxiliary frequency source range during superposition	0 - 150	100	%	☆
P0-21	0xA015	Main frequency coefficient	0 - 100.00	10.00	%	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P0-22	0xA016	Auxiliary frequency coefficient	0 - 100.00	10.00	%	☆
P0-23	0xA017	Frequency resolution	0: 0.01 Hz 1: 0.1 Hz	0	-	★
P0-24	0xA018	Select offset frequency source	0: Digital setting (P0-25) 1: AI1 2: AI2 3: AI3 4: AI4 5: AI5 6: Pulse reference 7: Multi-segment reference 8: Simple PLC 9: PID 11: Communication	0	-	☆
P0-25	0xA019	Digital setting of offset frequency	0 - Maximum frequency (P0-08)	0.00	Hz	☆
P0-26	0xA01A	Memorize digital correction frequency upon stop	0: Disabled 1: Enabled	0	-	☆

5.3.2 Group P1–Motor Parameter Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P1-00	0xA100	Auto-tuning mode	0: No operation 1: Static auto-tuning on some parameters of a non-loaded synchronous motor 2: Auto-tuning on all parameters of non-loaded synchronous motor, rotating in reverse direction at high speed 3: Static auto-tuning on some parameters of loaded synchronous motor 4: Auto-tuning on all parameters of non-loaded synchronous motor, rotating in forward direction at high speed 11: Static auto-tuning on some parameters of asynchronous motor 12: Dynamic auto-tuning on all parameters of asynchronous motor	0	-	★
P1-01	0xA101	Motor type	0: Asynchronous motor 1: Synchronous motor 2: Reluctance motor	0	-	★
P1-02	0xA102	Rated power	0.1 - 600.0	11.0	kW	★
P1-03	0xA103	Rated voltage	1 - 690	380	V	★
P1-04	0xA104	Rated current	0.01 - 655.35	22.40	A	★
P1-05	0xA105	Rated frequency	0.01 - P0-08	50.00	Hz	★
P1-06	0xA106	Rated speed	1 - 65535	1460	rpm	★
P1-07	0xA107	Stator resistance	0.001 - 65.535	0.340	Ω	★
P1-08	0xA108	Asynchronous motor rotor resistance	0.001 - 65.535	0.185	Ω	★
P1-09	0xA109	Asynchronous motor leakage inductance	0.01 - 655.35	3.04	mH	★
P1-10	0xA10A	Asynchronous motor mutual inductance	0.1 - 6553.5	77.4	mH	★
P1-11	0xA10B	Asynchronous motor no-load current	0.01 - P1-04	8.41	A	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P1-12	0xA10C	Mutual inductance saturation coefficient 0 of asynchronous motor	0 - 10000	132.9	%	★
P1-13	0xA10D	Mutual inductance saturation coefficient 1 of asynchronous motor	0 - 10000	117.7	%	★
P1-14	0xA10E	Mutual inductance saturation coefficient 2 of asynchronous motor	0 - 10000	84.1	%	★
P1-15	0xA10F	Mutual inductance saturation coefficient 3 of asynchronous motor	0 - 10000	72.0	%	★
P1-16	0xA110	Synchronous motor D-axis inductance	0.00 - 655.35	3.34	mh	★
P1-17	0xA111	Synchronous motor Q-axis inductance	0.00 - 655.35	6.09	mh	★
P1-18	0xA112	Synchronous motor back EMF	0.0 - 6553.5	327.0	V	★
P1-19	0xA113	Motor moment of inertia	0.1 - 6553.5	200.0	kg/cm ²	★
P1-20	0xA114	Total moment of inertia ratio	1 - 65535	100	%	★
P1-21	0xA115	Friction torque	0.0 - 100.0	0.0	%	★
P1-23	0xA117	High-current coefficient for magnetic pole position auto-tuning	20 - 400	120	%	★
P1-24	0xA118	Low-current coefficient for magnetic pole position auto-tuning	20 - 400	80	%	★
P1-25	0xA119	Compensation angle in magnetic pole position auto-tuning	0 - 3599	0.0	°	★
P1-26	0xA11A	Moment of inertia auto-tuning	0: Disabled 1: Enabled	0	-	★
P1-27	0xA11B	Command word of moment of inertia auto-tuning	Bit0: Reserved Bit1: Auto adjustment of acceleration/ deceleration 0: Disabled 1: Enabled	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P1-28	0xA11C	Acceleration/ Deceleration time in moment of inertia auto- tuning	0.1 - 1000.0	1.0	s	★
P1-29	0xA11D	Mean speed in moment of inertia auto-tuning	-100 - 100	40	%	★
P1-30	0xA11E	Speed change amplitude in moment of inertia auto-tuning	0 - 100	40	%	★
P1-31	0xA11F	Speed loop bandwidth in moment of inertia auto- tuning	0.1 - 500.0	5.0	Hz	★
P1-35	0xA123	D/Q-axis inductance curve auto-tuning	0: Disabled 1: Enabled	1	-	★
P1-37	0xA125	Maximum current for D/ Q-axis inductance curve auto-tuning	0 - 500	150	%	★
P1-38	0xA126	Enabling frequency of D/Q-axis inductance curve auto-tuning	0 - 1000	6.0	kHz	★
P1-39	0xA127	D-axis inductance 0	0 - 10000	100.0	%	★
P1-40	0xA128	D-axis inductance 1	0 - 10000	100.0	%	★
P1-41	0xA129	D-axis inductance 2	0 - 10000	100.0	%	★
P1-42	0xA12A	D-axis inductance 3	0 - 10000	100.0	%	★
P1-43	0xA12B	D-axis inductance 4	0 - 10000	100.0	%	★
P1-44	0xA12C	D-axis inductance 5	0 - 10000	100.0	%	★
P1-45	0xA12D	D-axis inductance 6	0 - 10000	100.0	%	★
P1-46	0xA12E	D-axis inductance 7	0 - 10000	100.0	%	★
P1-47	0xA12F	D-axis inductance 8	0 - 10000	100.0	%	★
P1-48	0xA130	D-axis inductance 9	0 - 10000	100.0	%	★
P1-49	0xA131	D-axis inductance 10	0 - 10000	100.0	%	★
P1-50	0xA132	D-axis inductance 11	0 - 10000	100.0	%	★
P1-51	0xA133	D-axis inductance 12	0 - 10000	100.0	%	★
P1-52	0xA134	Q-axis inductance 0	0 - 10000	100.0	%	★
P1-53	0xA135	Q-axis inductance 1	0 - 10000	100.0	%	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P1-54	0xA136	Q-axis inductance 2	0 - 10000	100.0	%	★
P1-55	0xA137	Q-axis inductance 3	0 - 10000	100.0	%	★
P1-56	0xA138	Q-axis inductance 4	0 - 10000	100.0	%	★
P1-57	0xA139	Q-axis inductance 5	0 - 10000	100.0	%	★

5.3.3 Group P2–Vector Control Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P2-01	0xA201	Speed loop position lock proportional gain	1 - 300	60	-	☆
P2-02	0xA202	Speed loop position lock integral time	0.001 - 10.000	0.500	s	☆
P2-03	0xA203	Speed loop low-speed proportional gain	1 - 300	30	-	☆
P2-04	0xA204	Speed loop low-speed integral time	0.001 - 10.000	0.500	s	☆
P2-05	0xA205	Speed loop high-speed proportional gain	1 - 300	20	-	☆
P2-06	0xA206	Speed loop high-speed integral time	0.001 - 10.000	1.000	s	☆
P2-07	0xA207	Position lock switching frequency of speed loop parameters	0 - 650.00	0.00	Hz	☆
P2-08	0xA208	Low switching frequency of speed loop parameters	0 - 650.00	5.00	Hz	☆
P2-09	0xA209	High switching frequency of speed loop parameters	0 - 650.00	10.00	Hz	☆
P2-28	0xA21C	Speed loop torque feedforward coefficient	0.1 - 1000.0	0.0	-	☆
P2-29	0xA21D	Feedforward bandwidth expansion coefficient of speed controller	1 - 10000	100	%	☆
P2-30	0xA21E	Overvoltage suppression in vector control mode	0: Disabled 1: Enabled	1	-	☆
P2-31	0xA21F	Electric torque upper limit for speed control	0.0 - 400.0	150.0	%	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P2-32	0xA220	Source of electric torque upper limit for speed control	0: Parameter setting (P2-31) 1: AI1 2: AI2 3: AI3 4: AI4 5: AI5 6: PULSE 7: Communication (0x770A) 8: Min (AI1, AI2) 9: Max (AI1, AI2) (full range of options 1 to 7 specified by P2-31)	0	-	☆
P2-33	0xA221	Power generation torque upper limit for speed control	0.0 - 400.0	150.0	%	☆
P2-34	0xA222	Source of power generation torque upper limit for speed control	0: Parameter setting (P2-33) 1: AI1 2: AI2 3: AI3 4: AI4 5: AI5 6: PULSE 7: Communication (0x770A) 8: Min (AI1, AI2) 9: Max (AI1, AI2) (full range of options 1 to 7 specified by P2-33)	0	-	☆
P2-35	0xA223	Motoring power upper limit	0.0 - 400.0	400.0	%	☆
P2-36	0xA224	Regenerative power upper limit	0.0 - 400.0	400.0	%	☆
P2-37	0xA225	Overvoltage suppression action voltage in vector control mode	350 - 1000	770	V	★
P2-38	0xA226	Overvoltage suppression gain in vector control mode	0 - 1000	10	Hz	★
P2-41	0xA229	Max. current coefficient for the drive	0.1 to 200.0% of rated AC drive current	180.0	%	★
P2-42	0xA22A	Integral time constant of voltage outer loop	0.001 - 65.000	0.050	s	★
P2-43	0xA22B	Weak magnetic frequency coefficient in vector control	10 - 10000	100	%	★
P2-44	0xA22C	Voltage adjustment coefficient in vector control	0 - 1000	0	%	★
P2-45	0xA22D	Torque filter time constant	0 - 650.00	1.00	ms	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P2-46	0xA22E	Excitation current injection in vector control	-1000% to 1000% (rated motor current)	0	%	☆
P2-47	0xA22F	Maximum torque current coefficient in vector control	0 - 200	100	%	★
P2-49	0xA231	Synchronous motor MTPA	0: Disabled 1: Enabled	1	-	☆
P2-50	0xA232	Filter time constant of synchronous motor MTPA	0 - 650.00	1.00	s	★
P2-54	0xA236	Flux linkage closed loop of asynchronous motor	0: Disabled 1: Enabled	1	-	★
P2-55	0xA237	Filter time constant reference of asynchronous motor flux linkage	0.001 - 1.000	0.010	s	★
P2-56	0xA238	Flux linkage closed loop bandwidth of asynchronous motor	0.01 - 100.00	2.00	Hz	★
P2-60	0xA23C	Current controller proportional gain	0.01 - 10.00	1.00	%	★
P2-61	0xA23D	Current controller integral gain	0.01 - 10.00	1.00	%	★
P2-62	0xA23E	Current controller decoupling gain	0.01 - 10.00	1.00	%	★
P2-63	0xA23F	Current controller bandwidth coefficient	0.01 - 1.00	0.60	%	★
P2-64	0xA240	Active damping coefficient of current controller	0.00 - 1.00	0.10	%	★
P2-65	0xA241	Low-speed carrier frequency of motor model	0 - 120	2.0	kHz	★
P2-67	0xA243	Filter time constant of synchronous motor model	0.0000 - 0.1000	0.0100	s	★
P2-68	0xA244	Low-speed D-axis current of synchronous motor model	0 - 150	30	%	★
P2-75	0xA24B	Filter time constant of asynchronous motor model	0.0000 - 0.1000	0.0100	s	★
P2-78	0xA24E	Low-pass filter time of current reference	0.0000 - 0.0600	0.0010	s	☆
P2-79	0xA24F	Center frequency of notch 1	0 - 8000	8000	Hz	☆
P2-80	0xA250	Bandwidth coefficient of notch 1	0.0 - 50.0	2.0	%	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P2-81	0xA251	Depth coefficient of notch 1	0.0 - 100.0	0.0	%	☆
P2-82	0xA252	Center frequency of notch 2	0 - 8000	8000	Hz	☆
P2-83	0xA253	Bandwidth coefficient of notch 2	0.0 - 50.0	2.0	%	☆
P2-84	0xA254	Depth coefficient of notch 2	0.0 - 100.0	0.0	%	☆

5.3.4 Group P3–V/f Control Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P3-01	0xA301	V/f curve type	0: Linear V/f curve 1: Multi-point V/f curve 2: V/f separated curve	0	-	★
P3-03	0xA303	V/f torque boost voltage	0.0 - 30.0	2.0	%	☆
P3-04	0xA304	V/f torque boost cutoff frequency	0.00 - P0-08	50.00	Hz	★
P3-06	0xA306	V/f oscillation suppression	0: Disabled 1: Enabled	1	-	★
P3-07	0xA307	V/f oscillation suppression cutoff frequency	0.00 - 10.00	2.50	Hz	☆
P3-08	0xA308	V/f oscillation suppression gain	0 - 1000	50	-	☆
P3-09	0xA309	Enable V/f overcurrent stall suppression	0: Disabled 1: Enabled	1	-	★
P3-10	0xA30A	V/f overcurrent stall suppression current	50 - 250	150	%	★
P3-11	0xA30B	V/f overcurrent stall suppression gain	0 - 100	20	-	☆
P3-12	0xA30C	V/f overcurrent stall suppression current compensation coefficient	0 - 500	50	%	★
P3-13	0xA30D	Enable V/f overvoltage stall suppression	0: Disabled 1: Enabled	1	-	★
P3-14	0xA30E	V/f overvoltage stall suppression voltage	350.0 - 800.0	750.0	V	★
P3-15	0xA30F	V/f overvoltage stall suppression frequency gain	0 - 100	30	-	☆
P3-16	0xA310	V/f overvoltage stall suppression voltage gain	0 - 100	30	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P3-17	0xA311	Maximum rising frequency for V/f overvoltage stall suppression	0 - 50.00	5.00	Hz	★
P3-18	0xA312	V/f slip compensation mode	0: Disabled 1: Standard mode 2: Reserved	0	-	★
P3-19	0xA313	V/f slip compensation gain	0.0 - 200.0	100.0	%	☆
P3-20	0xA314	Multi-point V/f frequency 1	0.00 - P3-22	0.00	Hz	☆
P3-21	0xA315	Multi-point V/f voltage 1	0.0 - 100.0	0.0	%	☆
P3-22	0xA316	Multi-point V/f frequency 2	P3-20 - P3-24	0.00	Hz	☆
P3-23	0xA317	Multi-point V/f voltage 2	0.0 - 100.0	0.0	%	☆
P3-24	0xA318	Multi-point V/f frequency 3	P3-22 - P1-05	0.00	Hz	☆
P3-25	0xA319	Multi-point V/f voltage 3	0.0 - 100.0	0.0	%	☆
P3-26	0xA31A	Voltage source for V/f separation	0: Digital setting 1: AI1 2: AI2 3: AI3 4: AI4 5: AI5 6: PULSE 7: Multi-segment reference 8: Communication (0x770C)	0	-	★
P3-27	0xA31B	Voltage digital setting for V/f separation	0 - P1-03	0	V	☆
P3-28	0xA31C	Voltage rise time for V/f separation	0.0 - 1000.0	0.0	s	★
P3-29	0xA31D	Voltage decline time for V/f separation	0.0 - 1000.0	0.0	s	★
P3-30	0xA31E	V/f separation stop mode	0: Frequency and voltage declining to 0 independently 1: Frequency declining after voltage declines to 0	0	-	★
P3-32	0xA320	V/f over-excitation control gain	0 - 200	0	%	☆

5.3.5 Group P4–DIO Parameter Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-00	0xA400	DI filter time	0 - 1000	10	ms	☆
P4-01	0xA401	Select DI1 hardware source	0: No function assigned 1: DI1 of AC drive 2: DI2 of AC drive 3: DI3 of AC drive 4: DIO1 of AC drive 5: HDI1 of AC drive 6: DI1 of expansion I/O card 7: DI2 of expansion I/O card 8: DI3 of expansion I/O card 9: DI4 of expansion I/O card 10: DI5 of expansion I/O card 11: DI1 of expansion relay I/O card 12: DI2 of expansion relay I/O card 13: DI3 of expansion relay I/O card 14: DI4 of expansion relay I/O card 15: DI5 of expansion relay I/O card 16: DI1 of expansion injection molding card 17: DI2 of expansion injection molding card 18: Reserved 19: Reserved 20: Reserved	1	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-02	0xA402	Function of DI1	See the description below	1	-	★
0: No function assigned			43: PID parameter switching			
1: Forward running (FWD)			44: User-defined fault 1			
2: Reverse running (REV)			45: User-defined fault 2			
3: Three-wire running control			46: Switching between speed and torque control			
4: Positive jog			47: Emergency stop			
5: Reverse jog			48: External terminal stop (deceleration time set to 4, which is effective at any time)			
6: Coast to stop			49: Deceleration DC braking			
7: Quick stop			50: Reset current running time			
8: Coast to stop (stop mode)			51: Two-wire/three-wire switching			
9: Fault reset			52: Terminal Up			
10: Pause			53: Terminal Down			
11: External fault input (NO)			54: Reset roll diameter			
12: Multi-segment reference terminal 1			55: Initial terminal 1			
13: Multi-segment reference terminal 2			56: Initial terminal 2			
14: Multi-segment reference terminal 3			57: Pre-drive			
15: Multi-segment reference terminal 4			58: Switching between winding and unwinding modes			
16: Terminal 1 for acceleration/deceleration selection			59: Stop roll diameter calculation			
17: Terminal 2 for acceleration/deceleration selection			60: Exit tension mode			
18: Frequency source switching			61: Disable reverse running			
19: Up/Down setting reset			62: Reserved			
20: Run command switching terminal			63: Reserved			
21: Acceleration/Deceleration disabling			64: Run in position control mode			
22: PID pause			65: Reserved			
23: PLC status reset			66: Incremental position			
24: Vibration frequency pause			67: Decremental position			
25: Counter input (DIO1)			68: Reserved			
26: Counter reset			69: Home signal			
27: Length counter input (DIO1)			70: Home search			
28: Length counter reset			71: Pause position control			
29: Torque control disabling			72: Reserved			
30: Pulse input			73: Reserved			
31: Incremental encoder pulse count clearing			74: Reserved			
32: DC braking			75: Reserved			
33: External fault input (NC)			76: Multi-segment position 1			
34: Frequency reference activation			77: Multi-segment position 2			
35: Reverse direction of PID action			78: Multi-segment position 3			
36: External parking terminal, valid only for panel control			79: Multi-segment position 4			
37: Command source switching terminal 2			80: Multi-segment position 5			
38: PID integral pause terminal			81: Positive limit			
39: Switching between main frequency source X and preset frequency			82: Reverse limit			
40: Switching between auxiliary frequency source Y and preset frequency			83: Switch to relative position control in position control mode			
41: Motor selection			84: Switch to absolute position control in position control mode			
42: Position lock			85: Switch to segmented position control in position control mode			

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
86: Switch to fixed-length segmented position control in position control mode			88: Reserved			
87: Re-trigger position control (incremental or fixed-length segmented)			89: Reserved			
			90: Set material thickness to 1			
			91: Set material thickness to 2			
P4-03	0xA403	Select DI2 hardware source	0: No function assigned 1: DI1 of AC drive 2: DI2 of AC drive 3: DI3 of AC drive 4: DIO1 of AC drive 5: HDI1 of AC drive 6: DI1 of expansion I/O card 7: DI2 of expansion I/O card 8: DI3 of expansion I/O card 9: DI4 of expansion I/O card 10: DI5 of expansion I/O card 11: DI1 of expansion relay I/O card 12: DI2 of expansion relay I/O card 13: DI3 of expansion relay I/O card 14: DI4 of expansion relay I/O card 15: DI5 of expansion relay I/O card 16: DI1 of expansion injection molding card 17: DI2 of expansion injection molding card 18-20: Reserved	2	-	★
P4-04	0xA404	Function of DI2	Same as those of P4-02	4	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-05	0xA405	Select DI3 hardware source	0: No function assigned 1: DI1 of AC drive 2: DI2 of AC drive 3: DI3 of AC drive 4: DIO1 of AC drive 5: HDI1 of AC drive 6: DI1 of expansion I/O card 7: DI2 of expansion I/O card 8: DI3 of expansion I/O card 9: DI4 of expansion I/O card 10: DI5 of expansion I/O card 11: DI1 of expansion relay I/O card 12: DI2 of expansion relay I/O card 13: DI3 of expansion relay I/O card 14: DI4 of expansion relay I/O card 15: DI5 of expansion relay I/O card 16: DI1 of expansion injection molding card 17: DI2 of expansion injection molding card 18-20: Reserved	3	-	★
P4-06	0xA406	Function of DI3	Same as those of P4-02	9	-	★
P4-07	0xA407	Select DI4 hardware source	0: No function assigned 1: DI1 of AC drive 2: DI2 of AC drive 3: DI3 of AC drive 4: DIO1 of AC drive 5: HDI1 of AC drive 6: DI1 of expansion I/O card 7: DI2 of expansion I/O card 8: DI3 of expansion I/O card 9: DI4 of expansion I/O card 10: DI5 of expansion I/O card 11: DI1 of expansion relay I/O card 12: DI2 of expansion relay I/O card 13: DI3 of expansion relay I/O card 14: DI4 of expansion relay I/O card 15: DI5 of expansion relay I/O card 16: DI1 of expansion injection molding card 17: DI2 of expansion injection molding card 18-20: Reserved	0	-	★
P4-08	0xA408	Function of DI4	Same as those of P4-02	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-09	0xA409	Select DI5 hardware source	0: No function assigned 1: DI1 of AC drive 2: DI2 of AC drive 3: DI3 of AC drive 4: DIO1 of AC drive 5: HDI1 of AC drive 6: DI1 of expansion I/O card 7: DI2 of expansion I/O card 8: DI3 of expansion I/O card 9: DI4 of expansion I/O card 10: DI5 of expansion I/O card 11: DI1 of expansion relay I/O card 12: DI2 of expansion relay I/O card 13: DI3 of expansion relay I/O card 14: DI4 of expansion relay I/O card 15: DI5 of expansion relay I/O card 16: DI1 of expansion injection molding card 17: DI2 of expansion injection molding card 18-20: Reserved	0	-	★
P4-10	0xA40A	Function of DI5	Same as those of P4-02	0	-	★
P4-11	0xA40B	Select DI6 hardware source	0: No function assigned 1: DI1 of AC drive 2: DI2 of AC drive 3: DI3 of AC drive 4: DIO1 of AC drive 5: HDI1 of AC drive 6: DI1 of expansion I/O card 7: DI2 of expansion I/O card 8: DI3 of expansion I/O card 9: DI4 of expansion I/O card 10: DI5 of expansion I/O card 11: DI1 of expansion relay I/O card 12: DI2 of expansion relay I/O card 13: DI3 of expansion relay I/O card 14: DI4 of expansion relay I/O card 15: DI5 of expansion relay I/O card 16: DI1 of expansion injection molding card 17: DI2 of expansion injection molding card 18-20: Reserved	0	-	★
P4-12	0xA40C	Function of DI6	Same as those of P4-02	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-13	0xA40D	Select DI7 hardware source	0: No function assigned 1: DI1 of AC drive 2: DI2 of AC drive 3: DI3 of AC drive 4: DIO1 of AC drive 5: HDI1 of AC drive 6: DI1 of expansion I/O card 7: DI2 of expansion I/O card 8: DI3 of expansion I/O card 9: DI4 of expansion I/O card 10: DI5 of expansion I/O card 11: DI1 of expansion relay I/O card 12: DI2 of expansion relay I/O card 13: DI3 of expansion relay I/O card 14: DI4 of expansion relay I/O card 15: DI5 of expansion relay I/O card 16: DI1 of expansion injection molding card 17: DI2 of expansion injection molding card 18-20: Reserved	0	-	★
P4-14	0xA40E	Function of DI7	Same as those of P4-02	0	-	★
P4-15	0xA40F	Select DI8 hardware source	0: No function assigned 1: DI1 of AC drive 2: DI2 of AC drive 3: DI3 of AC drive 4: DIO1 of AC drive 5: HDI1 of AC drive 6: DI1 of expansion I/O card 7: DI2 of expansion I/O card 8: DI3 of expansion I/O card 9: DI4 of expansion I/O card 10: DI5 of expansion I/O card 11: DI1 of expansion relay I/O card 12: DI2 of expansion relay I/O card 13: DI3 of expansion relay I/O card 14: DI4 of expansion relay I/O card 15: DI5 of expansion relay I/O card 16: DI1 of expansion injection molding card 17: DI2 of expansion injection molding card 18-20: Reserved	0	-	★
P4-16	0xA410	Function of DI8	Same as those of P4-02	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-17	0xA411	Select DI9 hardware source	0: No function assigned 1: DI1 of AC drive 2: DI2 of AC drive 3: DI3 of AC drive 4: DIO1 of AC drive 5: HDI1 of AC drive 6: DI1 of expansion I/O card 7: DI2 of expansion I/O card 8: DI3 of expansion I/O card 9: DI4 of expansion I/O card 10: DI5 of expansion I/O card 11: DI1 of expansion relay I/O card 12: DI2 of expansion relay I/O card 13: DI3 of expansion relay I/O card 14: DI4 of expansion relay I/O card 15: DI5 of expansion relay I/O card 16: DI1 of expansion injection molding card 17: DI2 of expansion injection molding card 18-20: Reserved	0	-	★
P4-18	0xA412	Function of DI9	Same as those of P4-02	0	-	★
P4-19	0xA413	Select DI10 hardware source	0: No function assigned 1: DI1 of AC drive 2: DI2 of AC drive 3: DI3 of AC drive 4: DIO1 of AC drive 5: HDI1 of AC drive 6: DI1 of expansion I/O card 7: DI2 of expansion I/O card 8: DI3 of expansion I/O card 9: DI4 of expansion I/O card 10: DI5 of expansion I/O card 11: DI1 of expansion relay I/O card 12: DI2 of expansion relay I/O card 13: DI3 of expansion relay I/O card 14: DI4 of expansion relay I/O card 15: DI5 of expansion relay I/O card 16: DI1 of expansion injection molding card 17: DI2 of expansion injection molding card 18-20: Reserved	0	-	★
P4-20	0xA414	Function of DI10	Same as those of P4-02	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-21	0xA415	Select DI11 hardware source	0: No function assigned 1: DI1 of AC drive 2: DI2 of AC drive 3: DI3 of AC drive 4: DIO1 of AC drive 5: HDI1 of AC drive 6: DI1 of expansion I/O card 7: DI2 of expansion I/O card 8: DI3 of expansion I/O card 9: DI4 of expansion I/O card 10: DI5 of expansion I/O card 11: DI1 of expansion relay I/O card 12: DI2 of expansion relay I/O card 13: DI3 of expansion relay I/O card 14: DI4 of expansion relay I/O card 15: DI5 of expansion relay I/O card 16: DI1 of expansion injection molding card 17: DI2 of expansion injection molding card 18-20: Reserved	0	-	★
P4-22	0xA416	Function of DI11	Same as those of P4-02	0	-	★
P4-23	0xA417	Select DI12 hardware source	0: No function assigned 1: DI1 of AC drive 2: DI2 of AC drive 3: DI3 of AC drive 4: DIO1 of AC drive 5: HDI1 of AC drive 6: DI1 of expansion I/O card 7: DI2 of expansion I/O card 8: DI3 of expansion I/O card 9: DI4 of expansion I/O card 10: DI5 of expansion I/O card 11: DI1 of expansion relay I/O card 12: DI2 of expansion relay I/O card 13: DI3 of expansion relay I/O card 14: DI4 of expansion relay I/O card 15: DI5 of expansion relay I/O card 16: DI1 of expansion injection molding card 17: DI2 of expansion injection molding card 18-20: Reserved	0	-	★
P4-24	0xA418	Function of DI12	Same as those of P4-02	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-25	0xA419	Select DI13 hardware source	0: No function assigned 1: DI1 of AC drive 2: DI2 of AC drive 3: DI3 of AC drive 4: DIO1 of AC drive 5: HDI1 of AC drive 6: DI1 of expansion I/O card 7: DI2 of expansion I/O card 8: DI3 of expansion I/O card 9: DI4 of expansion I/O card 10: DI5 of expansion I/O card 11: DI1 of expansion relay I/O card 12: DI2 of expansion relay I/O card 13: DI3 of expansion relay I/O card 14: DI4 of expansion relay I/O card 15: DI5 of expansion relay I/O card 16: DI1 of expansion injection molding card 17: DI2 of expansion injection molding card 18-20: Reserved	0	-	★
P4-26	0xA41A	Function of DI13	Same as those of P4-02	0	-	★
P4-27	0xA41B	Select DI14 hardware source	0: No function assigned 1: DI1 of AC drive 2: DI2 of AC drive 3: DI3 of AC drive 4: DIO1 of AC drive 5: HDI1 of AC drive 6: DI1 of expansion I/O card 7: DI2 of expansion I/O card 8: DI3 of expansion I/O card 9: DI4 of expansion I/O card 10: DI5 of expansion I/O card 11: DI1 of expansion relay I/O card 12: DI2 of expansion relay I/O card 13: DI3 of expansion relay I/O card 14: DI4 of expansion relay I/O card 15: DI5 of expansion relay I/O card 16: DI1 of expansion injection molding card 17: DI2 of expansion injection molding card 18-20: Reserved	0	-	★
P4-28	0xA41C	Function of DI14	Same as those of P4-02	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-29	0xA41D	Select DI15 hardware source	0: No function assigned 1: DI1 of AC drive 2: DI2 of AC drive 3: DI3 of AC drive 4: DIO1 of AC drive 5: HDI1 of AC drive 6: DI1 of expansion I/O card 7: DI2 of expansion I/O card 8: DI3 of expansion I/O card 9: DI4 of expansion I/O card 10: DI5 of expansion I/O card 11: DI1 of expansion relay I/O card 12: DI2 of expansion relay I/O card 13: DI3 of expansion relay I/O card 14: DI4 of expansion relay I/O card 15: DI5 of expansion relay I/O card 16: DI1 of expansion injection molding card 17: DI2 of expansion injection molding card 18-20: Reserved	0	-	★
P4-30	0xA41E	Function of DI15	Same as those of P4-02	0	-	★
P4-31	0xA41F	Active state of terminals DI1 to DI5	Ones digit: DI1 0: High level 1: Low level Tens digit: DI2 0: High level 1: Low level Hundreds digit: DI3 0: High level 1: Low level Thousands digit: DI4 0: High level 1: Low level Ten-thousands digit: DI5 0: High level 1: Low level	11111	-	☆
P4-32	0xA420	Active state of terminals DI6 to DI10	Ones digit: DI6 0: High level 1: Low level Tens digit: DI7 0: High level 1: Low level Hundreds digit: DI8 0: High level 1: Low level Thousands digit: DI9 0: High level 1: Low level Ten-thousands digit: DI10 0: High level 1: Low level	11111	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-33	0xA421	Active state of terminals DI11 to DI15	Ones digit: DI11 0: High level 1: Low level Tens digit: DI12 0: High level 1: Low level Hundreds digit: DI13 0: High level 1: Low level Thousands digit: DI14 0: High level 1: Low level Ten-thousands digit: DI15 0: High level 1: Low level	11111	-	☆
P4-34	0xA422	Delay in turning DI on	0.0 - 100.0	0.0	s	☆
P4-35	0xA423	Delay in turning DI off	0.0 - 100.0	0.0	s	☆
P4-36	0xA424	DI terminal control mode	0: Two-wire mode 1 1: Two-wire mode 2 2: Three-wire mode 1 3: Three-wire mode 2	0	-	☆
P4-37	0xA425	Select DO1 hardware source	0: No function assigned 1: DO1 of AC drive 2: TA/B/C of AC drive 3: DIO1 of AC drive 4: DO3 of expansion I/O card 5: DO4 of expansion I/O card 6: DO3 of expansion relay I/O card 7: RA/B/C of expansion relay I/O card 8: DO3 of expansion injection molding card	1	-	★
P4-38	0xA426	Function of DO1	See the description below	1	-	☆
0: No function assigned 1: Drive running 2: Drive failure (coast to stop failure) 3: FDT reached in frequency level detection 4: Frequency reference reached 5: Running at zero speed 6: Motor overload pre-warning 7: Drive overload pre-warning 8: Pulse reference reached 9: Specified pulse count reached 10: Length reached 11: PLC cycle completed 12: Running time reached 13: Frequency limiting 14: Torque limiting			15: Ready to run 16: AI1>AI2 17: Upper-limit frequency reached 18: Lower-limit frequency reached (no output in stop state) 19: Undervoltage state output 20: Communication control 21: Positioning completed 22: Proximity to position 23: Running at zero speed (active in stop state) 24: Power-on time reached 25: FDT1 reached in frequency level detection 26: Frequency reached 1 27: Frequency reached 2 28: Current reached 1 29: Current reached 2			

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
30: Scheduled time reached 31: AI1 input out of upper and lower limits 32: Load dropping 33: Speed direction 34: Zero-crossing detection completed 35: Module temperature reached 36: Software overcurrent 37: Lower-limit frequency reached (output available in both running and stop states) 38: Error occurred (direct output upon a fault or warning) 39: Motor over-temperature pre-warning 40: Running time reached			41: Fault output 2 (no output for undervoltage in the case of a coast to stop failure) 42: Fault output 3 (immediate output for coast to stop or decelerate to stop) 43: Position locked 44: Positioning completed 45: Proximity to position 46: Home position active 47: Home search end flag 48: RUN command state 49: Reserved 50: Machinery brake			
P4-39	0xA427	Select DO2 hardware source	0: No function assigned 1: DO1 of AC drive 2: TA/B/C of AC drive 3: DIO1 of AC drive 4: DO3 of expansion I/O card 5: DO4 of expansion I/O card 6: DO3 of expansion relay I/O card 7: RA/B/C of expansion relay I/O card 8: DO3 of expansion injection molding card	2	-	★
P4-40	0xA428	Function of DO2	Same as those of P4-38	2	-	☆
P4-41	0xA429	Select DO3 hardware source	0: No function assigned 1: DO1 of AC drive 2: TA/B/C of AC drive 3: DIO1 of AC drive 4: DO3 of expansion I/O card 5: DO4 of expansion I/O card 6: DO3 of expansion relay I/O card 7: RA/B/C of expansion relay I/O card 8: DO3 of expansion injection molding card	0	-	★
P4-42	0xA42A	Function of DO3	Same as those of P4-38	0	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-43	0xA42B	Select DO4 hardware source	0: No function assigned 1: DO1 of AC drive 2: TA/B/C of AC drive 3: DIO1 of AC drive 4: DO3 of expansion I/O card 5: DO4 of expansion I/O card 6: DO3 of expansion relay I/O card 7: RA/B/C of expansion relay I/O card 8: DO3 of expansion injection molding card	0	-	★
P4-44	0xA42C	Function of DO4	Same as those of P4-38	0	-	☆
P4-45	0xA42D	Select DO5 hardware source	0: No function assigned 1: DO1 of AC drive 2: TA/B/C of AC drive 3: DIO1 of AC drive 4: DO3 of expansion I/O card 5: DO4 of expansion I/O card 6: DO3 of expansion relay I/O card 7: RA/B/C of expansion relay I/O card 8: DO3 of expansion injection molding card	0	-	★
P4-46	0xA42E	Function of DO5	Same as those of P4-38	0	-	☆
P4-47	0xA42F	Select DO6 hardware source	0: No function assigned 1: DO1 of AC drive 2: TA/B/C of AC drive 3: DIO1 of AC drive 4: DO3 of expansion I/O card 5: DO4 of expansion I/O card 6: DO3 of expansion relay I/O card 7: RA/B/C of expansion relay I/O card 8: DO3 of expansion injection molding card	0	-	★
P4-48	0xA430	Function of DO6	Same as those of P4-38	0	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-49	0xA431	Select DO7 hardware source	0: No function assigned 1: DO1 of AC drive 2: TA/B/C of AC drive 3: DIO1 of AC drive 4: DO3 of expansion I/O card 5: DO4 of expansion I/O card 6: DO3 of expansion relay I/O card 7: RA/B/C of expansion relay I/O card 8: DO3 of expansion injection molding card	0	-	★
P4-50	0xA432	Function of DO7	Same as those of P4-38	0	-	☆
P4-51	0xA433	Select DO8 hardware source	0: No function assigned 1: DO1 of AC drive 2: TA/B/C of AC drive 3: DIO1 of AC drive 4: DO3 of expansion I/O card 5: DO4 of expansion I/O card 6: DO3 of expansion relay I/O card 7: RA/B/C of expansion relay I/O card 8: DO3 of expansion injection molding card	0	-	★
P4-52	0xA434	Function of DO8	Same as those of P4-38	0	-	☆
P4-53	0xA435	Select DO9 hardware source	0: No function assigned 1: DO1 of AC drive 2: TA/B/C of AC drive 3: DIO1 of AC drive 4: DO3 of expansion I/O card 5: DO4 of expansion I/O card 6: DO3 of expansion relay I/O card 7: RA/B/C of expansion relay I/O card 8: DO3 of expansion injection molding card	0	-	★
P4-54	0xA436	Function of DO9	Same as those of P4-38	0	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-55	0xA437	Select DO10 hardware source	0: No function assigned 1: DO1 of AC drive 2: TA/B/C of AC drive 3: DIO1 of AC drive 4: DO3 of expansion I/O card 5: DO4 of expansion I/O card 6: DO3 of expansion relay I/O card 7: RA/B/C of expansion relay I/O card 8: DO3 of expansion injection molding card	0	-	★
P4-56	0xA438	Function of DO10	Same as those of P4-38	0	-	☆
P4-57	0xA439	Select DO11 hardware source	0: No function assigned 1: DO1 of AC drive 2: TA/B/C of AC drive 3: DIO1 of AC drive 4: DO3 of expansion I/O card 5: DO4 of expansion I/O card 6: DO3 of expansion relay I/O card 7: RA/B/C of expansion relay I/O card 8: DO3 of expansion injection molding card	0	-	★
P4-58	0xA43A	Function of DO11	Same as those of P4-38	0	-	☆
P4-59	0xA43B	Select DO12 hardware source	0: No function assigned 1: DO1 of AC drive 2: TA/B/C of AC drive 3: DIO1 of AC drive 4: DO3 of expansion I/O card 5: DO4 of expansion I/O card 6: DO3 of expansion relay I/O card 7: RA/B/C of expansion relay I/O card 8: DO3 of expansion injection molding card	0	-	★
P4-60	0xA43C	Function of DO12	Same as those of P4-38	0	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-61	0xA43D	Active state of terminals DO1 to DO5	Ones digit: DO1 0: High level 1: Low level Tens digit: DO2 0: High level 1: Low level Hundreds digit: DO3 0: High level 1: Low level Thousands digit: DO4 0: High level 1: Low level Ten-thousands digit: DO5 0: High level 1: Low level	11111	-	☆
P4-62	0xA43E	Active state of terminals DO6 to DO10	Ones digit: DO6 0: High level 1: Low level Tens digit: DO7 0: High level 1: Low level Hundreds digit: DO8 0: High level 1: Low level Thousands digit: DO9 0: High level 1: Low level Ten-thousands digit: DO10 0: High level 1: Low level	11111	-	☆
P4-63	0xA43F	Active state of terminals DO11 to DO12	ones digit: DO11 0: High level 1: Low level Tens digit: DO12 0: High level 1: Low level	11	-	☆
P4-64	0xA440	DO delay time	0.0 - 100.0	0.0	s	☆
P4-65	0xA441	Select AO1 source	0: No function assigned 1: AO1 of AC drive 2: AO2 of expansion I/O card 3: AO2 of expansion relay I/O card 4: AO2 of expansion injection molding card	1	-	★
P4-66	0xA442	AO1 type	0: Voltage signal 1: Current signal	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-67	0xA443	Function of AO1	0: No function assigned 1: Running frequency 2: Frequency reference 3: Output current (2 times the motor's rated current) 4: Output torque (2 times the motor's rated torque) 5: Output power 6: Output voltage (1.2 times the motor's rated voltage) 7: AI1 voltage 8: AI2 voltage 9: AI3 voltage 10: Feedback speed 11: Target frequency 12: Length 13: Count 14: Communication control (0x770D)	1	-	☆
P4-68	0xA444	Select AO2 source	0: No function assigned 1: AO1 of AC drive 2: AO2 of expansion I/O card 3: AO2 of expansion relay I/O card 4: AO2 of expansion injection molding card	0	-	★
P4-69	0xA445	AO2 type	0: Voltage signal 1: Current signal	0	-	★
P4-70	0xA446	Function of AO2	0: No function assigned 1: Running frequency 2: Frequency reference 3: Output current (2 times the motor's rated current) 4: Output torque (2 times the motor's rated torque) 5: Output power 6: Output voltage (1.2 times the motor's rated voltage) 7: AI1 voltage 8: AI2 voltage 9: AI3 voltage 10: Feedback speed 11: Target frequency 12: Length 13: Count 14: Communication control (0x770D)	0	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-71	0xA447	Select AO3 source	0: No function assigned 1: AO1 of AC drive 2: AO2 of expansion I/O card 3: AO2 of expansion relay I/O card 4: AO2 of expansion injection molding card	0	-	★
P4-72	0xA448	AO3 type	0: Voltage signal 1: Current signal	0	-	★
P4-73	0xA449	Function of AO3	0: No function assigned 1: Running frequency 2: Frequency reference 3: Output current (2 times the motor's rated current) 4: Output torque (2 times the motor's rated torque) 5: Output power 6: Output voltage (1.2 times the motor's rated voltage) 7: AI1 voltage 8: AI2 voltage 9: AI3 voltage 10: Feedback speed 11: Target frequency 12: Length 13: Count 14: Communication control (0x770D)	0	-	☆
P4-74	0xA44A	Select FM1 source	0: No function assigned 1: DO1 of AC drive 2: Reserved 3: Reserved 4: Reserved	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-75	0xA44B	Function of FM1	0: No function assigned 1: Running frequency 2: Frequency reference 3: Output current (2 times the motor's rated current) 4: Output torque (2 times the motor's rated torque) 5: Output power 6: Output voltage (1.2 times the motor's rated voltage) 7: AI1 voltage 8: AI2 voltage 9: AI3 voltage 10: Feedback speed 11: Target frequency 12: Length 13: Count 14: Communication control (0x770E)	0	-	☆
P4-76	0xA44C	Select FM2 source	0: No function assigned 1: DO1 of AC drive 2: Reserved 3: Reserved 4: Reserved	0	-	★
P4-77	0xA44D	Function of FM2	0: No function assigned 1: Running frequency 2: Frequency reference 3: Output current (2 times the motor's rated current) 4: Output torque (2 times the motor's rated torque) 5: Output power 6: Output voltage (1.2 times the motor's rated voltage) 7: AI1 voltage 8: AI2 voltage 9: AI3 voltage 10: Feedback speed 11: Target frequency 12: Length 13: Count 14: Communication control (0x770E)	0	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-78	0xA44E	Select AI1 source	0: No function assigned 1: AI1 of AC drive 2: AI2 of AC drive 3: AI3 of expansion I/O card 4: AI4 of expansion I/O card 5: AI5 of absolute PG card (first encoder) 6: AI6 of absolute PG card (first encoder) 7: AI5 of absolute PG card (second encoder) 8: AI6 of absolute PG card (second encoder) 9: AI5 of CANopen card 10: AI6 of CANopen card 11: AI3 of expansion relay I/O card 12: Reserved 13: Reserved 14: AI3 of expansion injection molding card	1	-	★
P4-79	0xA44F	AI1 type	0: Voltage signal 1: Current signal 1 2: Current signal 2	0	-	★
P4-80	0xA450	Select AI2 source	0: No function assigned 1: AI1 of AC drive 2: AI2 of AC drive 3: AI3 of expansion I/O card 4: AI4 of expansion I/O card 5: AI5 of absolute PG card (first encoder) 6: AI6 of absolute PG card (first encoder) 7: AI5 of absolute PG card (second encoder) 8: AI6 of absolute PG card (second encoder) 9: AI5 of CANopen card 10: AI6 of CANopen card 11: AI3 of expansion relay I/O card 12: Reserved 13: Reserved 14: AI3 of expansion injection molding card	2	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-81	0xA451	AI2 type	0: Voltage signal 1: Current signal 1 2: Current signal 2	0	-	★
P4-82	0xA452	Select AI3 source	0: No function assigned 1: AI1 of AC drive 2: AI2 of AC drive 3: AI3 of expansion I/O card 4: AI4 of expansion I/O card 5: AI5 of absolute PG card (first encoder) 6: AI6 of absolute PG card (first encoder) 7: AI5 of absolute PG card (second encoder) 8: AI6 of absolute PG card (second encoder) 9: AI5 of CANopen card 10: AI6 of CANopen card 11: AI3 of expansion relay I/O card 12: Reserved 13: Reserved 14: AI3 of expansion injection molding card	0	-	★
P4-83	0xA453	AI3 type	0: Voltage signal 1: Current signal 1 2: Current signal 2	0	-	★
P4-84	0xA454	Select AI4 source	0: No function assigned 1: AI1 of AC drive 2: AI2 of AC drive 3: AI3 of expansion I/O card 4: AI4 of expansion I/O card 5: AI5 of absolute PG card (first encoder) 6: AI6 of absolute PG card (first encoder) 7: AI5 of absolute PG card (second encoder) 8: AI6 of absolute PG card (second encoder) 9: AI5 of CANopen card 10: AI6 of CANopen card 11: AI3 of expansion relay I/O card 12: Reserved 13: Reserved 14: AI3 of expansion injection molding card	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-85	0xA455	AI4 type	0: Voltage signal 1: Current signal 1 2: Current signal 2	0	-	★
P4-86	0xA456	Select AI5 source	0: No function assigned 1: AI1 of AC drive 2: AI2 of AC drive 3: AI3 of expansion I/O card 4: AI4 of expansion I/O card 5: AI5 of absolute PG card (first encoder) 6: AI6 of absolute PG card (first encoder) 7: AI5 of absolute PG card (second encoder) 8: AI6 of absolute PG card (second encoder) 9: AI5 of CANopen card 10: AI6 of CANopen card 11: AI3 of expansion relay I/O card 12: Reserved 13: Reserved 14: AI3 of expansion injection molding card	0	-	★
P4-87	0xA457	AI5 type	0: Voltage signal 1: Current signal 1 2: Current signal 2	0	-	★
P4-88	0xA458	Select source for temperature measurement signal 1	0: No function assigned 1: Reserved 2: Reserved 3: AI3 of expansion I/O card 4: AI4 of expansion I/O card 5: Temp of absolute PG card (first encoder) 6: Reserved 7: Temp of absolute PG card (second encoder) 8: Reserved 9: Temp of CANopen card 10: Reserved 11: Reserved 12: Temp of resolver PG card (first encoder) 13: Temp of resolver PG card (second encoder)	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-89	0xA459	Type of temperature measurement signal 1	0: PT100 1: PT1000 2: KTY84	0	-	★
P4-90	0xA45A	Select source for temperature measurement signal 2	0: No function assigned 1: Reserved 2: Reserved 3: AI3 of expansion I/O card 4: AI4 of expansion I/O card 5: Temp of absolute PG card (first encoder) 6: Reserved 7: Temp of absolute PG card (second encoder) 8: Reserved 9: Temp of CANopen card 10: Reserved 11: Reserved 12: Temp of resolver PG card (first encoder) 13: Temp of resolver PG card (second encoder)	0	-	★
P4-91	0xA45B	Type of temperature measurement signal 2	0: PT100 1: PT1000 2: KTY84	0	-	★
P4-92	0xA45C	Select source for temperature measurement signal 3	0: No function assigned 1: Reserved 2: Reserved 3: AI3 of expansion I/O card 4: AI4 of expansion I/O card 5: Temp of absolute PG card (first encoder) 6: Reserved 7: Temp of absolute PG card (second encoder) 8: Reserved 9: Temp of CANopen card 10: Reserved 11: Reserved 12: Temp of resolver PG card (first encoder) 13: Temp of resolver PG card (second encoder)	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-93	0xA45D	Type of temperature measurement signal 3	0: PT100 1: PT1000 2: KTY84	0	-	★
P4-94	0xA45E	Select source for temperature measurement signal 4	0: No function assigned 1: Reserved 2: Reserved 3: AI3 of expansion I/O card 4: AI4 of expansion I/O card 5: Temp of absolute PG card (first encoder) 6: Reserved 7: Temp of absolute PG card (second encoder) 8: Reserved 9: Temp of CANopen card 10: Reserved 11: Reserved 12: Temp of resolver PG card (first encoder) 13: Temp of resolver PG card (second encoder)	0	-	★
P4-95	0xA45F	Type of temperature measurement signal 4	0: PT100 1: PT1000 2: KTY84	0	-	★
P4-96	0xA460	Select source for PTC1 over-temperature measurement	0: No function assigned 1: PTC of absolute PG card (first encoder) 2: PTC of absolute PG card (second encoder) 3: PT130 of resolver PG card (first encoder) 4: PT130 of resolver PG card (second encoder) 5: PTC of CANopen card	0	-	☆
P4-97	0xA461	Select source for PTC2 over-temperature measurement	0: No function assigned 1: PTC of absolute PG card (first encoder) 2: PTC of absolute PG card (second encoder) 3: PT130 of resolver PG card (first encoder) 4: PT130 of resolver PG card (second encoder) 5: PTC of CANopen card	0	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P4-98	0xA462	Select source for PTC3 over-temperature measurement	0: No function assigned 1: PTC of absolute PG card (first encoder) 2: PTC of absolute PG card (second encoder) 3: PT130 of resolver PG card (first encoder) 4: PT130 of resolver PG card (second encoder) 5: PTC of CANopen card	0	-	☆

5.3.6 Group P5—Auxiliary Function Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P5-00	0xA500	Jog frequency reference	Maximum frequency of reverse rotation (P0-09) to maximum frequency (P0-08)	5.00	Hz	☆
P5-01	0xA501	Jog acceleration time	0.00 - 655.35	20.00	s	☆
P5-02	0xA502	Jog deceleration time	0.00 - 655.35	20.00	s	☆
P5-03	0xA503	Deceleration time for quick stop	0.00 - 655.35	1.00	s	☆
P5-04	0xA504	Acceleration time 2	0.00 - 655.35	20.00	s	☆
P5-05	0xA505	Deceleration time 2	0.00 - 655.35	20.00	s	☆
P5-06	0xA506	Acceleration time 3	0.00 - 655.35	20.00	s	☆
P5-07	0xA507	Deceleration time 3	0.00 - 655.35	20.00	s	☆
P5-08	0xA508	Acceleration time 4	0.00 - 655.35	20.00	s	☆
P5-09	0xA509	Deceleration time 4	0.00 - 655.35	20.00	s	☆
P5-10	0xA50A	Start protection	0: Disabled 1: Enabled	0	-	☆
P5-11	0xA50B	Reverse frequency control	0: Reverse frequency allowed 1: Reverse running not allowed	0	-	☆
P5-12	0xA50C	Reserved	0 - 65535	0	-	☆
P5-13	0xA50D	Running mode when frequency reference is lower than lower limit	0: Frequency lower limit 1: Stop 2: Run at zero speed	0	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P5-14	0xA50E	Measured width when target frequency reached	0.0 - 100.0	1.0	%	☆
P5-15	0xA50F	Frequency reached detection frequency 1	0.00 - Maximum frequency (P0-08)	50.00	Hz	☆
P5-16	0xA510	Frequency reached hysteresis value of detection frequency 1	0.0 - 100.0	5.0	%	☆
P5-17	0xA511	Frequency reached detection frequency 2	0.00 - Maximum frequency (P0-08)	50.00	Hz	☆
P5-18	0xA512	Frequency reached hysteresis value of detection frequency 2	0.0 - 100.0	5.0	%	☆
P5-19	0xA513	Detection value 1 when any frequency reached	0.00 - Maximum frequency (P0-08)	50.00	Hz	☆
P5-20	0xA514	Measured width 1 when any frequency reached	0.0 - 100.0	0.0	%	☆
P5-21	0xA515	Detection value 2 when any frequency reached	0.00 - Maximum frequency (P0-08)	50.00	Hz	☆
P5-22	0xA516	Measured width 2 when any frequency reached	0.0 - 100.0	0.0	%	☆
P5-23	0xA517	Any current reached 1	0.0 - 300.0	100.0	%	☆
P5-24	0xA518	Width of any current reached 1	0.0 - 300.0	0.0	%	☆
P5-25	0xA519	Any current reached 2	0.0 - 300.0	100.0	%	☆
P5-26	0xA51A	Width of any current reached 2	0.0 - 300.0	0.0	%	☆
P5-27	0xA51B	Set cumulative power-on reach time	0 - 65535	0	h	☆
P5-28	0xA51C	Set cumulative running reach time	0 - 65535	0	h	☆
P5-29	0xA51D	Set current running time	0.0 - 6500.0	0.0	min	☆
P5-30	0xA51E	Zero current detection level	0.0 - 300.0	5.0	%	☆
P5-31	0xA51F	Zero current detection delay time	0.00 - 600.00	0.10	s	☆
P5-32	0xA520	Output current exceeding limit	0.0 - 300.0	200.0	%	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P5-33	0xA521	Detection delay time for output current overrun	0.00 - 600.00	0.00	s	☆
P5-34	0xA522	Whether jump frequency is valid during acceleration/ deceleration	0: Disabled 1: Enabled	0	-	☆
P5-35	0xA523	Jump frequency 1	0.00 - Maximum frequency (P0-08)	0.00	Hz	☆
P5-36	0xA524	Jump frequency 2	0.00 - Maximum frequency (P0-08)	0.00	Hz	☆
P5-37	0xA525	Jump frequency amplitude	0.00 - Maximum frequency (P0-08)	0.00	Hz	☆
P5-38	0xA526	Dead time of forward or reverse running	0.0 - 3000.0	0.0	s	☆
P5-39	0xA527	Switching frequency point between acceleration time 1 and acceleration time 2	0.00 - Maximum frequency (P0-08)	0.00	Hz	☆
P5-40	0xA528	Switching frequency point between deceleration time 1 and deceleration time 2	0.00 - Maximum frequency (P0-08)	0.00	Hz	☆
P5-41	0xA529	AI1 input protection lower limit	0.00 - P5-42	3.10	V	☆
P5-42	0xA52A	AI1 input protection upper limit	P5-41 - 11.00	6.80	V	☆
P5-43	0xA52B	Select timing feature	0: Disabled 1: Enabled	0	-	★
P5-44	0xA52C	Timing duration source	0: P5-45 setting 1: AI1 2: AI2 3: AI3 4: AI4 5: AI5 (AI range specified by P5-45)	0	-	★
P5-45	0xA52D	Timed running time	0.0 - 6500.0	0.0	min	★
P5-46	0xA52E	Reference length	0 - 65535	1000	m	☆
P5-47	0xA52F	Actual length	0 - 65535	0	m	●
P5-48	0xA530	Pulses per meter	0.1 - 6553.5	100.0	-	☆
P5-49	0xA531	Reference count	0 - 65535	1000	-	☆
P5-50	0xA532	Specified count	0 - 65535	1000	-	☆
P5-51	0xA533	Select HDI application	0: HDI1 1: HDI2	0	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P5-52	0xA534	Vibration frequency setting mode	0: Relative to center frequency 1: Relative to maximum frequency	0	-	☆
P5-53	0xA535	Vibration frequency amplitude	0.0 - 100.0	0.0	%	☆
P5-54	0xA536	Jump frequency amplitude	0.0 - 50.0	0.0	%	☆
P5-55	0xA537	Vibration frequency period	0.1 - 3000.0	10.0	s	☆
P5-56	0xA538	Triangle wave rise time of vibration frequency	0.1 - 100.0	50.0	%	☆
P5-57	0xA539	Continue running frequency source selection upon fault	0: Current running frequency 1: Frequency reference 2: Frequency upper limit 3: Frequency lower limit 4: Standby frequency	0	-	☆
P5-58	0xA53A	Abnormal standby frequency upon fault	0.0 - 100.0	100.0	%	☆
P5-59	0xA53B	Start alignment time	0.0 - 1000.0	0.0	s	★
P5-60	0xA53C	Fan speed governing temperature	40 - 60	40	°C	☆
P5-61	0xA53D	Temperature for full-speed running of the fan	Temperature triggering fan start to 100°C (active when P9-05 is set to 3)	70	°C	☆
P5-62	0xA53E	Initial fan speed	0 - 100 (active when P9-05 is set to 3)	50	%	☆
P5-63	0xA53F	Display on LED 1	0: Frequency reference 1: Running frequency 2: Feedback speed 3: Output current 4: Output voltage 5: Bus voltage 6: AI1 voltage 7: AI2 voltage 8: AI3 voltage 9: Output power 10: Output torque 11: PID reference 12: PID feedback 13: Simple PLC 14: HDI input frequency 15: Current running time 16: Input line voltage 17: Group U parameters	5	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P5-64	0xA540	Display on LED 2	0: Frequency reference 1: Running frequency 2: Feedback speed 3: Output current 4: Output voltage 5: Bus voltage 6: AI1 voltage 7: AI2 voltage 8: AI3 voltage 9: Output power 10: Output torque 11: PID reference 12: PID feedback 13: Simple PLC 14: HDI input frequency 15: Current running time 16: Input line voltage 17: Group U parameters	3	-	☆
P5-65	0xA541	Braking start voltage	600 - 800	720	V	☆
P5-66	0xA542	Braking hysteresis voltage	0 - 100	10	V	☆
P5-67	0xA543	Braking rate	0 - 100	50	%	☆
P5-68	0xA544	Braking cycle	0.1 - 12.0	1.0	kHz	☆
P5-69	0xA545	Braking start condition	0: Start in any state 1: Start in running state 2: Disable braking	1	-	☆
P5-70	0xA546	Wake-up frequency	Dormancy frequency to maximum frequency	0.00	Hz	☆
P5-71	0xA547	Wake-up delay	0.0 - 6500.0	0.0	s	☆
P5-72	0xA548	Dormancy frequency	0.00 to wake-up frequency	0.00	Hz	☆
P5-73	0xA549	Dormancy delay	0.0 - 6500.0	0.0	s	☆
P5-75	0xA54B	Enable brake control	0: Disabled 1: Enabled	0	-	★
P5-76	0xA54C	Forward brake release frequency	0.00 - 20.00	2.00	Hz	★
P5-77	0xA54D	Reverse brake release frequency	0.00 - 20.00	2.00	Hz	★
P5-78	0xA54E	Forward brake release current	0.0 - 200.0	30.0	%	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P5-79	0xA54F	Reverse brake release current	0.0 - 200.0	30.0	%	★
P5-80	0xA550	Forward brake release torque (vector control active)	0.0 - 200.0	30.0	%	★
P5-81	0xA551	Reverse brake release torque (vector control active)	0.0 - 200.0	30.0	%	★
P5-82	0xA552	Brake release timeout interval	0.00 - 5.00	2.00	s	★
P5-83	0xA553	Brake release action time	0.00 - 5.00	0.30	s	★
P5-84	0xA554	Forward brake apply frequency	0.00 - 20.00	2.00	Hz	★
P5-85	0xA555	Reverse brake apply frequency	0.00 - 20.00	2.00	Hz	★
P5-86	0xA556	Brake delay	0.00 - 60.00	0.00	s	★
P5-87	0xA557	Braking action time	0.00 - 5.00	0.30	s	★
P5-88	0xA558	Forward compensation torque (vector control active)	-200.0 - 200.0	30.0	%	★
P5-89	0xA559	Reverse compensation torque (vector control active)	-200.0 - 200.0	30.0	%	★
P5-90	0xA55A	Forward running time of fan	0 - 6000	300	s	☆
P5-91	0xA55B	Reverse running time of fan	0 - 6000	60	s	☆

5.3.7 Group P6—Start and Stop Parameter Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P6-00	0xA600	Start mode	0: Direct start 1: DC braking before start 2: Speed tracking start 3: Pre-excitation start	0	-	★
P6-01	0xA601	Stop mode	0: Coast to stop 1: Decelerate to stop 2: Quick stop	1	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P6-02	0xA602	Stop determination frequency source	0: Feedback frequency 1: Frequency reference	0	-	★
P6-03	0xA603	Stop determination frequency	0.01 Hz to maximum frequency	0.50	Hz	★
P6-04	0xA604	Start magnetic pole position auto-tuning of synchronous motor	0: Disabled 1: Magnetic pole position auto-tuning solution 2: CD signal calculation solution (supported by sin-cos encoder)	1	-	★
P6-05	0xA605	Speed tracking current threshold of synchronous motor	1 - 100	30	%	★
P6-06	0xA606	Maximum frequency expansion coefficient of synchronous motor speed tracking	1 - 1000	500	%	★
P6-07	0xA607	V/f speed tracking speed of asynchronous motor	2 - 100	5	-	☆
P6-08	0xA608	V/f speed tracking current of asynchronous motor	30 - 150	50	%	★
P6-09	0xA609	Speed tracking demagnetization time of asynchronous motor	0.00 - 10.00	1.00	s	★
P6-10	0xA60A	Asynchronous motor pre-excitation command word	Bit 0: Pre-excitation exiting mode 0: Time reached 1: Flux linkage reached Bit 1: Enable stator resistance auto-tuning 0: Disabled 1: Enabled	0	-	★
P6-11	0xA60B	Asynchronous motor pre-excitation time	0.03 - 10.00	0.10	s	★
P6-12	0xA60C	Asynchronous motor pre-excitation current	0 - 1000	100	%	★
P6-13	0xA60D	Pre-torque	0: Disabled 1: Enabled	0	-	☆
P6-14	0xA60E	Pre-torque torque reference	0 - 1000	0	%	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P6-15	0xA60F	Pre-torque torque rise time	0.001 - 0.500	0.001	s	☆
P6-16	0xA610	Pre-torque time	0.000 - 60.000	0.600	s	☆
P6-17	0xA611	Pre-torque speed control bandwidth	0.1 - 1000.0	5.0	Hz	☆
P6-18	0xA612	Pre-torque speed control damping	0.10 - 10.00	0.70	-	☆
P6-19	0xA613	Pre-torque speed control integral time	0.001 - 60.000	0.062	s	☆
P6-20	0xA614	Pre-torque torque drop time	0: Disabled 0.001 - 60.00: Enabled	0.000	s	☆
P6-21	0xA615	I/F operation	0: Disabled 1: Enabled	0	-	☆
P6-22	0xA616	I/F operation switching frequency	0.01 Hz to maximum frequency	10.00	Hz	☆
P6-23	0xA617	I/F operation acceleration time	0.00 - 655.35	20.00	s	☆
P6-24	0xA618	I/F operation deceleration time	0.00 - 655.35	20.00	s	☆
P6-25	0xA619	I/F operation target current	0 - 1000	100	%	☆
P6-26	0xA61A	I/F operation current angle	0.0 - 360.0	0.0	°	☆
P6-29	0xA61D	Acceleration/ Deceleration mode	0: Linear acceleration/deceleration 1: S-curve acceleration/deceleration	0	-	★
P6-30	0xA61E	S-curve start segment ratio	0.0 - 100.0	30.0	%	★
P6-31	0xA61F	S-curve end segment ratio	0.0 - 100.0	30.0	%	★
P6-32	0xA620	Start frequency	0.00 - 10.00	0.00	Hz	☆
P6-33	0xA621	Start frequency holding time	0.0 - 100.0	0.0	s	☆
P6-34	0xA622	Start DC braking current	0 - 150	50	%	☆
P6-35	0xA623	Start DC braking time	0.00 - 100.00	2.00	s	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P6-36	0xA624	Start frequency for stop DC braking	0.05 to maximum frequency (frequency unit: 0.01) 0.5 to maximum frequency (frequency unit: 0.1)	0.05	Hz	☆
P6-37	0xA625	Waiting time before stop DC braking	0.0 - 100.0	0.0	s	☆
P6-38	0xA626	Stop DC braking current	0 - 1000	50	%	☆
P6-39	0xA627	Stop DC braking time	0.00 - 650.00	2.00	s	☆
P6-40	0xA628	Enable DC braking	0: Disabled 1: Enabled (solution 1) 2: Enabled (solution 2)	0	-	★
P6-41	0xA629	Condition for transition from DC braking to running	0: Timer expiration 1: AI1 2: AI2 3: AI3 4: AI4 5: AI5	0	-	★
P6-42	0xA62A	AI threshold for transition from DC braking to running	0.0 - 100.0	50.0	%	★
P6-43	0xA62B	Custom start/stop command source	0: Disabled 1: Stop at zero frequency, allow start upon AI reach	0	-	★
P6-44	0xA62C	Demagnetization time of asynchronous motor	0.00 - 5.00	0.50	s	★

5.3.8 Group P7–AI Parameter Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P7-00	0xA700	AI1 filter time	0.000 - 10.000	0.004	s	☆
P7-01	0xA701	AI1 minimum input	-10.000V to AI1 maximum input	0.000	(0.001) V/ (0.01) mA	☆
P7-02	0xA702	Setpoint at AI1 minimum input	-100.0 - 100.0	0.0	%	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P7-03	0xA703	AI1 maximum input	AI1 minimum input to 10.000V	10.000	(0.001) V/ (0.01) mA	☆
P7-04	0xA704	Setpoint at AI1 maximum input	0.0 - 100.0	100.0	%	☆
P7-05	0xA705	AI2 filter time	0.000 - 10.000	0.004	s	☆
P7-06	0xA706	AI2 minimum input	0.000V to AI2 maximum input	0.000	(0.001) V/ (0.01) mA	☆
P7-07	0xA707	Setpoint at AI2 minimum input	0.0 - 100.0	0.0	%	☆
P7-08	0xA708	AI2 maximum input	AI2 minimum input to 10.000V	10.000	(0.001) V/ (0.01) mA	☆
P7-09	0xA709	Setpoint at AI2 maximum input	0.0 - 100.0	100.0	%	☆
P7-10	0xA70A	AI3 filter time	0.000 - 10.000	0.004	s	☆
P7-11	0xA70B	AI3 minimum input	0.000V to AI3 maximum input	0.000	(0.001) V/ (0.01) mA	☆
P7-12	0xA70C	Setpoint at AI3 minimum input	0.0 - 100.0	0.0	%	☆
P7-13	0xA70D	AI3 maximum input	AI3 minimum input to 10.000V	10.000	(0.001) V/ (0.01) mA	☆
P7-14	0xA70E	Setpoint at AI3 maximum input	0.0 - 100.0	100.0	%	☆
P7-15	0xA70F	AI4 filter time	0.000 - 10.000	0.004	s	☆
P7-16	0xA710	AI4 minimum input	0.000V to AI4 maximum input	0.000	(0.001) V/ (0.01) mA	☆
P7-17	0xA711	Setpoint at AI4 minimum input	0.0 - 100.0	0.0	%	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P7-18	0xA712	AI4 maximum input	AI4 minimum input to 10.000V	10.000	(0.001) V/ (0.01) mA	☆
P7-19	0xA713	Setpoint at AI4 maximum input	0.0 - 100.0	100.0	%	☆
P7-20	0xA714	AI5 filter time	0.000 - 10.000	0.004	s	☆
P7-21	0xA715	AI5 minimum input	0.000V to AI5 maximum input	0.000	(0.001) V/ (0.01) mA	☆
P7-22	0xA716	Setpoint at AI5 minimum input	0.0 - 100.0	0.0	%	☆
P7-23	0xA717	AI5 maximum input	AI5 minimum input to 10.000V	10.000	V	☆
P7-24	0xA718	Setpoint at AI5 maximum input	0.0 - 100.0	100.0	%	☆
P7-25	0xA719	Minimum pulse input	0.00 - 100.00	0.000	kHz	☆
P7-26	0xA71A	Setpoint at minimum pulse input	-100.0 - 100.0	0.0	%	☆
P7-27	0xA71B	Maximum pulse input	0.00 - 100.00	50.00	kHz	☆
P7-28	0xA71C	Setpoint at maximum pulse input	-100.0 - 100.0	100.0	%	☆
P7-29	0xA71D	AO1 zero offset coefficient	-100.0 - 100.0	0.0	%	☆
P7-30	0xA71E	AO1 gain	-10.00 - 10.00	1.00	-	☆
P7-31	0xA71F	AO2 zero offset coefficient	-100.0 - 100.0	0.0	%	☆
P7-32	0xA720	AO2 gain	-10.00 - 10.00	1.00	-	☆
P7-33	0xA721	AO3 zero offset coefficient	-100.0 - 100.0	0.0	%	☆
P7-34	0xA722	AO3 gain	-10.00 - 10.00	1.00	-	☆

5.3.9 Group P8—Torque Control Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P8-00	0xA800	Torque control mode	0: Speed mode 1: Torque mode	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P8-01	0xA801	Torque reference source (full range of options 1 to 5 specified by P8-02)	0: Digital setting 1: AI1 2: AI2 3: AI3 4: AI4 5: AI5 6: Pulse 7: Communication	0		★
P8-02	0xA802	Torque digital setting	-300.0 - 300.0	100.0	%	☆
P8-03	0xA803	Filter time for torque reference setting	0 - 5.000	0.000	s	☆
P8-04	0xA804	Speed limit source	0: Digital setting 1: Frequency source setting	0		★
P8-05	0xA805	Speed limit	Maximum frequency of reverse rotation (P0-09) to maximum frequency (P0-08)	10.00	Hz	☆
P8-06	0xA806	Speed limit offset	0.00 Hz to maximum frequency (P0-08)	5.00	Hz	★
P8-07	0xA807	Frequency of switching from maximum electric torque to power generation torque	0 - 20	10.00	Hz	★
P8-08	0xA808	Speed limit offset mode	0: Bidirectional offset 1: Unidirectional offset	0		★
P8-09	0xA809	Speed limit frequency acceleration time	0.00 - 650.00	1.00	s	☆
P8-10	0xA80A	Speed limit frequency deceleration time	0.00 - 650.00	1.00	s	☆
P8-11	0xA80B	Stop mode in torque control	0: No switching 1: Switch to speed control upon shutdown 2: Switch to 0 upon shutdown	0		★
P8-12	0xA80C	Smooth transition time for switching from speed mode to torque mode	0.000 - 65.000	0.000	s	☆
P8-13	0xA80D	Rise time of torque reference ramp	0.00 - 650.00	0.00	s	☆
P8-14	0xA80E	Drop time of torque reference ramp	0.00 - 650.00	0.00	s	☆

5.3.10 Group P9–Fault and Protection Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P9-00	0xA900	Short-to-ground protection	0: Disabled 1: Enabled	1	-	☆
P9-01	0xA901	Pre-charge resistor protection	0: Disabled 1: Enabled	1	-	☆
P9-02	0xA902	Input phase-loss protection	0: Disabled 1: Enabled	1	-	☆
P9-03	0xA903	Output phase-loss protection	Ones digit: Output phase-loss protection during running 0: Disabled 1: Enabled Tens digit: Output phase-loss protection before running 0: Disabled 1: Enabled	1	-	☆
P9-05	0xA905	Fan working mode	0: Always running 1: Run with the device 2: Run above specified temperature 3: Regulated run	1	-	☆
P9-07	0xA907	System undervoltage point	0.0 - 400.0	350.0	-	★
P9-08	0xA908	Allowed running duration of braking unit	0.0 - 3600.0	0.0	-	☆
P9-09	0xA909	Fan fault detection	0: Disabled 1: Enabled	1	-	☆
P9-10	0xA90A	Temperature for module over-temperature warning	0 to IGBT module over-temperature point	80	°C	☆
P9-11	0xA90B	Motor overload protection	0: Disabled 1: Enabled	1	-	☆
P9-12	0xA90C	Motor overload protection gain	0.20 - 10.00	100	%	☆
P9-13	0xA90D	Pre-warning coefficient for motor overload protection	50 - 100	80	%	☆
P9-14	0xA90E	Fan direction control	0: Disabled 1: Enabled	0	-	★
P9-15	0xA90F	Threshold for motor overspeed detection	0.0 - 50.0 (maximum frequency P0-08)	5.0	%	☆
P9-16	0xA910	Motor overspeed detection time	0.0: No detection 0.1 - 60.0	1.0	s	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P9-17	0xA911	Motor speed deviation upper threshold	1.0 - 50.0 (maximum frequency P0-08)	20.0	%	☆
P9-18	0xA912	Time of excessive motor speed deviation	0.0: No detection 0.1 - 60.0	5.0	s	☆
P9-19	0xA913	Motor load drop detection level	0.0 - 100.0	10.0	%	☆
P9-20	0xA914	Motor load drop detection time	0.0: No detection 0.1 - 60.0	0.0	s	☆
P9-21	0xA915	24 V power exception detection	0: Disabled 1: Enabled	1	-	☆
P9-22	0xA916	Set KTY temperature upper limit	0 - 300	130	°C	☆
P9-23	0xA917	Software overcurrent protection point	0 - 1000 0 Software overcurrent protection disabled	300	%	★
P9-24	0xA918	Power-on initialization timeout threshold	0.0: No detection 0.1 - 60.0	0.0	s	☆
P9-26	0xA91A	Motor locked-rotor detection current	20 - 200	100	%	☆
P9-27	0xA91B	Motor locked-rotor detection frequency	0 - 100	5	%	☆
P9-28	0xA91C	Motor locked-rotor detection time	0.0: No detection 0.0 - 600.0	0.0	s	☆
P9-29	0xA91D	Black box sampling frequency	0: Slow 1: Medium 2: Fast	1	-	☆
P9-30	0xA91E	Motor over-temperature detection source 1	0: No detection 1: Temperature measurement signal 1 2: Temperature measurement signal 2 3: Temperature measurement signal 3 4: Temperature measurement signal 4	0	-	☆
P9-31	0xA91F	Motor over-temperature point 1	0 - 200	120	°C	☆
P9-32	0xA920	Motor over-temperature detection source 2	0: No detection 1: Temperature measurement signal 1 2: Temperature measurement signal 2 3: Temperature measurement signal 3 4: Temperature measurement signal 4	0	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P9-33	0xA921	Motor over-temperature point 2	0 - 200	120	°C	☆
P9-34	0xA922	First fault	0.00 - 655.35	0.00	-	●
P9-35	0xA923	Second fault	0.00 - 655.35	0.00	-	●
P9-36	0xA924	Third (latest) fault	0.00 - 655.35	0.00	-	●
P9-37	0xA925	Frequency upon third (latest) fault	0.00 - 655.35	0.00	Hz	●
P9-38	0xA926	Current upon third (latest) fault	0.0 - 6553.5	0.0	A	●
P9-39	0xA927	Bus voltage upon third (latest) fault	0 - 65535	0	V	●
P9-40	0xA928	Input terminal status upon third (latest) fault	0 - 65535	0	-	●
P9-41	0xA929	Output terminal status upon third (latest) fault	0 - 65535	0	-	●
P9-42	0xA92A	Status upon third (latest) fault	0 - 65535	0	-	●
P9-43	0xA92B	Power-on time upon third (latest) fault	0 - 65535	0	min	●
P9-44	0xA92C	Running time upon third (latest) fault	0 - 65535	0	min	●
P9-45	0xA92D	IGBT temperature upon third (latest) fault	-32768 - 32767	0	°C	●
P9-46	0xA92E	Subcode of third (latest) fault	0 - 65535	0	-	●
P9-47	0xA92F	Frequency upon second fault	0.00 - 655.35	0.00	Hz	●
P9-48	0xA930	Current upon second fault	0.0 - 6553.5	0.0	A	●
P9-49	0xA931	Bus voltage upon second fault	0 - 65535	0	V	●
P9-50	0xA932	Input terminal status upon second fault	0 - 65535	0	-	●
P9-51	0xA933	Output terminal status upon second fault	0 - 65535	0	-	●
P9-52	0xA934	Status upon second fault	0 - 65535	0	-	●

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P9-53	0xA935	Power-on time upon second fault	0 - 65535	0	min	●
P9-54	0xA936	Running time upon second fault	0.0 - 6553.5	0.0	min	●
P9-55	0xA937	IGBT temperature upon second fault	-32768 - 32767	0	°C	●
P9-56	0xA938	Subcode of second fault	0 - 65535	0	-	●
P9-57	0xA939	Frequency upon first fault	0.00 - 655.35	0.00	Hz	●
P9-58	0xA93A	Current upon first fault	0.0 - 6553.5	0.0	A	●
P9-59	0xA93B	Bus voltage upon first fault	0 - 65535	0	V	●
P9-60	0xA93C	Output terminal status upon first fault	0 - 65535	0	-	●
P9-61	0xA93D	Power-on time upon first fault	0 - 65535	0	-	●
P9-62	0xA93E	Status upon first fault	0 - 65535	0	-	●
P9-63	0xA93F	Power-on time upon first fault	0 - 65535	0	min	●
P9-64	0xA940	Running time upon first fault	0.0 - 6553.5	0.0	min	●
P9-65	0xA941	IGBT temperature upon first fault	-32768 - 32767	0	°C	●
P9-66	0xA942	Subcode of first fault	0 - 65535	0	-	●
P9-67	0xA943	Auto fault resets	0: Auto fault reset disabled 0 - 5	0	-	★
P9-68	0xA944	Select faulty DO action during auto fault reset	0: Do not output a fault signal 1: Output a fault signal	0	-	★
P9-69	0xA945	Auto fault reset interval	0.1 - 100.0	1.0	s	★
P9-70	0xA946	Restart interval for fault reset	0.0 - 100.0	1.0	s	★
P9-71	0xA947	Allow restarting fault code 1 upon fault reset	0: Unavailable 2: Overcurrent 3: Overvoltage 4: Undervoltage 5: AC drive overload 7: Motor overload 13: STO fault 15: External fault 21: Motor control exception	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P9-72	0xA948	Allow restarting fault code 2 upon fault reset	0: Unavailable 2: Overcurrent 3: Overvoltage 4: Undervoltage 5: AC drive overload 7: Motor overload 13: STO fault 15: External fault 21: Motor control exception	0	-	★
P9-73	0xA949	Allow restarting fault code 3 upon fault reset	0: Unavailable 2: Overcurrent 3: Overvoltage 4: Undervoltage 5: AC drive overload 7: Motor overload 13: STO fault 15: External fault 21: Motor control exception	0	-	★
P9-74	0xA94A	Allow restarting fault code 4 upon fault reset	0: Unavailable 2: Overcurrent 3: Overvoltage 4: Undervoltage 5: AC drive overload 7: Motor overload 13: STO fault 15: External fault 21: Motor control exception	0	-	★
P9-75	0xA94B	Allow restarting fault code 5 upon fault reset	0: Unavailable 2: Overcurrent 3: Overvoltage 4: Undervoltage 5: AC drive overload 7: Motor overload 13: STO fault 15: External fault 21: Motor control exception	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P9-76	0xA94C	Fault code 1 whose fault level is to be changed	0: Unavailable 7: Motor overload 8: Motor over-temperature 11: Encoder fault 12: Fan fault 13: STO fault 15: External fault 16: Communication fault 17: Output phase loss 18: Overspeed 19: Excessive speed deviation 20: Load loss 21: Motor control exception 23: Cumulative running time reached 24: Cumulative power-on time reached 25: User-defined fault 1 26: User-defined fault 2 33: PID feedback loss	0	-	★
P9-77	0xA94D	Fault level of fault code 1	0: Coast to stop 1: Emergency stop 2: Stop according to the stop mode 3: Reserved 4: Warning 5: Fault removed	0	-	★
P9-78	0xA94E	Fault code 2 whose fault level is to be changed	0: Unavailable 7: Motor overload 8: Motor over-temperature 11: Encoder fault 12: Fan fault 13: STO fault 15: External fault 16: Communication fault 17: Output phase loss 18: Overspeed 19: Excessive speed deviation 20: Load loss 21: Motor control exception 23: Cumulative running time reached 24: Cumulative power-on time reached 25: User-defined fault 1 26: User-defined fault 2 33: PID feedback loss	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P9-79	0xA94F	Fault level of fault code 2	0: Coast to stop 1: Emergency stop 2: Stop according to the stop mode 3: Reserved 4: Warning 5: Fault removed	0	-	★
P9-80	0xA950	Fault code 3 whose fault level is to be changed	0: Unavailable 7: Motor overload 8: Motor over-temperature 11: Encoder fault 12: Fan fault 13: STO fault 15: External fault 16: Communication fault 17: Output phase loss 18: Overspeed 19: Excessive speed deviation 20: Load loss 21: Motor control exception 23: Cumulative running time reached 24: Cumulative power-on time reached 25: User-defined fault 1 26: User-defined fault 2 33: PID feedback loss	0	-	★
P9-81	0xA951	Fault level of fault code 3	0: Coast to stop 1: Emergency stop 2: Stop according to the stop mode 3: Reserved 4: Warning 5: Fault removed	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P9-82	0xA952	Fault code 4 whose fault level is to be changed	0: Unavailable 7: Motor overload 8: Motor over-temperature 11: Encoder fault 12: Fan fault 13: STO fault 15: External fault 16: Communication fault 17: Output phase loss 18: Overspeed 19: Excessive speed deviation 20: Load loss 21: Motor control exception 23: Cumulative running time reached 24: Cumulative power-on time reached 25: User-defined fault 1 26: User-defined fault 2 33: PID feedback loss	0	-	★
P9-83	0xA953	Fault level of fault code 4	0: Coast to stop 1: Emergency stop 2: Stop according to the stop mode 3: Reserved 4: Warning 5: Fault removed	0	-	★
P9-84	0xA954	Fault code 5 whose fault level is to be changed	0: Unavailable 7: Motor overload 8: Motor over-temperature 11: Encoder fault 12: Fan fault 13: STO fault 15: External fault 16: Communication fault 17: Output phase loss 18: Overspeed 19: Excessive speed deviation 20: Load loss 21: Motor control exception 23: Cumulative running time reached 24: Cumulative power-on time reached 25: User-defined fault 1 26: User-defined fault 2 33: PID feedback loss	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P9-85	0xA955	Fault level of fault code 5	0: Coast to stop 1: Emergency stop 2: Stop according to the stop mode 3: Reserved 4: Warning 5: Fault removed	0	-	★
P9-86	0xA956	Fault code 6 whose fault level is to be changed	0: Unavailable 7: Motor overload 8: Motor over-temperature 11: Encoder fault 12: Fan fault 13: STO fault 15: External fault 16: Communication fault 17: Output phase loss 18: Overspeed 19: Excessive speed deviation 20: Load loss 21: Motor control exception 23: Cumulative running time reached 24: Cumulative power-on time reached 25: User-defined fault 1 26: User-defined fault 2 33: PID feedback loss	0	-	★
P9-87	0xA957	Fault level of fault code 6	0: Coast to stop 1: Emergency stop 2: Stop according to the stop mode 3: Reserved 4: Warning 5: Fault removed	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P9-88	0xA958	Fault code 7 whose fault level is to be changed	0: Unavailable 7: Motor overload 8: Motor over-temperature 11: Encoder fault 12: Fan fault 13: STO fault 15: External fault 16: Communication fault 17: Output phase loss 18: Overspeed 19: Excessive speed deviation 20: Load loss 21: Motor control exception 23: Cumulative running time reached 24: Cumulative power-on time reached 25: User-defined fault 1 26: User-defined fault 2 33: PID feedback loss	0	-	★
P9-89	0xA959	Fault level of fault code 7	0: Coast to stop 1: Emergency stop 2: Stop according to the stop mode 3: Reserved 4: Warning 5: Fault removed	0	-	★
P9-90	0xA95A	Fault code 8 whose fault level is to be changed	0: Unavailable 7: Motor overload 8: Motor over-temperature 11: Encoder fault 12: Fan fault 13: STO fault 15: External fault 16: Communication fault 17: Output phase loss 18: Overspeed 19: Excessive speed deviation 20: Load loss 21: Motor control exception 23: Cumulative running time reached 24: Cumulative power-on time reached 25: User-defined fault 1 26: User-defined fault 2 33: PID feedback loss	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P9-91	0xA95B	Fault level of fault code 8	0: Coast to stop 1: Emergency stop 2: Stop according to the stop mode 3: Reserved 4: Warning 5: Fault removed	0	-	★
P9-92	0xA95C	Fault code 9 whose fault level is to be changed	0: Unavailable 7: Motor overload 8: Motor over-temperature 11: Encoder fault 12: Fan fault 13: STO fault 15: External fault 16: Communication fault 17: Output phase loss 18: Overspeed 19: Excessive speed deviation 20: Load loss 21: Motor control exception 23: Cumulative running time reached 24: Cumulative power-on time reached 25: User-defined fault 1 26: User-defined fault 2 33: PID feedback loss	0	-	★
P9-93	0xA95D	Fault level of fault code 9	0: Coast to stop 1: Emergency stop 2: Stop according to the stop mode 3: Reserved 4: Warning 5: Fault removed	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
P9-94	0xA95E	Fault code 10 whose fault level is to be changed	0: Unavailable 7: Motor overload 8: Motor over-temperature 11: Encoder fault 12: Fan fault 13: STO fault 15: External fault 16: Communication fault 17: Output phase loss 18: Overspeed 19: Excessive speed deviation 20: Load loss 21: Motor control exception 23: Cumulative running time reached 24: Cumulative power-on time reached 25: User-defined fault 1 26: User-defined fault 2 33: PID feedback loss	0	-	★
P9-95	0xA95F	Fault level of fault code 10	0: Coast to stop 1: Emergency stop 2: Stop according to the stop mode 3: Reserved 4: Warning 5: Fault removed	0	-	★

5.3.11 Group PA–Auxiliary Control Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
PA-00	0xAA00	Flux decreased depth for maximum voltage	0 - 100	5	%	★
PA-01	0xAA01	Dead zone compensation	0: Disabled 1: Actual current 2: Predicted current	1	-	★
PA-02	0xAA02	Narrow pulse limiting	0: Disabled 1: Enabled	1	-	★
PA-03	0xAA03	DPWM switchover frequency	0.00 - 300.00 (0: Disabled)	12.00	Hz	☆
PA-04	0xAA04	SVC low speed carrier frequency	1.2 - 12.0	3.0	kHz	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
PA-05	0xAA05	Reduce carrier frequency upon over-temperature	0: Disabled 1: Enabled	0	-	★
PA-06	0xAA06	DC braking carrier frequency	0.0 - 12.0	1.0	kHz	★
PA-07	0xAA07	Zero speed carrier frequency	0.0 - 3.0 (0: Disabled)	1.0	kHz	★
PA-08	0xAA08	Select power dip ride-through mode	0: Disabled 1: Decelerate 2: Decelerate to stop 3: Suppress voltage dip	0	-	☆
PA-09	0xAA09	Voltage for power dip ride-through	400 - 500	450	V	☆
PA-10	0xAA0A	Rising value of voltage for power dip ride-through	PA-09 - 750	650	V	☆
PA-11	0xAA0B	Voltage dip suppression time	100 - 5000	300	ms	☆
PA-12	0xAA0C	Filter time constant of interrupt speed feedback	0.0000 - 6.0000	0.0040	s	★
PA-15	0xAA0F	Deceleration time for power dip ride-through	0.00 - 300.00	10.00	s	☆
PA-23	0xAA17	Reluctance motor model command word	Bit 0: Enable open loop 0: Disabled 1: Enabled Bit 1: Enable online inductance 0: Disabled 1: Enabled Bit 2 to bit 3: Model selection 0: Solution 1 1: Solution 2 3: Solution 3 Bit 4: Current sampling delay 0: No delay 1: Delayed by one beat	11	-	☆
PA-24	0xAA18	Reluctance motor model switching frequency	0.0 - 100.0	5.0	Hz	☆
PA-25	0xAA19	Speed observation gain of reluctance motor	1 - 1000	50	Hz	☆
PA-27	0xAA1B	Reluctance motor control frequency 1	0.0 - 1000.0	1.0	Hz	☆
PA-28	0xAA1C	Reluctance motor control frequency 2	0.0 - 1000.0	10.0	Hz	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
PA-29	0xAA1D	Reluctance motor control command word	Bit 0 to bit 1: MTPA solution 0: Solution 1 1: Solution 2 3: Solution 3 Bit 2: Enable online inductance 0: Disabled 1: Enabled Bit 3: Enable open loop 0: Disabled 1: Enabled	0	-	☆
PA-30	0xAA1E	Flux linkage control gain of reluctance motor control	0 - 1000	20	Hz	☆
PA-31	0xAA1F	Flux linkage control integral time of reluctance motor control	0 - 1000	100	%	☆
PA-32	0xAA20	Torque control gain of reluctance motor control	0 - 1000	40	Hz	☆
PA-33	0xAA21	Torque control integral time of reluctance motor control	0 - 1000	200	%	☆
PA-34	0xAA22	Coefficient of control point 1 with optimal efficiency	0 - 200	65	%	☆
PA-35	0xAA23	Coefficient of control point 2 with optimal efficiency	0 - 200	85	%	☆
PA-36	0xAA24	Coefficient of control point 3 with optimal efficiency	0 - 200	95	%	☆
PA-37	0xAA25	Coefficient of control point 4 with optimal efficiency	0 - 200	100	%	☆
PA-38	0xAA26	MTPA control coefficient	0 - 100	90	%	☆

5.3.12 Group PB–PID Parameter Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
PB-00	0xAB00	PID reference source	0: PID numeric reference 1: AI1 2: AI2 3: AI3 4: AI4 5: AI5 6: Communication 7: Torque reference	0	-	★
PB-01	0xAB01	PID numeric reference	0.0 - 100.0	50.0	%	☆
PB-02	0xAB02	PID feedback source	0: AI1 1: AI2 2: AI3 3: AI4 4: AI5 5: Communication 6: Torque feedback	0	-	☆
PB-03	0xAB03	PID action direction	0: Forward 1: Reverse	0	-	★
PB-04	0xAB04	PID reference feedback range	0 - 65535	1000	-	☆
PB-05	0xAB05	Proportional gain P	0.0 - 1000.0	10.0	-	☆
PB-06	0xAB06	Integral time I	0.00 - 10.00	1.50	s	☆
PB-07	0xAB07	Derivative time D	0.000 - 10.000	0.000	s	☆
PB-08	0xAB08	PID reverse cutoff frequency	0.00 - 50.00	0.00	Hz	☆
PB-09	0xAB09	PID deviation limit	0.0 - 100.0	0.0	%	☆
PB-10	0xAB0A	PID differential limiting	0.00 - 100.00	1.50	%	☆
PB-11	0xAB0B	PID reference change time	0.00 - 650.00	5.00	s	☆
PB-12	0xAB0C	PID feedback filter time	0.00 - 60.00	0.0..	s	☆
PB-13	0xAB0D	PID output filter time	0.00 - 60.00	0.00	s	☆
PB-14	0xAB0E	PID proportional gain P2	0.0 - 1000.0	10.0		☆
PB-15	0xAB0F	PID integral time I2	0.00 - 10.00	1.50	s	☆
PB-16	0xAB10	PID derivative time D2	0.000 - 10.000	0.000	s	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
PB-17	0xAB11	PID parameter switching condition	0: No switching 1: Switching via DI 2: Auto switching based on deviation 3: Auto switching based on running frequency 6: Auto adjustment based on roll diameter 7: Adjustment based on maximum roll diameter percentage	0	-	☆
PB-18	0xAB12	PID parameter switching deviation 1	0.0 to PID parameter switching deviation 2	20.0	%	☆
PB-19	0xAB13	PID parameter switching deviation 2	PID parameter switching deviation 1 - 100.0%	80.0	%	☆
PB-20	0xAB14	PID initial value	0.0 - 100.0	0.0	%	☆
PB-21	0xAB15	PID initial value holding time	0.00 - 650.00	0.00	s	☆
PB-22	0xAB16	Maximum deviation between two PID outputs	0.00 - 100.00	1.00	%	☆
PB-23	0xAB17	Minimum deviation between two PID outputs	0.00 - 100.00	1.00	%	☆
PB-24	0xAB18	PID integral attribute	0: Continue integral 1: Stop integral	0	-	☆
PB-25	0xAB19	PID feedback loss detection value	0.0: No detection 0.1 - 100.0	0.0	%	☆
PB-26	0xAB1A	PID feedback loss detection time	0.0 - 20.0	0.0	s	☆
PB-27	0xAB1B	PID output limiting mode	0: Revert the integral when the output limit is reached 1: Revert the integral and assign a value when the output limit is reached	1	-	☆
PB-28	0xAB1C	PID integral initial value	0: Set the integral initial value of the first beat to 0 1: Revert the integral initial value of the first beat based on the output	1	-	☆

5.3.13 Group PC—Communication Parameter Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
PC-00	0xAC00	Local station number	1 - 247	1	-	☆
PC-01	0xAC01	Modbus baud rate	0: 2400bps 1: 4800bps 2: 9600bps 3: 19200bps 4: 38400bps 5: 57600bps 6: 115200bps	2	-	☆
PC-02	0xAC02	Modbus data format	0: No check (8-N-2) 1: Even check(8-E-1) 2: Odd check(8-O-1) 3: No check (8-N-1) 4: Even check(8-E-2) 5: Odd check(8-O-2) 6: ASCII – not checked (7-N-2) 7: ASCII – not checked (8-N-2) 8: ASCII – not checked (8-N-1) 9: ASCII – even checked (8-E-1) 10: ASCII – odd checked (8-O-1) 11: ASCII – even checked (8-E-2) 12: ASCII – odd checked (8-O-2)	0	-	☆
PC-03	0xAC03	Modbus response delay	0 - 20	2	ms	☆
PC-04	0xAC04	Modbus timeout period	0.0 - 60.0	0.0	s	☆
PC-05	0xAC05	CANopen master node ID	1 - 127	127	-	★
PC-06	0xAC06	CANopen communication status	0: Initializing 1: Application layer reset 2: Communication reset 3: Pre-operation 4: Operating 5: Stop	0	-	●
PC-07	0xAC07	Canopen baud rate	0: 20kbps 1: 50kbps 2: 100bps 3: 125kbps 4: 250kbps 5: 500kbps 6: 800kbps 7: 1Mbps	5	-	★
PC-08	0xAC08	Auto reset of Modbus timeout fault	0: Disabled 1: Enabled	0	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
PC-10	0xAC0A	Interaction frequency of expansion communication card (effective after power cycling)	0: 50Hz 1: 100Hz 2: 200Hz 3: 500Hz 4: 1000Hz 5: 2000Hz 6: 5000Hz	4	Hz	★
PC-11	0xAC0B	Tick count for frame loss determination in cycle control	2 - 20	10	-	☆
PC-12	0xAC0C	EtherCAT station alias	0 - 65535	0	-	☆
PC-13	0xAC0D	Saving parameters written during communication	0: Disabled 1: Enabled	0	-	☆
PC-14	0xAC0E	Communication parameter mapping	0: Disabled 1: Enabled	0	-	☆
PC-23	0xAC17	Disconnection determination time for expansion card 1	0.0 - 500.0 (0: Disconnection protection disabled)	50.0	ms	☆
PC-24	0xAC18	Disconnection determination time for expansion card 2	0.0 - 500.0 (0: Disconnection protection disabled)	50.0	ms	☆
PC-25	0xAC19	Disconnection determination time for expansion card 3	0.0 - 500.0 (0: Disconnection protection disabled)	50.0	ms	☆
PC-30	0xAC1E	Master station lifecycle	0 - 2000 (0: Master online monitoring disabled)	200	ms	☆
PC-31	0xAC1F	Number of master station disconnections	1 - 20	3	-	☆
PC-32	0xAC20	Select communication function for RJ45 port	0: External operating panel 1: Modbus	0	-	☆
PC-33	0xAC21	RJ45 – local station number	1 - 247	1	-	☆
PC-34	0xAC22	RJ45 – Modbus baud rate	0: 2400bps 1: 4800bps 2: 9600bps 3: 19200bps 4: 38400bps 5: 57600bps 6: 115200bps	2	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
PC-35	0xAC23	RJ45 – Modbus data format	0: No check (8-N-2) 1: Even check(8-E-1) 2: Odd check(8-O-1) 3: No check (8-N-1) 4: Even check(8-E-2) 5: Odd check(8-O-2) 6: ASCII – not checked (7-N-2) 7: ASCII – not checked (8-N-2) 8: ASCII – not checked (8-N-1) 9: ASCII – even checked (8-E-1) 10: ASCII – odd checked (8-O-1) 11: ASCII – even checked (8-E-2) 12: ASCII – odd checked (8-O-2)	0	-	☆
PC-36	0xAC24	RJ45 – Modbus response delay	0 - 20	2	ms	☆
PC-37	0xAC25	RJ45 – Modbus timeout period	0.0 - 60.0	0.0	s	☆

5.3.14 Group PD–Multi-segment References of Simple PLC Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
PD-00	0xAD00	Multi-segment reference 0	-100.0 - 100.0	0.0	%	☆
PD-01	0xAD01	Multi-segment reference 1	-100.0 - 100.0	0.0	%	☆
PD-02	0xAD02	Multi-segment reference 2	-100.0 - 100.0	0.0	%	☆
PD-03	0xAD03	Multi-segment reference 3	-100.0 - 100.0	0.0	%	☆
PD-04	0xAD04	Multi-segment reference 4	-100.0 - 100.0	0.0	%	☆
PD-05	0xAD05	Multi-segment reference 5	-100.0 - 100.0	0.0	%	☆
PD-06	0xAD06	Multi-segment reference 6	-100.0 - 100.0	0.0	%	☆
PD-07	0xAD07	Multi-segment reference 7	-100.0 - 100.0	0.0	%	☆
PD-08	0xAD08	Multi-segment reference 8	-100.0 - 100.0	0.0	%	☆
PD-09	0xAD09	Multi-segment reference 9	-100.0 - 100.0	0.0	%	☆
PD-10	0xAD0A	Multi-segment reference 10	-100.0 - 100.0	0.0	%	☆
PD-11	0xAD0B	Multi-segment reference 11	-100.0 - 100.0	0.0	%	☆
PD-12	0xAD0C	Multi-segment reference 12	-100.0 - 100.0	0.0	%	☆
PD-13	0xAD0D	Multi-segment reference 13	-100.0 - 100.0	0.0	%	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
PD-14	0xAD0E	Multi-segment reference 14	-100.0 - 100.0	0.0	%	☆
PD-15	0xAD0F	Multi-segment reference 15	-100.0 - 100.0	0.0	%	☆
PD-16	0xAD10	Simple PLC operation mode	0: Stop after running for one cycle 1: Keep final values after running for one cycle 2: Repeat after running for one cycle	0	-	★
PD-17	0xAD11	Select retention upon power failure for simple PLC	Ones digit: Retentive upon power failure 0: Disabled 1: Enabled Tens digit: Retentive upon stop 0: Disabled 1: Enabled	0	-	☆
PD-18	0xAD12	Segment 0 running time of simple PLC	0.0 - 6553.5	0.0	s(h)	☆
PD-19	0xAD13	Select segment 0 acceleration/deceleration time for simple PLC	0 - 3	0	-	☆
PD-20	0xAD14	Segment 1 running time of simple PLC	0.0 - 6553.5	0.0	s(h)	☆
PD-21	0xAD15	Select segment 1 acceleration/deceleration time for simple PLC	0 - 3	0	-	☆
PD-22	0xAD16	Segment 2 running time of simple PLC	0.0 - 6553.5	0.0	s(h)	☆
PD-23	0xAD17	Select segment 2 acceleration/deceleration time for simple PLC	0 - 3	0	-	☆
PD-24	0xAD18	Segment 3 running time of simple PLC	0.0 - 6553.5	0.0	s(h)	☆
PD-25	0xAD19	Select segment 3 acceleration/deceleration time for simple PLC	0 - 3	0	-	☆
PD-26	0xAD1A	Segment 4 running time of simple PLC	0.0 - 6553.5	0.0	s(h)	☆
PD-27	0xAD1B	Select segment 4 acceleration/deceleration time for simple PLC	0 - 3	0	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
PD-28	0xAD1C	Segment 5 running time of simple PLC	0.0 - 6553.5	0.0	s(h)	☆
PD-29	0xAD1D	Select segment 5 acceleration/deceleration time for simple PLC	0 - 3	0	-	☆
PD-30	0xAD1E	Segment 6 running time of simple PLC	0.0 - 6553.5	0.0	s(h)	☆
PD-31	0xAD1F	Select segment 6 acceleration/deceleration time for simple PLC	0 - 3	0	-	☆
PD-32	0xAD20	Segment 7 running time of simple PLC	0.0 - 6553.5	0.0	s(h)	☆
PD-33	0xAD21	Select segment 7 acceleration/deceleration time for simple PLC	0 - 3	0	-	☆
PD-34	0xAD22	Segment 8 running time of simple PLC	0.0 - 6553.5	0.0	s(h)	☆
PD-35	0xAD23	Select segment 8 acceleration/deceleration time for simple PLC	0 - 3	0	-	☆
PD-36	0xAD24	Segment 9 running time of simple PLC	0.0 - 6553.5	0.0	s(h)	☆
PD-37	0xAD25	Select segment 9 acceleration/deceleration time for simple PLC	0 - 3	0	-	☆
PD-38	0xAD26	Segment 10 running time of simple PLC	0.0 - 6553.5	0.0	s(h)	☆
PD-39	0xAD27	Select segment 10 acceleration/deceleration time for simple PLC	0 - 3	0	-	☆
PD-40	0xAD28	Segment 11 running time of simple PLC	0.0 - 6553.5	0.0	s(h)	☆
PD-41	0xAD29	Select segment 11 acceleration/deceleration time for simple PLC	0 - 3	0	-	☆
PD-42	0xAD2A	Segment 12 running time of simple PLC	0.0 - 6553.5	0.0	s(h)	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
PD-43	0xAD2B	Select segment 12 acceleration/deceleration time for simple PLC	0 - 3	0	-	☆
PD-44	0xAD2C	Segment 13 running time of simple PLC	0.0 - 6553.5	0.0	s(h)	☆
PD-45	0xAD2D	Select segment 13 acceleration/deceleration time for simple PLC	0 - 3	0	-	☆
PD-46	0xAD2E	Segment 14 running time of simple PLC	0.0 - 6553.5	0.0	s(h)	☆
PD-47	0xAD2F	Select segment 14 acceleration/deceleration time for simple PLC	0 - 3	0	-	☆
PD-48	0xAD30	Segment 15 running time of simple PLC	0.0 - 6553.5	0.0	s(h)	☆
PD-49	0xAD31	Select segment 15 acceleration/deceleration time for simple PLC	0 - 3	0	-	☆
PD-50	0xAD32	Running time unit of simple PLC	0: s 1: h	0	-	☆
PD-51	0xAD33	Reference mode of multi-segment reference 0	0: Parameter PD-00 setting 1: AI1 2: AI2 3: AI3 4: AI4 5: AI5 6: Pulse 7: PID 8: Preset frequency reference	0	-	★

5.3.15 Group PE–Parameter Management Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
PE-00	0xAE00	Parameter initialization	0: No operation 1: Restore some parameters to factory defaults 2: Clear records 3: Save current parameter 10: Restore all parameters to factory defaults	0	-	★
PE-01	0xAE01	User password	0 - 65535	0	-	☆
PE-02	0xAE02	Operating panel display mode	0: Unavailable 1: Basic mode 2: User mode 3: Verification mode	0	-	☆
PE-03	0xAE03	RUN button working mode	0: High-speed running mode 1: Jog mode	0	-	★
PE-04	0xAE04	Parameter setting display on keypad LED	Bit 0: Frequency reference 0: Not displayed; 1: Displayed Bit 1: Running frequency 0: Not displayed; 1: Displayed Bit 2: Feedback speed 0: Not displayed; 1: Displayed Bit 3: Output current 0: Not displayed; 1: Displayed Bit 4: Output voltage 0: Not displayed; 1: Displayed Bit 5: Bus voltage 0: Not displayed; 1: Displayed Bit 6: All voltage 0: Not displayed; 1: Displayed Bit 9: Output power 0: Not displayed; 1: Displayed Bit 10: Output torque 0: Not displayed; 1: Displayed Bit 11: PID reference 0: Not displayed; 1: Displayed Bit 12: PID feedback 0: Not displayed; 1: Displayed Bit 13: PLC stage 0: Not displayed; 1: Displayed Bit 14: Input pulse frequency 0: Not displayed; 1: Displayed Bit 15: Current running time 0: Not displayed; 1: Displayed	511	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
PE-05	0xAE05	Select group U parameters displayed on LED 1	0 - 89: U0-00 - U0-89 100 - 149: U1-00 - U0-49 200 - 269: U2-00 - U2-69	0	-	☆
PE-06	0xAE06	Select group U parameters displayed on LED 2	0 - 89: U0-00 - U0-89 100 - 149: U1-00 - U0-49 200 - 269: U2-00 - U2-69	1	-	☆

5.3.16 Group B0—Process Data Address Mapping Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B0-00	0xB000	High bits of address mapping RPDO1 sub-index 1	0 - 65535	0	-	★
B0-01	0xB001	Low bits of address mapping RPDO1 sub-index 1	0 - 65535	0	-	★
B0-02	0xB002	High bits of address mapping RPDO1 sub-index 2	0 - 65535	0	-	★
B0-03	0xB003	Low bits of address mapping RPDO1 sub-index 2	0 - 65535	0	-	★
B0-04	0xB004	High bits of address mapping RPDO1 sub-index 3	0 - 65535	0	-	★
B0-05	0xB005	Low bits of address mapping RPDO1 sub-index 3	0 - 65535	0	-	★
B0-06	0xB006	High bits of address mapping RPDO1 sub-index 4	0 - 65535	0	-	★
B0-07	0xB007	Low bits of address mapping RPDO1 sub-index 4	0 - 65535	0	-	★
B0-08	0xB008	High bits of address mapping RPDO2 sub-index 1	0 - 65535	0	-	★
B0-09	0xB009	Low bits of address mapping RPDO2 sub-index 1	0 - 65535	0	-	★
B0-10	0xB00A	High bits of address mapping RPDO2 sub-index 2	0 - 65535	0	-	★
B0-11	0xB00B	Low bits of address mapping RPDO2 sub-index 2	0 - 65535	0	-	★
B0-12	0xB00C	High bits of address mapping RPDO2 sub-index 3	0 - 65535	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B0-13	0xB00D	Low bits of address mapping RPDO2 sub-index 3	0 - 65535	0	-	★
B0-14	0xB00E	High bits of address mapping RPDO2 sub-index 4	0 - 65535	0	-	★
B0-15	0xB00F	Low bits of address mapping RPDO2 sub-index 4	0 - 65535	0	-	★
B0-16	0xB010	High bits of address mapping RPDO3 sub-index 1	0 - 65535	0	-	★
B0-17	0xB011	Low bits of address mapping RPDO3 sub-index 1	0 - 65535	0	-	★
B0-18	0xB012	High bits of address mapping RPDO3 sub-index 2	0 - 65535	0	-	★
B0-19	0xB013	Low bits of address mapping RPDO3 sub-index 2	0 - 65535	0	-	★
B0-20	0xB014	High bits of address mapping RPDO3 sub-index 3	0 - 65535	0	-	★
B0-21	0xB015	Low bits of address mapping RPDO3 sub-index 3	0 - 65535	0	-	★
B0-22	0xB016	High bits of address mapping RPDO3 sub-index 4	0 - 65535	0	-	★
B0-23	0xB017	Low bits of address mapping RPDO3 sub-index 4	0 - 65535	0	-	★
B0-24	0xB018	High bits of address mapping RPDO4 sub-index 1	0 - 65535	0	-	★
B0-25	0xB019	Low bits of address mapping RPDO4 sub-index 1	0 - 65535	0	-	★
B0-26	0xB01A	High bits of address mapping RPDO4 sub-index 2	0 - 65535	0	-	★
B0-27	0xB01B	Low bits of address mapping RPDO4 sub-index 2	0 - 65535	0	-	★
B0-28	0xB01C	High bits of address mapping RPDO4 sub-index 3	0 - 65535	0	-	★
B0-29	0xB01D	Low bits of address mapping RPDO4 sub-index 3	0 - 65535	0	-	★
B0-30	0xB01E	High bits of address mapping RPDO4 sub-index 4	0 - 65535	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B0-31	0xB01F	Low bits of address mapping RPDO4 sub-index 4	0 - 65535	0	-	★
B0-32	0xB020	High bits of address mapping TPDO1 sub-index 1	0 - 65535	0	-	★
B0-33	0xB021	Low bits of address mapping TPDO1 sub-index 1	0 - 65535	0	-	★
B0-34	0xB022	High bits of address mapping TPDO1 sub-index 2	0 - 65535	0	-	★
B0-35	0xB023	Low bits of address mapping TPDO1 sub-index 2	0 - 65535	0	-	★
B0-36	0xB024	High bits of address mapping TPDO1 sub-index 3	0 - 65535	0	-	★
B0-37	0xB025	Low bits of address mapping TPDO1 sub-index 3	0 - 65535	0	-	★
B0-38	0xB026	High bits of address mapping TPDO1 sub-index 4	0 - 65535	0	-	★
B0-39	0xB027	Low bits of address mapping TPDO1 sub-index 4	0 - 65535	0	-	★
B0-40	0xB028	High bits of address mapping TPDO2 sub-index 1	0 - 65535	0	-	★
B0-41	0xB029	Low bits of address mapping TPDO2 sub-index 1	0 - 65535	0	-	★
B0-42	0xB02A	High bits of address mapping TPDO2 sub-index 2	0 - 65535	0	-	★
B0-43	0xB02B	Low bits of address mapping TPDO2 sub-index 2	0 - 65535	0	-	★
B0-44	0xB02C	High bits of address mapping TPDO2 sub-index 3	0 - 65535	0	-	★
B0-45	0xB02D	Low bits of address mapping TPDO2 sub-index 3	0 - 65535	0	-	★
B0-46	0xB02E	High bits of address mapping TPDO2 sub-index 4	0 - 65535	0	-	★
B0-47	0xB02F	Low bits of address mapping TPDO2 sub-index 4	0 - 65535	0	-	★
B0-48	0xB030	High bits of address mapping TPDO3 sub-index 1	0 - 65535	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B0-49	0xB031	Low bits of address mapping TPDO3 sub-index 1	0 - 65535	0	-	★
B0-50	0xB032	High bits of address mapping TPDO3 sub-index 2	0 - 65535	0	-	★
B0-51	0xB033	Low bits of address mapping TPDO3 sub-index 2	0 - 65535	0	-	★
B0-52	0xB034	High bits of address mapping TPDO3 sub-index 3	0 - 65535	0	-	★
B0-53	0xB035	Low bits of address mapping TPDO3 sub-index 3	0 - 65535	0	-	★
B0-54	0xB036	High bits of address mapping TPDO3 sub-index 4	0 - 65535	0	-	★
B0-55	0xB037	Low bits of address mapping TPDO3 sub-index 4	0 - 65535	0	-	★
B0-56	0xB038	High bits of address mapping TPDO4 sub-index 1	0 - 65535	0	-	★
B0-57	0xB039	Low bits of address mapping TPDO4 sub-index 1	0 - 65535	0	-	★
B0-58	0xB03A	High bits of address mapping TPDO4 sub-index 2	0 - 65535	0	-	★
B0-59	0xB03B	Low bits of address mapping TPDO4 sub-index 2	0 - 65535	0	-	★
B0-60	0xB03C	High bits of address mapping TPDO4 sub-index 3	0 - 65535	0	-	★
B0-61	0xB03D	Low bits of address mapping TPDO4 sub-index 3	0 - 65535	0	-	★
B0-62	0xB03E	High bits of address mapping TPDO4 sub-index 4	0 - 65535	0	-	★
B0-63	0xB03F	Low bits of address mapping TPDO4 sub-index 4	0 - 65535	0	-	★
B0-66	0xB042	Number of active RPDOs	0 - 65535	0	-	●
B0-67	0xB043	Number of active TPDOs	0 - 65535	0	-	●

5.3.17 Group B1—Tension Control Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B1-00	0xB100	Tension control mode	0: Disabled 1: Open-loop tension torque control 2: Closed-loop tension speed control 3: Closed-loop tension torque control 4: Constant linear velocity control	0	-	☆
B1-01	0xB101	Curl mode	0: Winding 1: Unwinding	0	-	☆
B1-02	0xB102	Allow reverse winding	0: No 0.1 to 20.0: Linear velocity for reverse winding	0.0	m/min	☆
B1-03	0xB103	Mechanical transmission ratio	0.01 - 300.00	1.00	-	☆
B1-04	0xB104	Linear velocity source	0: No input 1: AI1 2: AI2 3: AI3 4: AI4 5: AI5 6: Pulse input 7: Communication (7707H)	0	-	★
B1-05	0xB105	Maximum linear velocity	0.1 - 6500.0	1000.0	m/min	☆
B1-06	0xB106	Calculate minimum linear velocity for winding and unwinding	0.1 - 6500.0	20.0	m/min	☆
B1-07	0xB107	Roll diameter-based calculation method	0: Calculation by linear velocity 1: Cumulative calculation by thickness 2: AI1 input 3: AI2 input 4: AI3 input 5: AI4 input 6: AI5 input 7: Pulse input (HDI) 8: Communication input (0x7710)	0	-	★
B1-08	0xB108	Maximum roll diameter	0.1 - 6000.0	500.0	mm	☆
B1-09	0xB109	Reel diameter	0.1 - 6000.0	100.0	mm	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B1-10	0xB10A	Initial winding and unwinding source	0: Digital setting/Specified by DI 1: AI1 2: AI2 3: AI3 4: AI4 5: AI5	0	-	★
B1-11	0xB10B	Initial winding and unwinding 1	0.1 - 6000.0	100.0	mm	☆
B1-12	0xB10C	Initial winding and unwinding 2	0.1 - 6000.0	100.0	mm	☆
B1-13	0xB10D	Initial winding and unwinding 3	0.1 - 6000.0	100.0	mm	☆
B1-14	0xB10E	Current winding and unwinding	0.1 - 6000.0	100.0	mm	☆
B1-15	0xB10F	Winding and unwinding filter time	0.00 - 10.00	5.00	s	☆
B1-16	0xB110	Roll diameter change rate	0: Disabled 0.1 - 10.0	1.0	mm	☆
B1-17	0xB111	Direction limit of roll diameter change	0: Disabled 1: Deny decrease for winding and increase for unwinding	0	-	☆
B1-18	0xB112	Allow roll diameter reset during running	0: Disabled 1: Enabled	0	-	☆
B1-19	0xB113	Pre-drive speed gain	-100.0 - 100.0	0.0	%	☆
B1-20	0xB114	Pre-drive torque limiting source	0: Digital setting 1: By tension	1	-	★
B1-21	0xB115	Pre-drive torque limiting	-100.0 - 100.0	0.0	%	☆
B1-22	0xB116	Pre-drive roll diameter calculation delay	0.1 - 6500.0	10.0	s	☆
B1-23	0xB117	Pre-drive acceleration time	0.1 - 6500.0	0.0	s	☆
B1-24	0xB118	Pre-drive deceleration time	0.1 - 6500.0	0.0	s	☆
B1-25	0xB119	Allow pre-drive roll diameter calculation	0: Disabled 1: Enabled	0	-	☆
B1-26	0xB11A	Winding frequency limit	0.0 - 100.0	50.0	%	☆
B1-27	0xB11B	Winding frequency limit offset	B1-00=2: 0.00Hz - 100.00Hz B1-00≠2: 0.00% - 100.0%	5.00	Hz/%	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B1-28	0xB11C	Winding frequency limit	B1-00 = 2: Select PID adjustment limit for speed closed-loop B1-00 ≠ 2: Enable speed limiting	0	-	☆
B1-29	0xB11D	Pulses per circle	1 - 60000	1	-	☆
B1-30	0xB11E	Circles per layer	1 - 10000	100	-	☆
B1-31	0xB11F	Material thickness setting source	0: Digital setting/Specified by DI 1: AI1 2: AI2 3: AI3 4: AI4 5: AI5	0	-	☆
B1-32	0xB120	Material thickness 0	0.01 - 100.00	0.01	mm	☆
B1-33	0xB121	Material thickness 1	0.01 - 100.00	0.01	mm	☆
B1-34	0xB122	Material thickness 2	0.01 - 100.00	0.01	mm	☆
B1-35	0xB123	Material thickness 3	0.01 - 100.00	0.01	mm	☆
B1-36	0xB124	Maximum thickness	0.01 - 100.00	1.00	mm	☆
B1-37	0xB125	Mode free of roll diameter reset	0 - 65535	0	-	☆
B1-38	0xB126	Pulse source for thickness-based roll diameter calculation	0: HDI1 1: Reserved	0	-	☆

5.3.18 Group B2–Tension Setting Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B2-00	0xB200	Tension setting source	0: B2-01 1: AI1 2: AI2 3: AI3 4: AI4 5: AI5 6: Pulse 7: Communication (0x7706)	0	-	★
B2-01	0xB201	Tension digital setting	0.0 - 6500.0	50.0	N	☆
B2-02	0xB202	Maximum tension reference	0.0 - 6500.0	200.0	N	☆
B2-03	0xB203	Zero-speed threshold	0.0 - 20.0 (Maximum frequency)	0.0	%	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B2-04	0xB204	Zero-speed tension boost	0.0 - 1000.0	0.0	%	☆
B2-05	0xB205	Torque frequency acceleration time	0.0 - 6500.0	0.0	s	☆
B2-06	0xB206	Torque frequency deceleration time	0.0 - 6500.0	0.0	s	☆
B2-07	0xB207	Friction compensation coefficient	0.0 - 50.0	0.0	%	☆
B2-08	0xB208	Mechanical inertia compensation	0 - 65535	0	N*m ²	☆
B2-09	0xB209	Acceleration inertia compensation correction	0.0 - 200.0	100.0	%	☆
B2-10	0xB20A	Deceleration inertia compensation correction	0.0 - 200.0	100.0	%	☆
B2-11	0xB20B	Material density	0 - 60000	0	Kg/M ³	☆
B2-12	0xB20C	Material width	0 - 60000	0	mm	☆
B2-13	0xB20D	Inertia compensation exit delay	0 - 1000	0	ms	☆
B2-15	0xB20F	Start and end linear velocity for torque open-loop anti-shake	0.1 - 6500.0	0.0	m/min	☆
B2-16	0xB210	Torque closed-loop torque limiting	0.0 - 100.0	50.0	-	☆
B2-17	0xB211	Friction correction coefficient	-50.0 - 50.0	0.0	-	☆
B2-18	0xB212	Friction compensation curve	0: Frequency 1: Linear velocity 2: Multi-segment friction compensation curve 1 3: Multi-segment friction compensation curve 2 4: Roll diameter	0	-	★
B2-19	0xB213	Multi-friction compensation torque 1	0.0 - 50.0	0.0	%	☆
B2-20	0xB214	Multi-friction compensation torque 2	0.0 - 50.0	0.0	%	☆
B2-21	0xB215	Multi-friction compensation torque 3	0.0 - 50.0	0	%	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B2-22	0xB216	Multi-friction compensation torque 4	0.0 - 50.0	0.0	%	☆
B2-23	0xB217	Multi-friction compensation torque 5	0.0 - 50.0	0.0	%	☆
B2-24	0xB218	Multi-friction compensation torque 6	0.0 - 50.0	0.0	%	☆
B2-25	0xB219	Multi-segment friction compensation inflection point 1	0.00 Hz to maximum frequency	0.00	Hz	☆
B2-26	0xB21A	Multi-segment friction compensation inflection point 2	0.00 Hz to maximum frequency	0.00	Hz	☆
B2-27	0xB21B	Multi-segment friction compensation inflection point 3	0.00 Hz to maximum frequency	0.00	Hz	☆
B2-28	0xB21C	Multi-segment friction compensation inflection point 4	0.00 Hz to maximum frequency	0.00	Hz	☆
B2-29	0xB21D	Multi-segment friction compensation inflection point 5	0.00 Hz to maximum frequency	0.00	Hz	☆
B2-30	0xB21E	Multi-segment friction compensation inflection point 6	0.00 Hz to maximum frequency	0.00	Hz	☆

5.3.19 Group B3—Tension Taper Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B3-00	0xB300	Taper mode	0: Curve 1: Straight line	0	-	★
B3-01	0xB301	Select tension taper source	0: B3-02 1: AI1 2: AI2 3: AI3 4: AI4 5: AI5	0	-	★
B3-02	0xB302	Tension taper	0.0 - 100.0	0.0	%	☆
B3-03	0xB303	Taper compensation correction	0mm - 10000mm	0	mm	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B3-04	0xB304	Set closed-loop tension taper to valid	0: Disabled 1: Enabled	0	-	★
B3-05	0xB305	Taper output source	0: B3-06 1: AI1 2: AI2 3: AI3 4: AI4 5: AI5	0	-	★
B3-06	0xB306	External taper digital setting	0 - 1000	0	-	☆
B3-07	0xB307	Reserved	0 - 5	5	-	☆
B3-08	0xB308	Taper value of minimum roll	0.1 - 100.0	100.0	%	☆
B3-09	0xB309	Linear taper switching point 1	B1-09 - B1-08	150.0	mm	☆
B3-10	0xB30A	Taper value of switching point 1	0.1 - 100.0	100.0	%	☆
B3-11	0xB30B	Linear taper switching point 2	B3-09 - B1-08	200.0	mm	☆
B3-12	0xB30C	Taper value of switching point 2	0.1 - 100.0	90.0	%	☆
B3-13	0xB30D	Linear taper switching point 3	B3-11 - B1-08	250.0	mm	☆
B3-14	0xB30E	Taper value of switching point 3	0.1 - 100.0	80.0	%	☆
B3-15	0xB30F	Linear taper switching point 4	B3-13 - B1-08	300.0	mm	☆
B3-16	0xB310	Taper value of switching point 4	0.1 - 100.0	70.0	%	☆
B3-17	0xB311	Linear taper switching point 5	B3-15 - B1-08	400.0	mm	☆
B3-18	0xB312	Taper value of switching point 5	0.1 - 100.0	50.0	%	☆
B3-19	0xB313	Taper value of maximum roll diameter	0.1 - 100.0	30.0	%	☆
B3-20	0xB314	Reserved	0 - 65535	0	-	☆

5.3.20 Group B5—Load Synchronization Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B5-00	0xB500	Select master or slave for load distribution	0: Disabled 1: Master 2: Slave	0	-	★
B5-01	0xB501	Synchronization mode	0 - 65535	0	-	★
B5-02	0xB502	Synchronize master station number (valid for slave)	1 - 124	1	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B5-03	0xB503	Select master control	0 - 65535	0	-	★
B5-04	0xB504	Select slave control (load distribution)	0: Do not follow master start/stop command 1: Follow master start/stop command	1	-	★
B5-05	0xB505	Torque acceleration time	0.000 - 60.000	0.000	s	☆
B5-06	0xB506	Torque deceleration time	0.000 - 60.000	0.000	s	☆
B5-07	0xB507	Frequency gain	-10.00 - 10.00	1.00	-	★
B5-08	0xB508	Frequency offset	-100.00 - 100.00	0.00	-	★
B5-09	0xB509	Torque gain	-10.00 - 10.00	1.00	-	★
B5-10	0xB50A	Torque offset	-100.00 - 100.00	0.00	-	★
B5-11	0xB50B	Select transmission frequency source for master (valid for master)	0: Feedback frequency 1: Running frequency 2: Operating frequency if lower than value specified by B5-06, feedback frequency otherwise	0	-	☆
B5-12	0xB50C	Transmission frequency switching point of master	0.00 - 600.00	5.00	Hz	☆

5.3.21 Group B6—Droop Control Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B6-00	0xB600	Select master and slave for droop control	0: Disabled 1: Master 2: Slave 3: Self-adaptive droop	0	-	★
B6-01	0xB601	Select droop control mode	2: Master and slave droop	2	-	★
B6-02	0xB602	Synchronize master station number (valid for slave)	1 - 124	1	-	★
B6-03	0xB603	Select master control	0 - 65535	0	-	★
B6-04	0xB604	Slave configuration parameters (droop control)	0: Do not follow master start/stop command 1: Follow master start/stop command	0	-	★
B6-05	0xB605	Acceleration time	0 - 65535	0	-	★
B6-06	0xB606	Deceleration time	0 - 65535	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B6-07	0xB607	Droop rate	0.00 - 15.00	5.00	%	☆
B6-08	0xB608	Frequency gain	0 - 65535	0	-	★
B6-09	0xB609	Frequency offset	0 - 65535	0	-	★
B6-10	0xB60A	Torque gain	0 - 65535	0	-	★
B6-11	0xB60B	Torque offset	0 - 65535	0	-	★

5.3.22 Group B9—Virtual DIO Parameter Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B9-00	0xB900	Function of virtual terminal VDI1	See the description below	0	-	★
B9-01	0xB901	Function of virtual terminal VDI2		0	-	★
B9-02	0xB902	Function of virtual terminal VDI3		0	-	★
B9-03	0xB903	Function of virtual terminal VDI4		0	-	★
B9-04	0xB904	Function of virtual terminal VDI5		0	-	★
0: No function assigned			25: Counter input (DIO1)			
1: Forward running (FWD)			26: Counter reset			
2: Reverse running (REV)			27: Length counter input (DIO1)			
3: Three-wire running control			28: Length counter reset			
4: Positive jog			29: Torque control disabling			
5: Reverse jog			30: Pulse input			
6: Coast to stop			31: Reserved			
7: Quick stop			32: DC braking			
8: Coast to stop (block PWM output)			33: External fault input (NC)			
9: Fault reset			34: Frequency reference activation			
10: Pause			35: Reverse direction of PID action			
11: External fault input (NO)			36: External parking terminal, valid only for panel control			
12: Multi-segment reference terminal 1			37: Command source switching terminal 2			
13: Multi-segment reference terminal 2			38: PID integral pause terminal			
14: Multi-segment reference terminal 3			39: Switching between main frequency source X and preset frequency			
15: Multi-segment reference terminal 4			40: Switching between auxiliary frequency source Y and preset frequency			
16: Terminal 1 for acceleration/deceleration selection			41: Motor selection			
17: Terminal 2 for acceleration/deceleration selection			42: Position lock			
18: Frequency source switching			43: PID parameter switching			
19: Up/Down setting reset			44: User-defined fault 1			
20: Run command switching terminal			45: User-defined fault 2			
21: Acceleration/Deceleration disabling			46: Switching between speed and torque control			
22: PID pause			47: Emergency stop			
23: PLC status reset						
24: Vibration frequency pause						

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
48: External terminal stop (deceleration time set to 4, which is effective at any time)			70: Home search			
49: Deceleration DC braking			71: Pause position control			
50: Reset current running time			72: Reserved			
51: Two-wire/three-wire switching			73: Reserved			
52: Terminal Up			74: Reserved			
53: Terminal Down			75: Reserved			
54: Reset roll diameter			76: Multi-segment position 1			
55: Initial terminal 1			77: Multi-segment position 2			
56: Initial terminal 2			78: Multi-segment position 3			
57: Pre-drive			79: Multi-segment position 4			
58: Switching between winding and unwinding modes			80: Multi-segment position 5			
59: Stop roll diameter calculation			81: Positive limit			
60: Exit tension mode			82: Reverse limit			
61: Disable reverse running			83: Switch to relative position control in position control mode			
62: Reserved			84: Switch to absolute position control in position control mode			
63: Reserved			85: Switch to segmented position control in position control mode			
64: Run in position control mode			86: Switch to fixed-length segmented position control in position control mode			
65: Reserved			87: Re-trigger position control (incremental or fixed-length segmented)			
B9-05	0xB905	VDI terminal status setting mode	Ones digit: VDI1 0: Parameter setting (B9-06) 1: DO status 2: DI status Tens digit: VDI2 0: Parameter setting (B9-06) 1: DO status 2: DI status Hundreds digit: VDI3 0: Parameter setting (B9-06) 1: DO status 2: DI status Thousands digit: VDI4 0: Parameter setting (B9-06) 1: DO status 2: DI status Ten-thousands digit: VDI5 0: Parameter setting (B9-06) 1: DO status 2: DI status	0	-	★

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B9-06	0xB906	VDI terminal status	Ones digit: VDI1 0: Disabled 1: Enabled Tens digit: VDI2 0: Disabled 1: Enabled Hundreds digit: VDI3 0: Disabled 1: Enabled Thousands digit: VDI4 0: Disabled 1: Enabled Ten-thousands digit: VDI5 0: Disabled 1: Enabled	0	-	☆
B9-11	0xB90B	Function of VDO1	See the description below	0	-	★
B9-12	0xB90C	Function of VDO2		0	-	★
B9-13	0xB90D	Function of VDO3		0	-	★
B9-14	0xB90E	Function of VDO4		0	-	★
B9-15	0xB90F	Function of VDO5		0	-	★
B9-16	0xB910	Function of VDO6		0	-	★
B9-17	0xB911	Function of VDO7		0	-	★
B9-18	0xB912	Function of VDO8		0	-	★
B9-19	0xB913	Function of VDO9		0	-	★
B9-20	0xB914	Function of VDO10		0	-	★
0: No function assigned 1: Drive running 2: Drive failure (coast to stop failure) 3: FDT reached in frequency level detection 4: Frequency reference reached 5: Running at zero speed 6: Motor overload pre-warning 7: Drive overload pre-warning 8: Pulse reference reached 9: Specified pulse count reached 10: Length reached 11: PLC cycle completed 12: Running time reached 13: Frequency limiting 14: Torque limiting 15: Ready to run			16: AI1>AI2 17: Upper-limit frequency reached 18: Lower-limit frequency reached (no output in stop state) 19: Undervoltage state output 20: Communication control 21: Positioning completed 22: Proximity to position 23: Running at zero speed (active in stop state) 24: Power-on time reached 25: FDT1 reached in frequency level detection 26: Frequency reached 1 27: Frequency reached 2 28: Current reached 1 29: Current reached 2 30: Scheduled time reached 31: AI1 input out of upper and lower limits			

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
32: Load dropping 33: Speed direction 34: Zero-crossing detection completed 35: Module temperature reached 36: Software overcurrent 37: Lower-limit frequency reached (output available in both running and stop states) 38: Error occurred (direct output upon a fault or warning) 39: Motor over-temperature pre-warning 40: Running time reached			41: Fault output 2 (no output for undervoltage in the case of a coast to stop failure) 42: Fault output 3 (immediate output for coast to stop or decelerate to stop) 43: Position locked 44: Positioning completed 45: Proximity to position 46: Home position active 47: Home search end flag 48: RUN command state			
B9-21	0xB915	Active state of terminals VDO1 to VDO5	Ones digit: VDO1 0: Positive logic 1: Negative logic Tens digit: VDO2 0: Positive logic 1: Negative logic Hundreds digit: VDO3 0: Positive logic 1: Negative logic Thousands digit: VDO4 0: Positive logic 1: Negative logic Ten-thousands digit: VDO5 0: Positive logic 1: Negative logic	0	-	★
B9-22	0xB916	Active state of terminals VDO6 to VDO10	Ones digit: VDO6 0: Positive logic 1: Negative logic Tens digit: VDO7 0: Positive logic 1: Negative logic Hundreds digit: VDO8 0: Positive logic 1: Negative logic Thousands digit: VDO9 0: Positive logic 1: Negative logic Ten-thousands digit: VDO10 0: Positive logic 1: Negative logic	0	-	★
B9-23	0xB917	VDO status	0 - 65535	0	-	●
B9-30	0xB91E	PLC mapping address 1	0 - 65535	0	-	☆
B9-31	0xB91F	PLC mapping address 2	0 - 65535	0	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
B9-32	0xB920	PLC mapping address 3	0 - 65535	0	-	☆
B9-33	0xB921	PLC mapping address 4	0 - 65535	0	-	☆
B9-34	0xB922	PLC mapping address 5	0 - 65535	0	-	☆
B9-35	0xB923	PLC mapping address 6	0 - 65535	0	-	☆
B9-36	0xB924	PLC mapping address 7	0 - 65535	0	-	☆
B9-37	0xB925	PLC mapping address 8	0 - 65535	0	-	☆
B9-38	0xB926	PLC mapping address 9	0 - 65535	0	-	☆
B9-39	0xB927	PLC mapping address 10	0 - 65535	0	-	☆
B9-40	0xB928	PLC mapping address 11	0 - 65535	0	-	☆
B9-41	0xB929	PLC mapping address 12	0 - 65535	0	-	☆
B9-42	0xB92A	PLC mapping address 13	0 - 65535	0	-	☆
B9-43	0xB92B	PLC mapping address 14	0 - 65535	0	-	☆
B9-44	0xB92C	PLC mapping address 15	0 - 65535	0	-	☆
B9-45	0xB92D	PLC mapping address 16	0 - 65535	0	-	☆
B9-46	0xB92E	PLC mapping address 17	0 - 65535	0	-	☆
B9-47	0xB92F	PLC mapping address 18	0 - 65535	0	-	☆
B9-48	0xB930	PLC mapping address 19	0 - 65535	0	-	☆
B9-49	0xB931	PLC mapping address 20	0 - 65535	0	-	☆

5.3.23 Group BA—User-defined Parameter Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
BA-00	0xBA00	User-defined parameter 0	0 - 65535	0	-	☆
BA-01	0xBA01	User-defined parameter 1	0 - 65535	0	-	☆
BA-02	0xBA02	User-defined parameter 2	0 - 65535	0	-	☆
BA-03	0xBA03	User-defined parameter 3	0 - 65535	0	-	☆
BA-04	0xBA04	User-defined parameter 4	0 - 65535	0	-	☆
BA-05	0xBA05	User-defined parameter 5	0 - 65535	0	-	☆
BA-06	0xBA06	User-defined parameter 6	0 - 65535	0	-	☆
BA-07	0xBA07	User-defined parameter 7	0 - 65535	0	-	☆
BA-08	0xBA08	User-defined parameter 8	0 - 65535	0	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
BA-09	0xBA09	User-defined parameter 9	0 - 65535	0	-	☆
BA-10	0xBA0A	User-defined parameter 10	0 - 65535	0	-	☆
BA-11	0xBA0B	User-defined parameter 11	0 - 65535	0	-	☆
BA-12	0xBA0C	User-defined parameter 12	0 - 65535	0	-	☆
BA-13	0xBA0D	User-defined parameter 13	0 - 65535	0	-	☆
BA-14	0xBA0E	User-defined parameter 14	0 - 65535	0	-	☆
BA-15	0xBA0F	User-defined parameter 15	0 - 65535	0	-	☆
BA-16	0xBA10	User-defined parameter 16	0 - 65535	0	-	☆
BA-17	0xBA11	User-defined parameter 17	0 - 65535	0	-	☆
BA-18	0xBA12	User-defined parameter 18	0 - 65535	0	-	☆
BA-19	0xBA13	User-defined parameter 19	0 - 65535	0	-	☆
BA-20	0xBA14	User-defined parameter 20	0 - 65535	0	-	☆
BA-21	0xBA15	User-defined parameter 21	0 - 65535	0	-	☆
BA-22	0xBA16	User-defined parameter 22	0 - 65535	0	-	☆
BA-23	0xBA17	User-defined parameter 23	0 - 65535	0	-	☆
BA-24	0xBA18	User-defined parameter 24	0 - 65535	0	-	☆
BA-25	0xBA19	User-defined parameter 25	0 - 65535	0	-	☆
BA-26	0xBA1A	User-defined parameter 26	0 - 65535	0	-	☆
BA-27	0xBA1B	User-defined parameter 27	0 - 65535	0	-	☆
BA-28	0xBA1C	User-defined parameter 28	0 - 65535	0	-	☆
BA-29	0xBA1D	User-defined parameter 29	0 - 65535	0	-	☆
BA-30	0xBA1E	User-defined parameter 30	0 - 65535	0	-	☆
BA-31	0xBA1F	User-defined parameter 31	0 - 65535	0	-	☆
BA-32	0xBA20	User-defined parameter 32	0 - 65535	0	-	☆
BA-33	0xBA21	User-defined parameter 33	0 - 65535	0	-	☆
BA-34	0xBA22	User-defined parameter 34	0 - 65535	0	-	☆
BA-35	0xBA23	User-defined parameter 35	0 - 65535	0	-	☆
BA-36	0xBA24	User-defined parameter 36	0 - 65535	0	-	☆
BA-37	0xBA25	User-defined parameter 37	0 - 65535	0	-	☆
BA-38	0xBA26	User-defined parameter 38	0 - 65535	0	-	☆
BA-39	0xBA27	User-defined parameter 39	0 - 65535	0	-	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
BA-40	0xBA28	User-defined parameter 40	0 - 65535	0	-	☆
BA-41	0xBA29	User-defined parameter 41	0 - 65535	0	-	☆
BA-42	0xBA2A	User-defined parameter 42	0 - 65535	0	-	☆
BA-43	0xBA2B	User-defined parameter 43	0 - 65535	0	-	☆
BA-44	0xBA2C	User-defined parameter 44	0 - 65535	0	-	☆
BA-45	0xBA2D	User-defined parameter 45	0 - 65535	0	-	☆
BA-46	0xBA2E	User-defined parameter 46	0 - 65535	0	-	☆
BA-47	0xBA2F	User-defined parameter 47	0 - 65535	0	-	☆
BA-48	0xBA30	User-defined parameter 48	0 - 65535	0	-	☆
BA-49	0xBA31	User-defined parameter 49	0 - 65535	0	-	☆

5.3.24 Group C1—Drive Parameter Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
C1-00	0xC100	Main version of software	0.00 - 655.35	0.00	-	●
C1-01	0xC101	Debugging version of software	0.00 - 655.35	0	-	●
C1-02	0xC102	Non-standard version of software	0 - 65535	0	-	●
C1-03	0xC103	Software firmware version	0 - 65535	0	-	●
C1-04	0xC104	Current sampling delay	0.00 - 100.00	1.00	us	★
C1-05	0xC105	Enable current sampling gain deviation auto-tuning	0: Disabled 1: Enabled	1	-	★
C1-06	0xC106	Inverter non-linear auto-tuning command word	Bit 0: Enable inverter non-linear auto-tuning 0: Disabled 1: Enabled Bit 1: Enable high-precision Rs auto-tuning 0: Disabled 1: Enabled	1	-	★
C1-07	0xC107	High enabling frequency of inverter non-linear auto-tuning	0.0 - 16.0	6.0	kHz	★

Parameter	Communica- tion Address	Name	Setting Range	Default	Unit	Operation
C1-08	0xC108	Low enabling frequency of inverter non-linear auto- tuning	0.0 - 16.0	1.0	kHz	★
C1-09	0xC109	Equivalent delay 1	0.0 - 200.0	51.7	%	☆
C1-10	0xC10A	Equivalent delay 2	0.0 - 200.0	71.9	%	☆
C1-11	0xC10B	Equivalent delay 3	0.0 - 200.0	78.3	%	☆
C1-12	0xC10C	Equivalent delay 4	0.0 - 200.0	79.9	%	☆
C1-13	0xC10D	Equivalent delay 5	0.0 - 200.0	80.2	%	☆
C1-14	0xC10E	Equivalent delay 6	0.0 - 200.0	80.2	%	☆
C1-15	0xC10F	Equivalent delay 7	0.0 - 200.0	80.2	%	☆
C1-16	0xC110	Equivalent delay 8	0.0 - 200.0	80.2	%	☆
C1-17	0xC111	Equivalent delay 9	0.0 - 200.0	80.3	%	☆
C1-18	0xC112	Equivalent delay 10	0.0 - 200.0	81.0	%	☆
C1-19	0xC113	Equivalent delay 11	0.0 - 200.0	82.2	%	☆
C1-20	0xC114	Equivalent delay 12	0.0 - 200.0	83.0	%	☆
C1-21	0xC115	Equivalent delay 13	0.0 - 200.0	83.7	%	☆
C1-22	0xC116	Equivalent delay 14	0.0 - 200.0	84.2	%	☆
C1-23	0xC117	Equivalent delay 15	0.0 - 200.0	84.7	%	☆
C1-24	0xC118	Equivalent delay 16	0.0 - 200.0	85.2	%	☆
C1-25	0xC119	Equivalent delay 17	0.0 - 200.0	85.8	%	☆
C1-26	0xC11A	Equivalent delay 18	0.0 - 200.0	86.2	%	☆
C1-27	0xC11B	Equivalent delay 19	0.0 - 200.0	86.5	%	☆
C1-28	0xC11C	Equivalent delay 20	0.0 - 200.0	86.7	%	☆
C1-29	0xC11D	Equivalent voltage drop 1	0.000 - 2.000	0.498	V	☆
C1-30	0xC11E	Equivalent voltage drop 2	0.000 - 2.000	0.498	V	☆
C1-31	0xC11F	Equivalent voltage drop 3	0.000 - 2.000	0.555	V	☆
C1-32	0xC120	Equivalent voltage drop 4	0.000 - 2.000	0.623	V	☆
C1-33	0xC121	Equivalent voltage drop 5	0.000 - 2.000	0.679	V	☆
C1-34	0xC122	Equivalent voltage drop 6	0.000 - 2.000	0.728	V	☆
C1-35	0xC123	Equivalent voltage drop 7	0.000 - 2.000	0.767	V	☆
C1-36	0xC124	Equivalent voltage drop 8	0.000 - 2.000	0.800	V	☆
C1-37	0xC125	Equivalent voltage drop 9	0.000 - 2.000	0.824	V	☆

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
C1-38	0xC126	Equivalent voltage drop 10	0.000 - 2.000	0.837	V	☆
C1-39	0xC127	Equivalent voltage drop 11	0.000 - 2.000	0.837	V	☆
C1-40	0xC128	Equivalent voltage drop 12	0.000 - 2.000	0.841	V	☆
C1-41	0xC129	Equivalent voltage drop 13	0.000 - 2.000	0.846	V	☆
C1-42	0xC12A	Equivalent voltage drop 14	0.000 - 2.000	0.855	V	☆
C1-43	0xC12B	Equivalent voltage drop 15	0.000 - 2.000	0.857	V	☆
C1-44	0xC12C	Equivalent voltage drop 16	0.000 - 2.000	0.860	V	☆
C1-45	0xC12D	Equivalent voltage drop 17	0.000 - 2.000	0.860	V	☆
C1-46	0xC12E	Equivalent voltage drop 18	0.000 - 2.000	0.864	V	☆
C1-47	0xC12F	Equivalent voltage drop 19	0.000 - 2.000	0.869	V	☆
C1-48	0xC130	Equivalent voltage drop 20	0.000 - 2.000	0.871	V	☆
C1-49	0xC131	Select G/P model	0: G model 1: P model	0		★

5.3.25 Group U0—Observation Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
U0-00	0x7000	Running frequency	100 ms filter	0.00	Hz	●
U0-01	0x7001	Frequency reference	100 ms filter	0.00	Hz	●
U0-02	0x7002	Bus voltage	100 ms filter	0.0	V	●
U0-03	0x7003	Output voltage	100 ms filter	0.0	V	●
U0-04	0x7004	Output current	100 ms filter	0.00	A	●
U0-05	0x7005	Output power	100 ms filter	0.0	KW	●
U0-06	0x7006	Output torque	100 ms filter	0	%	●

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
U0-07	0x7007	DI active state 1	Bit0: DI1 Bit1: DI2 Bit2: DI3 Bit3: DI4 Bit4: DI5 Bit5: DI6 Bit6: DI7 Bit7: DI8 Bit8: DI9 Bit9: DI10 Bit10: DI11 Bit11: DI12 Bit12: DI13 Bit13: DI14 Bit14: DI15 Bit15: VDI1	0	-	●
U0-08	0x7008	DI active state 2	Bit0: DI1 Bit1: DI2 Bit2: DI3 Bit3: DI4 Bit4: DI5 Bit5: DI6 Bit6: DI7 Bit7: DI8 Bit8: DI9 Bit9: DI10 Bit10: DI11 Bit11: DI12 Bit12: DI13 Bit13: DI14 Bit14: DI15 Bit15: VDI1	0	-	●

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
U0-09	0x7009	DO state 1	Bit0: DO1 Bit1: DO2 Bit2: DO3 Bit3: DO4 Bit4: DO5 Bit5: DO6 Bit6: DO7 Bit7: DO8 Bit8: DO9 Bit9: DO10 Bit10: DO11 Bit11: DO12 Bit12: VD01 Bit13: VDO2 Bit14: VDO3 Bit15: VDO4	0	-	●
U0-10	0x700A	DO state 2	Bit0: VDO5 Bit1: VDO6 Bit2: VDO7 Bit3: VDO8	0	-	●
U0-11	0x700B	Encoder feedback frequency	-	0.00	Hz	●
U0-12	0x700C	Motor speed	100 ms filter	0	rpm	●
U0-13	0x700D	Set speed	100 ms filter	0	rpm	●
U0-14	0x700E	Count	-	0	-	●
U0-15	0x700F	Length remainder	-	0.0	-	●
U0-16	0x7010	Error code	-	0	-	●
U0-17	0x7011	AI1 voltage (before correction)	-	0.000	(0.001)V/ (0.01)mA	●
U0-18	0x7012	AI1 voltage (after correction)	-	0.000	(0.001)V/ (0.01)mA	●
U0-19	0x7013	AI2 voltage (before correction)	-	0.000	(0.001)V/ (0.01)mA	●
U0-20	0x7014	AI2 voltage (after correction)	-	0.000	(0.001)V/ (0.01)mA	●
U0-21	0x7015	AI3 voltage (before correction)	-	0.000	(0.001)V/ (0.01)mA	●
U0-22	0x7016	AI3 voltage (after correction)	-	0.000	(0.001)V/ (0.01)mA	●

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
U0-23	0x7017	AI4 voltage (before correction)	-	0.000	(0.001)V/ (0.01)mA	●
U0-24	0x7018	AI4 voltage (after correction)	-	0.000	(0.001)V/ (0.01)mA	●
U0-25	0x7019	AI5 voltage (before correction)	-	0.000	(0.001)V/ (0.01)mA	●
U0-26	0x701A	AI5 voltage (after correction)	-	0.000	(0.001)V/ (0.01)mA	●
U0-27	0x701B	PID reference	-	0	-	●
U0-28	0x701C	PID feedback	-	0	-	●
U0-29	0x701D	PID output	-	0.0	%	●
U0-30	0x701E	Drive unit temperature	-40 - 200	0	°C	●
U0-31	0x701F	Cumulative power-on time (s)	0 - 65535	0	s	●
U0-32	0x7020	Cumulative power-on time (h)	0 - 65535	0	h	●
U0-33	0x7021	Cumulative running time (s)	0 - 65535	0	s	●
U0-34	0x7022	Cumulative running time (h)	0 - 65535	0	h	●
U0-35	0x7023	Current running time (min)	0 - 65535	0.0	min	●
U0-36	0x7024	Current power-on time (min)	0 - 65535	0	min	●
U0-37	0x7025	Main frequency display	Main frequency	0.00	Hz	●
U0-38	0x7026	Auxiliary frequency display	Auxiliary frequency	0.00	Hz	●
U0-39	0x7027	Low bits in encoder pulse incremental display	0 - 65535	0	-	●
U0-40	0x7028	High bits in encoder pulse incremental display	0 - 65535	0	-	●
U0-41	0x7029	Input pulse frequency	0 - 65535	0.00	kHz	●
U0-42	0x702A	Drive status display	0: Not initialized 1: Initializing 2: Standby 3: Start 4: Running 5: Decelerate to stop 6: Auto-tuning 7: Faulty	0	-	●
U0-43	0x702B	10 ms execution cycle	0 - 65535	0.0	ms	●
U0-44	0x702C	Low bits of HDI pulse count	0 - 65535	0	-	●

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
U0-45	0x702D	High bits of HDI pulse count	0 - 65535	0	-	●
U0-46	0x702E	Rectifier unit temperature	-40 - 200	0	°C	●
U0-47	0x702F	AO1 output	0 - 10.000V/20.00mA	0.000	(0.001)V/ (0.01)mA	●
U0-48	0x7030	AO2 output	0 - 10.000V/20.00mA	0.000	(0.001)V/ (0.01)mA	●
U0-49	0x7031	AO3 output	0 - 10.000V/20.00mA	0.000	(0.001)V/ (0.01)mA	●
U0-50	0x7032	Value of temperature measurement signal 1	-40 - 200	0	°C	●
U0-51	0x7033	Value of temperature measurement signal 2	-40 - 200	0	°C	●
U0-52	0x7034	Value of temperature measurement signal 3	-40 - 200	0	°C	●
U0-53	0x7035	Value of temperature measurement signal 4	-40 - 200	0	°C	●
U0-54	0x7036	Voltage measured by PTC1	0 - 65535	0.000	V	●
U0-55	0x7037	Voltage measured by PTC2	0 - 65535	0.000	V	●
U0-56	0x7038	Voltage measured by PTC3	0 - 65535	0.000	V	●

5.3.26 Group U2—Observation Group

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
U2-00	0x7200	Current linear velocity	0.0 - 6553.5	1000.0	m/min	●
U2-01	0x7201	Current roll diameter	0.0 - 6553.5	100.0	mm	●
U2-02	0x7202	Linear velocity matching frequency	0.00 - 655.35	0.00	Hz	●
U2-03	0x7203	Current tension reference	0.0 - 6553.5	0.0	N	●
U2-04	0x7204	Tension after taper calculation	0.0 - 6553.5	0.0	N	●
U2-05	0x7205	Tension computation torque	0.0 - 6553.5	0.0	%	●
U2-06	0x7206	Acceleration/Deceleration compensation torque	0.0 - 6553.5	0.0	%	●
U2-07	0x7207	Dynamic friction compensation torque	0.0 - 6553.5	0.0	%	●

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
U2-08	0x7208	Static friction compensation torque	0.0 - 6553.5	0.0	%	●
U2-09	0x7209	Open-loop torque output	0.0 - 6553.5	0.0	%	●
U2-10	0x720A	Torque PID reference	0.0 - 6553.5	0.0	%	●
U2-11	0x720B	Torque PID feedback	0.0 - 6553.5	0.0	%	●
U2-12	0x720C	Torque PID output	0.0 - 6553.5	0.0	%	●
U2-13	0x720D	Open-loop torque and PID output	0.0 - 6553.5	0.0	%	●
U2-14	0x720E	Frequency limiting in tension torque mode	0.00 - 655.35	0.00	Hz	●
U2-15	0x720F	Frequency PID reference	0.0 - 6553.5	0.0	%	●
U2-16	0x7210	Frequency PID feedback	0.0 - 6553.5	0.0	%	●
U2-17	0x7211	Frequency PID output	0.00 - 655.35	0.00	Hz	●
U2-18	0x7212	Roll diameter frequency and PID output	0.00 - 655.35	0.00	Hz	●
U2-19	0x7213	Tension control command word	0 - 65535	0	-	●
U2-20	0x7214	Tension control status word	0 - 65535	0	-	●
U2-21	0x7215	Linear acceleration	0.0 - 6553.5	0.0	m/min/s	●
U2-22	0x7216	Tension taper output	0.0 - 6553.5	0.0	%	●
U2-31	0x721F	Communication control command word	0 - 65535	0	-	●
U2-32	0x7220	Main frequency set via communication	-300.00 - 300.00	0.00	Hz	●
U2-33	0x7221	Auxiliary frequency set via communication	-300.00 - 300.00	0.00	Hz	●
U2-34	0x7222	Torque set via communication	-100.00 - 100.00	0.00	%	●
U2-35	0x7223	PID reference set via communication	-100.00 - 100.00	0.00	%	●
U2-36	0x7224	PID feedback set via communication	-100.00 - 100.00	0.00	%	●
U2-37	0x7225	Tension set via communication	-100.00 - 100.00	0.00	%	●
U2-38	0x7226	Linear velocity set via communication	-100.00 - 100.00	0.00	%	●

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
U2-39	0x7227	DO communication control	0 - 255	0	-	●
U2-40	0x7228	Communication setting of upper-limit frequency	-100.00 - 100.00	0.00	%	●
U2-41	0x7229	Communication setting of electric torque upper limit	-100.00 - 100.00	0.00	%	●
U2-42	0x722A	Communication setting of power generation torque upper limit	-100.00 - 100.00	0.00	%	●
U2-43	0x722B	Communication setting of V/f separation voltage	0.00 - 100.00	0.00	%	●
U2-44	0x722C	Communication setting of AO output	0.00 - 100.00	0.00	%	●
U2-45	0x722D	Communication setting of HDO output	0.00 - 100.00	0.00	%	●
U2-46	0x722E	Communication setting of offset frequency	0.00 - 100.00	0.00	%	●
U2-47	0x722F	Communication setting of roll diameter	0.00 - 100.00	0.00	%	●
U2-61	0x723D	Status of expansion card in slot 1	0: Offline 1: Online	0	-	●
U2-62	0x723E	Type of communication expansion card in slot 1	0: Not identified 1: I/O expansion card 2: PROFINET communication card 3: EtherCAT communication card 4: CANopen communication 5: Air compressor process card 6: Resolver PG card 7: Incremental PG card 8: Absolute PG card 9: Relay I/O card 10: Communication card of injection molding machine	0	-	●
U2-63	0x723F	Software version of communication expansion card in slot 1	0.00 - 655.35	0.00	-	●
U2-64	0x7240	Status of expansion card in slot 2	0: Offline 1: Online	0	-	●

Parameter	Communication Address	Name	Setting Range	Default	Unit	Operation
U2-65	0x7241	Type of communication expansion card in slot 2	0: Not identified 1: I/O expansion card 2: PROFINET communication card 3: EtherCAT communication card 4: CANopen communication 5: Air compressor process card 6: Resolver PG card 7: Incremental PG card 8: Absolute PG card 9: Relay I/O card 10: Communication card of injection molding machine	0	-	●
U2-66	0x7242	Software version of communication expansion card in slot 2	0.00 - 655.35	0.00	-	●
U2-67	0x7243	Status of expansion card in slot 3	0: Offline 1: Online	0	-	●
U2-68	0x7244	Type of communication expansion card in slot 3	0: Not identified 1: I/O expansion card 2: PROFINET communication card 3: EtherCAT communication card 4: CANopen communication 5: Air compressor process card 6: Resolver PG card 7: Incremental PG card 8: Absolute PG card 9: Relay I/O card 10: Communication card of injection molding machine	0	-	●
U2-69	0x7245	Software version of communication expansion card in slot 3	0.00 - 655.35	0.00	-	●

6.1 Fault Alarms

The product provides multiple protection features and can terminate output upon exceptions. When a fault occurs, the red indicator turns on, and the LED panel displays an error code.



Figure 6-1 Faulty state indication

6.2 Troubleshooting and Reset

6.2.1 Checklist

- Are the parameters set correctly?
- Is an error code displayed?
- Has the supply voltage changed? Is there a power input? Is the power cable disconnected?
- Are motor cables connected properly? Is any cable disconnected?
- Are control signal cables connected properly? Is any cable disconnected?
- Does the motor generate noise?

6.2.2 Reset

- When a fault occurs:
Check the operating panel for the occurrence time and types of the last three faults, and data upon occurrence of the faults, including the frequency, current, bus voltage, DI/DO state, power-on time, and running duration.
- Before reset:
Find the cause of the fault based on the fault type displayed on the operating panel, and rectify the fault.
- Reset methods:
 - ① Assign the fault reset function to a DI terminal by setting any of P4-01 through P4-15 to 9, and enable this function on the terminal.
 - ② Press the Reset button.
 - ③ Power cycle the drive for the drive to reset automatically.
Cut off the power supply to the main circuit. When the error code on the operating panel disappears, resume the power supply.
 - ④ Reset the drive on the host controller (communication control).
When P0-01 is set to 2, you can set the 7700H communication address to 7 to reset the drive.

6.3 Solutions to Common Faults

Fault Description	Possible Cause	Solution
There is no display when the drive is powered on.	The mains voltage is absent or too low.	Check the power supply.
	The switch-mode power supply on the drive board is faulty.	Check that the 24 V and 10 V voltage outputs from the control board are normal.
	The control board is disconnected from the drive board or operating panel.	Reconnect the cables between these components.
	The snubber resistor of the drive is damaged.	Seek technical support.
	The control board or operating panel is faulty.	Seek technical support.
	The rectifier bridge is damaged.	Seek technical support.
Error code r01.01 is displayed when the drive is powered on.	The motor or power output cable is short-circuited to the ground.	Measure the insulation resistance of the motor and power output cable.
	The drive is damaged.	Seek technical support.

Fault Description	Possible Cause	Solution
Error code r06.00 (over-temperature) is displayed frequently.	The carrier frequency is too high.	Reduce the carrier frequency (P0-14).
	The fan is damaged or the cooling air channel is blocked.	Replace the fan or clean the air channel.
	The thermistor or other components in the drive are damaged.	Seek technical support.
	A wrong power value is selected for the drive.	Check whether the environmental conditions, such as the temperature and altitude, are suitable for the product model. If not, select a power value matching the environmental conditions for the drive.
The motor does not rotate when the drive is running.	Cables are not connected properly between the drive and motor.	Check that the cable connection between the drive and motor is correct.
	Motor parameters are not set properly.	Restore factory settings and re-set the related parameters properly:
		Check that the encoder parameters and motor ratings, such as rated frequency and rated speed, are set properly.
		Check that P0-00 (control mode) and P0-01 (running mode) are set properly.
		Adjust P3-03 (torque boost) in V/f control mode under heavy load.
	The cable between the drive board and control board is not securely connected.	Reconnect the cable securely.
A DI terminal fails.	The drive board is faulty.	Seek technical support.
	The relevant parameters are set incorrectly.	Check and modify relevant parameters in group P4.
	External signals are incorrect.	Reconnect external signal cables.
The drive reports overcurrent and overvoltage errors frequently.	The control board is faulty.	Seek technical support.
	The motor parameters are set incorrectly.	Set motor parameters or perform motor auto-tuning.
	The acceleration/deceleration time is improper.	Set a proper acceleration/deceleration time.
	Load fluctuates.	Seek technical support.

6.4 Error Code

No.	Fault Name	Possible Cause	Solution
r01.01	Short to ground	The output side is short-circuited to the ground.	Check whether the motor or intermediate contactor is short-circuited by measuring the grounding impedance of the cables with a multimeter.
r01.02	Current detection failure	The zero drift of three-phase output current is too large.	Seek technical support.
r01.03	Pre-charge resistor failure	Relay conduction timed out.	Seek technical support.
r01.04	EEPROM failure	● Parameter settings are out of allowed range. ● An error occurred while reading and writing functional parameters.	If the EEPROM chip is damaged, seek technical support.
r01.05	ADC sampling failure	ADC conversion overflow occurs.	If ARM is damaged, seek technical support.
r01.07	Bus short circuit	The bus is short-circuited and experiences overcurrent.	Seek technical support.
r01.08	24 V power failure	The 24 V power supply is abnormal.	Seek technical support.
r01.09	Inter-phase short circuit	An inter-phase short circuit occurs.	Check whether different phases are short-circuited.
r02.01	Hardware overcurrent in acceleration stage	● Short to ground, phase-to-phase short circuit, or abnormal load is detected on the output cable and motor of the drive. ● The model of the motor or drive is found incorrect in vector control, and parameter auto-tuning is not performed. ● The acceleration time is set improperly. ● The V/f overcurrent stall suppression parameter values are too large. ● The manual torque boost of the high-power motor is too large or the V/f curve is not reasonable. ● The motor is started while rotating. ● A control exception is caused by external interference.	● Use a multimeter to check the cable or contactor, and measure the inter-phase resistance and load of the motor to determine whether the motor is short-circuited. ● Restore the parameters to factory defaults, select the correct drive model, enter the motor parameters, and tune the motor parameters. ● Set P0-15 to an appropriate value. ● Make sure overcurrent stall suppression has been enabled, and P3-10 is set to a value in the recommended range of 120% to 160%. ● Adjust the manual torque boost or V/f curve. ● Enable speed tracking start or wait for the motor to stop before starting the drive. ● Seek technical support.

No.	Fault Name	Possible Cause	Solution
r02.02	Hardware overcurrent in deceleration stage	● Short to ground, phase-to-phase short circuit, or abnormal load is detected on the output cable and motor of the drive. ● The model of the motor or drive is found incorrect in vector control, and parameter auto-tuning is not performed. ● The deceleration time is set improperly. ● The V/f overcurrent stall suppression parameter values are too large. ● A control exception is caused by external interference.	● Use a multimeter to check the cable or contactor, and measure the inter-phase resistance and load of the motor to determine whether the motor is short-circuited. ● Restore the parameters to factory defaults, select the correct drive model, enter the motor parameters, and tune the motor parameters. ● Set P0-16 to an appropriate value. ● Make sure overcurrent stall suppression has been enabled, and P3-10 is set to a value in the recommended range of 120% to 160%. ● Seek technical support.
r02.03	Hardware overcurrent in constant speed stage	● Short to ground, phase-to-phase short circuit, or abnormal load is detected on the output cable and motor of the drive. ● The model of the motor or drive is found incorrect in vector control, and parameter auto-tuning is not performed. ● The V/f overcurrent stall suppression parameter values are too large. ● A control exception is caused by external interference.	● Use a multimeter to check the cable or contactor, and measure the inter-phase resistance and load of the motor to determine whether the motor is short-circuited. ● Restore the parameters to factory defaults, select the correct drive model, enter the motor parameters, and tune the motor parameters. ● Make sure overcurrent stall suppression has been enabled, and P3-10 is set to a value in the recommended range of 120% to 160%. ● Seek technical support.
r02.04	Software overcurrent	● Short to ground, phase-to-phase short circuit, or abnormal load is detected on the output cable and motor of the drive. ● The model of the motor or drive is found incorrect in vector control, and parameter auto-tuning is not performed. ● The V/f overcurrent stall suppression parameter values are too large. ● The motor selected has low power, or the motor parameters are set incorrectly.	● Use a multimeter to check the cable or contactor, and measure the inter-phase resistance and load of the motor to determine whether the motor is short-circuited. ● Restore the parameters to factory defaults, select the correct drive model, enter the motor parameters, and tune the motor parameters. ● Make sure overcurrent stall suppression has been enabled, and P3-10 is set to a value in the recommended range of 120% to 160%. ● Check whether the rated parameters of the motor are correct.

No.	Fault Name	Possible Cause	Solution
r03.01	Bus voltage overvoltage in acceleration stage	● The input voltage from the power grid is too high, or a step-up transformer is installed. ● An inertia disk is used for the acceleration process, leading to unstable control. ● The setting of overvoltage suppression is inappropriate. ● No braking resistor is installed, or the braking resistor is damaged.	● Stop the drive, use a multimeter to measure the inlet voltage, and adjust the voltage to the normal range. ● Seek technical support. ● Make sure the functions matching P2-30 and P3-13 are enabled, and the values of P2-37 and P3-14 are in the recommended range of 650 V to 770 V. ● Install or replace the braking resistor.
r03.02	Bus voltage overvoltage in deceleration stage	● The setting of overvoltage suppression is inappropriate. ● The deceleration time is short. ● No braking resistor is installed, or the braking resistor is damaged.	● Make sure the functions matching P2-30 and P3-13 are enabled, and the values of P2-37 and P3-14 are in the recommended range of 650 V to 770 V. ● Increase the deceleration time. ● Install or replace the braking resistor.
r03.03	Bus overvoltage in constant speed stage	● The setting of overvoltage suppression is inappropriate. ● The motor is equipped with a large inertia device, leading to unstable speed control.	● Make sure the functions matching P2-30 and P3-13 are enabled, and the values of P2-37 and P3-14 are in the recommended range of 650 V to 770 V. ● Seek technical support.
r04.00	Bus voltage undervoltage	● An instantaneous power failure occurs. ● The input voltage is excessively low. ● The motor accelerates instantaneously with heavy load. ● The rectifier module, drive board, or control board does not work properly.	● Enable the power dip ride-through feature (P2-30). ● Measure the input voltage with a multimeter. ● Set the acceleration time (P0-15) in a reasonable range. ● Seek technical support.
r04.01	Power dip ride-through	● An instantaneous power failure occurs. ● The input voltage is excessively low. ● The motor accelerates instantaneously with heavy load. ● The rectifier module, drive board, or control board does not work properly.	● Enable the power dip ride-through feature (P2-30). ● Measure the input voltage with a multimeter. ● Set the acceleration time (P0-15) in a reasonable range. ● Seek technical support.

No.	Fault Name	Possible Cause	Solution
r04.02	Power dip ride-through (V/f control)	● An instantaneous power failure occurs. ● The input voltage is excessively low. ● The motor accelerates instantaneously with heavy load. ● The rectifier module, drive board, or control board does not work properly.	● Enable the power dip ride-through feature (P2-30). ● Measure the input voltage with a multimeter. ● Set the acceleration time (P0-15) in a reasonable range. ● Seek technical support.
r05.00	Drive overload	● The load is too heavy or rotors of the motor are blocked. ● The power rating of the drive is too low. ● Loop parameters or motor parameters are found abnormal in vector control, and parameter auto-tuning is not performed. ● The control mode is V/f, and the torque boost is too large.	● Reduce the load and check the motor and machinery. ● Replace the drive with one that has higher power rating. ● Restore all parameters to factory defaults, set the motor parameters according to the motor nameplate, and auto-tune the motor parameters. ● Reduce the torque boost voltage (P3-01).
r06.00	Module over-temperature	● The device is working in a closed environment with high temperature. ● The air filter of the device is blocked. ● The fan of the device is damaged. ● The fan control logic is unreasonable. ● The thermistor of the module or the module itself is damaged.	● Lower the ambient temperature. ● Check whether the air filter is blocked by foreign objects such as cotton. ● Check whether the fan is running. If not, replace the fan. ● Change the fan running logic (P9-20) to always running. ● Seek technical support.
r07.00	Motor overload	● The load is too heavy or rotors of the motor are blocked. ● The power rating of the motor is too low. ● Loop parameters or motor parameters are found abnormal in vector control, and parameter auto-tuning is not performed. ● The control mode is V/f, and the torque boost is too large.	● Reduce the load and check the motor and machinery. ● Select a motor of a higher power rating. ● Restore all parameters to factory defaults, set the motor parameters according to the motor nameplate, and auto-tune the motor parameters. ● Reduce the torque boost voltage (P3-01).

No.	Fault Name	Possible Cause	Solution
r08.01	Motor over-temperature detected (over-temperature detection source 1)	● The cable of the temperature sensor becomes loose. ● The temperature of the motor is too high. ● The over-temperature protection threshold of the motor is too small.	● Check the temperature sensor connection. Re-connect the temperature sensor if necessary. ● Check whether the fan of the motor works properly. ● Increase the over-temperature protection threshold of the motor.
r08.02	Motor over-temperature detected (over-temperature detection source 2)	● The cable of the temperature sensor becomes loose. ● The temperature of the motor is too high. ● The over-temperature protection threshold of the motor is too small.	● Check the temperature sensor connection. Re-connect the temperature sensor if necessary. ● Check whether the fan of the motor works properly. ● Increase the over-temperature protection threshold of the motor.
r08.03	Motor over-temperature detected (measured by PTC1)	● The cable of the temperature sensor becomes loose. ● The temperature of the motor is too high. ● The over-temperature protection threshold of the motor is too small.	● Check the temperature sensor connection. Re-connect the temperature sensor if necessary. ● Check whether the fan of the motor works properly. ● Increase the over-temperature protection threshold of the motor.
r08.04	Motor over-temperature detected (measured by PTC2)	● The cable of the temperature sensor becomes loose. ● The temperature of the motor is too high. ● The over-temperature protection threshold of the motor is too small.	● Check the temperature sensor connection. Re-connect the temperature sensor if necessary. ● Check whether the fan of the motor works properly. ● Increase the over-temperature protection threshold of the motor.
r08.05	Motor over-temperature detected (measured by PTC3)	● The cable of the temperature sensor becomes loose. ● The temperature of the motor is too high. ● The over-temperature protection threshold of the motor is too small.	● Check the temperature sensor connection. Re-connect the temperature sensor if necessary. ● Check whether the fan of the motor works properly. ● Increase the over-temperature protection threshold of the motor.
r09.00	Input phase loss	The input three-phase voltage is out of phase.	● Check the three-phase power supply and ensure that it does not experience phase loss or unbalancing between the three phases. ● Check whether the terminal or cable is connected properly. ● Seek technical support.

No.	Fault Name	Possible Cause	Solution
r10.00	Braking transistor short-circuit upon stop	● The braking unit is short-circuited. ● The resistance of the braking resistor is too small. ● The braking unit does not work properly.	● Check that the braking unit is connected properly. ● Check whether the braking resistor is short-circuited to the ground. ● Check whether the resistance and power of the selected braking resistor are too small. ● Seek technical support.
r10.01	Braking transistor short-circuit during operation	● The braking unit is short-circuited. ● The resistance of the braking resistor is too small. ● The braking unit does not work properly.	● Check that the braking unit is connected properly. ● Check whether the braking resistor is short-circuited to the ground. ● Check whether the resistance and power of the selected braking resistor are too small. ● Seek technical support.
r10.02	Braking unit overload	● The resistance of the braking resistor is too small, and the braking current is too large. ● The braking process is triggered frequently.	● Check the parameter settings. ● Select an appropriate resistor model. ● Optimize the process.
r12.00	Fan failure	The fan status is abnormal.	● Check whether the fan is blocked. ● Seek technical support.
r15.01	External fault indicated by multi-functional DI (NO)	An external fault indicated by multi-function DI (NO) occurred.	Rectify the external fault, verify that the device can be restarted, and reset the device.
r15.02	External fault indicated by multi-functional DI (NC)	An external fault indicated by multi-function DI (NC) occurred.	Rectify the external fault, verify that the device can be restarted, and reset the device.
r15.06	External fault of process card input	External fault indicated by multi-functional DI (NO)	Rectify the external fault, confirm that the device can be restarted, and reset to run the device.
r16.00	Modbus communication timeout	Modbus communication timed out.	● Check that communication cables are connected properly. ● Check that the communication delay is set properly.
r16.01	CANopen communication timeout	CANopen communication timed out.	● Check that communication cables are connected properly. ● Check that the communication delay is set properly.

No.	Fault Name	Possible Cause	Solution
r16.02	EtherCAT communication timeout (Reserved)	EtherCAT communication timed out.	● Check that communication cables are connected properly. ● Check that the communication delay is set properly.
r16.03	PROFINET communication timeout (Reserved)	PROFINET communication timed out.	● Check that communication cables are connected properly. ● Check that the communication delay is set properly.
r17.01	Output phase loss during RS auto-tuning	● The motor fails. ● The cable between the drive and motor does not work normally. ● The three-phase outputs of the drive are unbalanced when the motor is running. ● The drive board or IGBT module does not work normally.	● Check the motor for short circuit. ● Check the cable between the drive and the motor. ● Check whether the motor three-phase winding is normal. If not, rectify the fault. ● Seek technical support.
r17.02	Output phase loss during normal operation	● The motor fails. ● The cable between the drive and motor does not work normally. ● The three-phase outputs of the drive are unbalanced when the motor is running. ● The drive board or IGBT module does not work normally.	● Check the motor for short circuit. ● Check the cable between the drive and the motor. ● Check whether the motor three-phase winding is normal. If not, rectify the fault. ● Seek technical support.
r17.03	Output phase loss (power-on detection)	● The motor fails. ● The cable between the drive and motor does not work normally. ● The three-phase outputs of the drive are unbalanced when the motor is running. ● The drive board or IGBT module does not work normally.	● Check the motor for short circuit. ● Check the cable between the drive and the motor. ● Check whether the motor three-phase winding is normal. If not, rectify the fault. ● Seek technical support.
r18.00	Overspeed fault	● Encoder parameters are set incorrectly. ● Motor auto-tuning is not performed. ● The motor overspeed check parameters (P9-15 and P9-16) are set improperly.	● Set the encoder parameters correctly. ● Perform motor auto-tuning. ● Properly set the check parameters (P9-15 and P9-16).

No.	Fault Name	Possible Cause	Solution
r19.00	Excessive speed deviation	- Encoder parameters are set incorrectly. - Motor auto-tuning is not performed. - The excessive speed deviation detection parameters are set incorrectly.	- Set the encoder parameters correctly. - Perform motor auto-tuning. - Properly set the check parameters.
r20.00	Load loss	The running current of the drive stays below the threshold for a certain period.	Check whether the load is removed and whether settings of P9-19 and P9-20 are suitable for the actual operating conditions.
r21.01	Motor stall	- The load is too heavy. - The motor parameters are set incorrectly.	- Check the motor load. - Check whether the motor parameters are set correctly and tuned.
r22.00	Auto-tuning not completed	Auto-tuning fails to be completed.	- Set the motor parameters based on the motor nameplate. - Check the connection between the drive and the motor. - Check the motor. - Seek technical support.
r22.01	Auto-tuning timeout	The auto-tuning process has timed out, and is not completed after a long period.	- Set the motor parameters based on the motor nameplate. - Check the connection between the drive and the motor. - Check the motor. - Seek technical support.
r22.02	Back EMF auto-tuning failure	The maximum frequency (P0-08), maximum reverse frequency (P0-09), and upper-limit frequency (P0-11) are small.	Increase the maximum frequency, upper-limit frequency, and maximum reverse frequency above the motor's rated frequency.
r22.03	Magnetic pole position auto-tuning failure of synchronous motor	The drive provides voltage output, but the current is small.	Make sure the motor is connected properly.
r22.04	Inductance curve auto-tuning failure of synchronous motor	The drive provides voltage output, but the current is small.	- Check the motor parameter settings. - Seek technical support.
r22.05	Moment of inertia auto-tuning failure	The maximum frequency (P0-08), maximum reverse frequency (P0-09), and upper-limit frequency (P0-11) are small.	Increase the maximum frequency, upper-limit frequency, and maximum reverse frequency above the motor's rated frequency.

No.	Fault Name	Possible Cause	Solution
r22.06	Drive non-linear auto-tuning failure	The deviation between the current reference and current feedback is too large.	Seek technical support.
r23.00	Cumulative running time reach fault	The cumulative running time reaches the reference value.	Use the parameter initialization feature (set PE-00 to 2) to clear the records.
r24.00	Cumulative power-on time reach fault	The cumulative power-on time reaches the reference value.	Use the parameter initialization feature (set PE-00 to 2) to clear the records.
r25.00	User-defined fault 1	Signals of user-defined fault 1 are injected through the multi-functional terminal DI feature.	-
r26.00	User-defined fault 2	Signals of user-defined fault 2 are injected through the multi-functional terminal DI feature.	-
r33.00	PID feedback loss	The PID feedback is smaller than the value specified by FA-26.	Check the PID feedback signal or set FA-26 to an appropriate value.
r34.00	Initial state timeout fault	The initial power-on logic is abnormal, and has not entered the normal ready start state after a long period.	● Check bus voltage. ● Check whether the rectifier slow-start contactor is sucked together.
r35.01	Same hardware source selected for different DIs	The same hardware source is selected for different DIs.	Check whether the same hardware source is selected for different DIs.
r35.02	Same expansion card inserted (except PG card)	The same card is inserted into slot 1 and slot 2.	Check whether the same card is inserted into slot 1 and slot 2.
r35.04	Repeated DI function assignments	The same function is assigned to different DI terminals.	Check whether different DI terminals are assigned the same function.
r35.05	Same hardware source selected for different AIs	The same hardware source is selected for different AIs.	Check whether the same hardware source is selected for different AIs.
r35.06	Same hardware source selected for temperature measurement	The same hardware source is selected for temperature measurement signals.	Check whether the same hardware source is selected for different temperature measurement signals.
r35.07	Same hardware source selected for AI and temperature measurement	The same hardware source is selected for AI and temperature measurement.	Check whether the same hardware source is selected for AI and temperature measurement.
r35.08	Same hardware source selected for different AOs	The same hardware source is selected for different AOs.	Check whether the same hardware source is selected for different AOs.
r35.09	Same hardware source selected for different DOs	The same hardware source is selected for different DOs.	Check whether the same hardware source is selected for different DOs.

No.	Fault Name	Possible Cause	Solution
r35.10	Same hardware source selected for different HDOs	The same hardware source is selected for different HDOs.	Check whether the same hardware source is selected for different HDOs.
r35.11	DIO used as the hardware source for both DI and DO	DIO is used as the hardware source for both DI and DO.	Check whether DIO is used as the hardware source for both DI and DO.
r36.00	Encoder type error	An incorrect encoder type is selected.	Adjust C1-01 to select the correct encoder type, and then power the encoder off and on.
r37.00	CANopen card communication interruption (slot 1)	CANopen card communication timed out.	● Check the cable connectivity. ● Check that the baud rate is set correctly.
r37.03	Process card communication interruption (slot 1)	Process card communication timed out.	● Check the cable connectivity. ● Check that the baud rate is set correctly.
r37.04	I/O card communication interruption (slot 1)	I/O card communication timed out.	● Check the cable connectivity. ● Check that the baud rate is set correctly.
r37.05	Resolver PG card communication interruption (slot 1)	Resolver PG card communication timed out.	● Check the cable connectivity. ● Check that the baud rate is set correctly.
r37.06	Incremental PG card communication interruption (slot 1)	Incremental PG card communication timed out.	● Check the cable connectivity. ● Check that the baud rate is set correctly.
r37.07	Absolute PG card communication interruption (slot 1)	Absolute PG card communication timed out.	● Check the cable connectivity. ● Check that the baud rate is set correctly.
r37.08	Relay I/O card communication interruption (slot 1)	Relay I/O card communication timed out.	● Check the cable connectivity. ● Check that the baud rate is set correctly.
r37.10	CANopen card communication interruption (slot 2)	CANopen card communication timed out.	● Check the cable connectivity. ● Check that the baud rate is set correctly.
r37.13	Process card communication interruption (slot 2)	Process card communication timed out.	● Check the cable connectivity. ● Check that the baud rate is set correctly.
r37.14	I/O card communication interruption (slot 2)	I/O card communication timed out.	● Check the cable connectivity. ● Check that the baud rate is set correctly.

No.	Fault Name	Possible Cause	Solution
r37.15	Resolver PG card communication interruption (slot 2)	Resolver PG card communication timed out.	● Check the cable connectivity. ● Check that the baud rate is set correctly.
r37.16	Incremental PG card communication interruption (slot 2)	Incremental PG card communication timed out.	● Check the cable connectivity. ● Check that the baud rate is set correctly.
r37.17	Absolute PG card communication interruption (slot 2)	Absolute PG card communication timed out.	● Check the cable connectivity. ● Check that the baud rate is set correctly.
r37.18	Relay I/O card communication interruption (slot 2)	Relay I/O card communication timed out.	● Check the cable connectivity. ● Check that the baud rate is set correctly.
r37.21	PROFINET card communication interruption (slot 3)	PROFINET card communication timed out.	● Check the cable connectivity. ● Check that the baud rate is set correctly.
r37.22	EtherCAT card communication interruption (slot 3)	EtherCAT card communication timed out.	● Check the cable connectivity. ● Check that the baud rate is set correctly.
r39.01	Brake opening error	Brake opening timed out.	● Check whether the mechanical brake is normal.
r42.01	Copying external keypad parameter interrupted	Parameter copy is interrupted.	● Check whether parameter copy is interrupted.
r42.02	Downloading external keypad parameter interrupted	Parameter download is interrupted.	● Check whether parameter download is interrupted.
r42.03	Downloading external keypad parameter timeout	Parameter download timed out.	● Check whether parameter download timed out.

Chapter 7

Expansion Card

7.1 Expansion Card Types

Name		Model	Installation Position
Functional expansion card	I/O expansion card (standard)	AMD760-IO-A1	Slot 1 or slot 2
	Relay I/O expansion card (push-in terminal wiring)	AMD760-IO-A3	
	Relay I/O expansion card (screw terminal wiring)	AMD760-IO-A4	
Communication expansion card	CAN communication card	AMD760-CAN-A1	

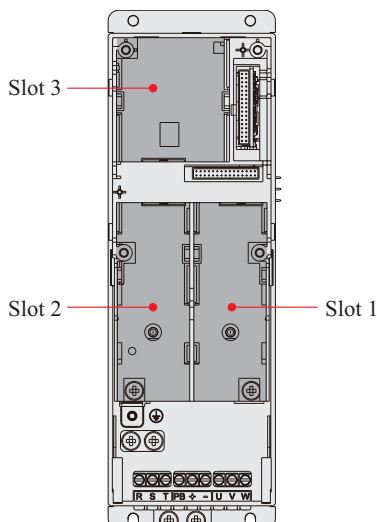


Figure 7-1 Positions of expansion card slots

NOTICE

- Figures in this chapter illustrate the slots and installation of expansion cards on SIZE B. The expansion card slots on other models are similar.

7.2 Installation Procedure

- ① Remove the cover following the steps in section 2.7 "Cover Removal and Installation", and then push the expansion card horizontally into slot 1 or slot 2 along the direction shown in the following figure, until the top terminals are completely embedded into the device.
- ② Use an M3 screw to secure the expansion card on the device (torque: 0.5-0.6 N·m).

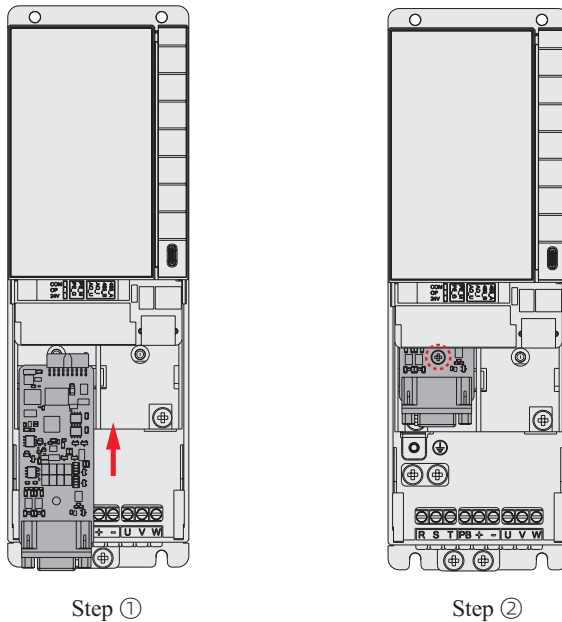


Figure 7-2 Installing an expansion card

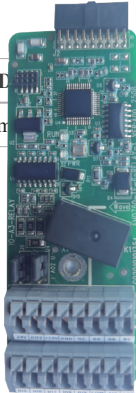
7.3 Terminal Description

■ CAN communication card										Identifier	Description																				
AMD760-CAN-A1  <table><tr><td>1</td><td>485+</td><td>485-</td><td>+5V_C</td><td>CGND</td><td>PTC</td><td>GND</td><td>TEMP</td><td>GND</td><td>8</td></tr><tr><td>1</td><td>PE</td><td>CGND</td><td>CANHB</td><td>CANLB</td><td>AI5</td><td>GND</td><td>AI6</td><td>GND</td><td>8</td></tr></table>										1	485+	485-	+5V_C	CGND	PTC	GND	TEMP	GND	8	1	PE	CGND	CANHB	CANLB	AI5	GND	AI6	GND	8	485+	Absolute encoder RS485 signal
										1	485+	485-	+5V_C	CGND	PTC	GND	TEMP	GND	8												
										1	PE	CGND	CANHB	CANLB	AI5	GND	AI6	GND	8												
										485-																					
										+5V_C	Absolute encoder 5 V power supply																				
										CGND	Encoder power grounding																				
										PTC	Motor PTC temperature signal																				
										GND	Analog ground																				
										TEMP	Temperature signal (compatible with PT100, PT1000, and KTY84)																				
										PE	Ground terminal																				
										CGND	CAN bus																				
										CANHB																					
										CANLB																					
										AI5	Analog input 5																				
GND	Analog ground																														
AI6	Analog input 6																														

1	485+	485-	+5V_C	CGND	PTC	GND	TEMP	GND	8
1	PE	CGND	CANHB	CANLB	AI5	GND	AI6	GND	8

 ■ I/O expansion card | | | | | | | | | | Identifier | Description || AMD760-IO-A1 | | | | | | | | | | | |---|-----|-----|-----|------|-----|-----|------|-----|----| | 1 | DI5 | DI6 | DI7 | +24V | OP | COM | +10V | GND | 15 | | 2 | DI8 | DI9 | COM | DO3 | DO4 | AI3 | AI4 | AO2 | 16 | | | | | | | | | | | DI5 | Digital input 5 |
DI6	Digital input 6
DI7	Digital input 7
+24V	Digital 24 V power output
OP	Common terminal for input digital signals
COM	Digital ground
+10V	10 V power output
GND	Analog ground
DI8	Digital input 8
DI9	Digital input 9
COM	Digital ground
DO3	Digital output 3
DO4	Digital output 4
AI3	Analog input 3
AI4	Analog input 4
AO2	Analog output 2

1	DI5	DI6	DI7	+24V	OP	COM	+10V	GND	15
2	DI8	DI9	COM	DO3	DO4	AI3	AI4	AO2	16

■ Relay I/O expansion card (push-in terminal wiring)		Identifier	Revision History	
			Description	
AMD760-IO-A3		24V	Digital 24 V power output	
		DO3	Digital output 3	
		+10V	10 V power output	
		Change	Version	Change Description
		AO2	-	First release
		RA	Relay output	
		RB		
		RC		
		DI5	Digital input 5	
		DI6	Digital input 6	
		DI7	Digital input 7	
		DI8	Digital input 8	
		DI9	Digital input 9	
		COM	Digital ground	
		AI3	Analog input 3	
		AO2	Analog output 2	

■ Relay I/O expansion card (screw terminal wiring)		Identifier	Description
AMD760-IO-A4		24V	Digital 24 V power output
		COM	Digital ground
		NC	-
		RA	Relay output (RA-RB is an NC terminal, and RA-RC is an NO terminal.)
		RB	
		RC	
		DI5	Digital input 5
		DI6	Digital input 6
		DI7	Digital input 7
		DI8	Digital input 8
		AO2	Analog output 2
		GND	Analog ground

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